Transforming Static Structural Topologies into Functional Reconfigurable Systems

by

Hardik Y. Patil

A dissertation submitted in partial fulfillment
of the requirements for the degree of
Doctor of Philosophy
(Civil Engineering and Scientific Computing)
in the University of Michigan
2026

Doctoral Committee:

Associate Professor Evgueni T. Filipov, Chair
Professor Jeff Scruggs
Professor Kevin Maki
Professor Lorenzo Valdevit

Hardik Y. Patil
hardikyp@umich.edu
ORCID iD: 0009–0006–9191–3738

DEDICATION

Dedicated to the loving memory of my grandpa, Liladhar G. Bendale, who would have been the happiest and proudest person on Earth to see the end of this adventure!

ACKNOWLEDGEMENTS

The research presented in this dissertation is based on work supported in part by the Automotive Research Center Cooperative Agreement W56HZV-19-2-0001, the National Science Foundation Grant No. 1943723, and the Offce of Naval Research Grant No. N00014-20-5-B001 #13209604. Any opinions, fndings, conclusions, or recommendations expressed in this dissertation are those of the author and do not necessarily refect the views of these institutions.

TABLE OF CONTENTS

LIST OF FIGURES

FIGURE

LIST OF TABLES

TABLE

LIST OF APPENDICES

APPENDIX

ABSTRACT

Most structural systems are designed to remain in a fixed configuration throughout their service lives. This fixed-form design philosophy has produced durable and reliable structures with well-defined load paths, efficient material use, and predictable load response. However, emerging applications such as post-disaster replacement infrastructure and compactly stowable systems for space exploration require structures that can change shape and withstand loads during storage, deployment, operation, or reuse. Existing deployable and reconfigurable systems offer several mechanisms for shape transformations, but many remain difficult to translate into large-scale load-bearing systems. Origami systems are difficult to scale, even with thick panels, because stress concentrations, panel buckling, and hinge failures can disrupt reliable load transfer. Mechanical metamaterials enable energy absorption, multistability, and frequency band gaps, but their reliance on specialized materials and small-scale architectures makes large-scale structural integration challenging. The central challenge, therefore, is to introduce reconfigurability into structural systems without sacrificing load-bearing capacity, stability, or application-specific mechanical response.

To address this challenge, this dissertation introduces methods for transforming static structural topologies into functional reconfigurable systems. The work focuses on trusses because they provide a fundamental structural topology with simple geometry, connectivity, load paths, and high stiffness-to-weight efficiency. The proposed design framework converts triangular units in static trusses into flat-foldable quadrilateral linkages by strategically adding joints in accordance with Grashof's flat-foldability criterion and member-force information from prescribed design loads. This transformation introduces kinematic degrees of freedom while preserving the load capacity of the original topology. The dissertation also develops a sequential kinematic analysis tool to evaluate deployment trajectories, actuation sequences, and packing efficiency. Fabrication-aware design strategies account for member thickness, joint clearances, and physical interference to prevent overlap and interlocking during deployment of prototypes. Load tests on proof-of-concept prototypes demonstrate that transformed trusses can achieve large geometric reconfiguration while retaining meaningful stiffness and load-carrying capacity.

Next, we introduce rotational stiffness at the folding joints of reconfigurable trusses to improve robustness against load reversal and enable functional applications such as controlled deployment, energy absorption, and multistability. A Three-node Torsional Spring (3NTS) element is formulated and validated against analytical solutions that is capable of capturing buckling instabilities.

We build on the 3NTS element to model nonlinearity in the spring for bi- and multi-stable hinges, including a magnetic hinge design that uses magnetic interactions to create multiple stable configurations. Reconfigurable trusses augmented with these hinges exhibit sequential collapse, multiple stable states, showing how programmable hinge behavior can support applications such as reusable impact absorption, controlled standoff in military vehicles, and path planning in robotics

Finally, we extend this dissertation's philosophy beyond civil engineering by shifting the focus towards curved-origami-inspired shape-morphing hulls. This work demonstrates how curved-origami principles can create rapidly deployable hulls from flat sheets. These curved origami hulls can be designed to match conventional hull forms and actively change shape to tune hydrodynamic performance on demand.

Together, the work presented in this dissertation provides a framework for moving from isolated reconfigurable mechanisms toward functional structural systems that combine controlled motion with programmable mechanical performance.

CHAPTER 1

Introduction

1.1 Motivation: From Fixed-form Structures to Functional Reconfigurable Systems

Historically, most load-bearing structural systems were conceived as fixed objects that remain in a prescribed configuration throughout their service lives. The geometry, connectivity, and primary function of buildings, bridges, towers, and other civil infrastructure were established during the design phase and remained unchanged after construction. This design philosophy reflected the core priorities of structural engineering, namely, durability, safety, and strength [1]. In this context, fixed geometry was not a limitation. It was a practical basis for designing structures with well-defined load paths and predictable service behavior under expected loads.

The persistence of fixed-form structural designs is closely tied to the development of structural engineering itself. Before the rise of modern engineering, resilience in structures was often achieved through practices such as oversizing members, introducing redundancy, and ensuring reparability. With the rise of calculation-based design, these practices were supplemented by methods aimed at achieving safety and durability with greater material efficiency [2]. Advances in materials, computational analysis, and construction and fabrication techniques have since expanded the efficiency, scale, and complexity of structural forms [3]. Despite these advances, most structures continue to be designed for single use cases, even though modern tools now allow engineers to explore complex geometries and mechanical behavior.

This fixed-form paradigm is becoming increasingly restrictive, especially for a range of emerging applications that require structures to change configuration during storage, deployment, operation, or reuse. Emergency infrastructure, compactly stowable systems, deployable shelters, adaptive load-bearing assemblies, marine structures, and future space-based construction all benefit from systems that can change shape or mechanical response in a controlled manner [4–12]. In these settings, the utility of a structure depends on transportability, deployability, adaptability, and performance across multiple configurations, in addition to strength and durability in any one configuration. These emerging demands motivate the study of functional reconfiguration, defined here as controlled changes in structural form or mechanical response that preserve, or enhance, the structure's load-bearing purpose. The central challenge, therefore, is to create structures that incorporate reconfigurability without sacrificing stability, load capacity, or application-specific mechanical response.

1.2 Current State of Reconfigurable Systems in Engineering

Over the past century, researchers have developed a wide range of reconfigurable systems for applications that require compact storage for efficient transportation, rapid deployment from stowed configuration, and controlled shape change for functional uses. These systems have largely emerged from automotive, aerospace, mechanical, and robotics engineering, where kinematic performance, packing efficiency, actuation, and adaptability are primary design drivers.

The resulting body of literature spans many distinct families of systems. Mechanical linkages, including Scissor mechanism and Ackerman linkage shown in Fig. 1.1 (a), use connected bars and

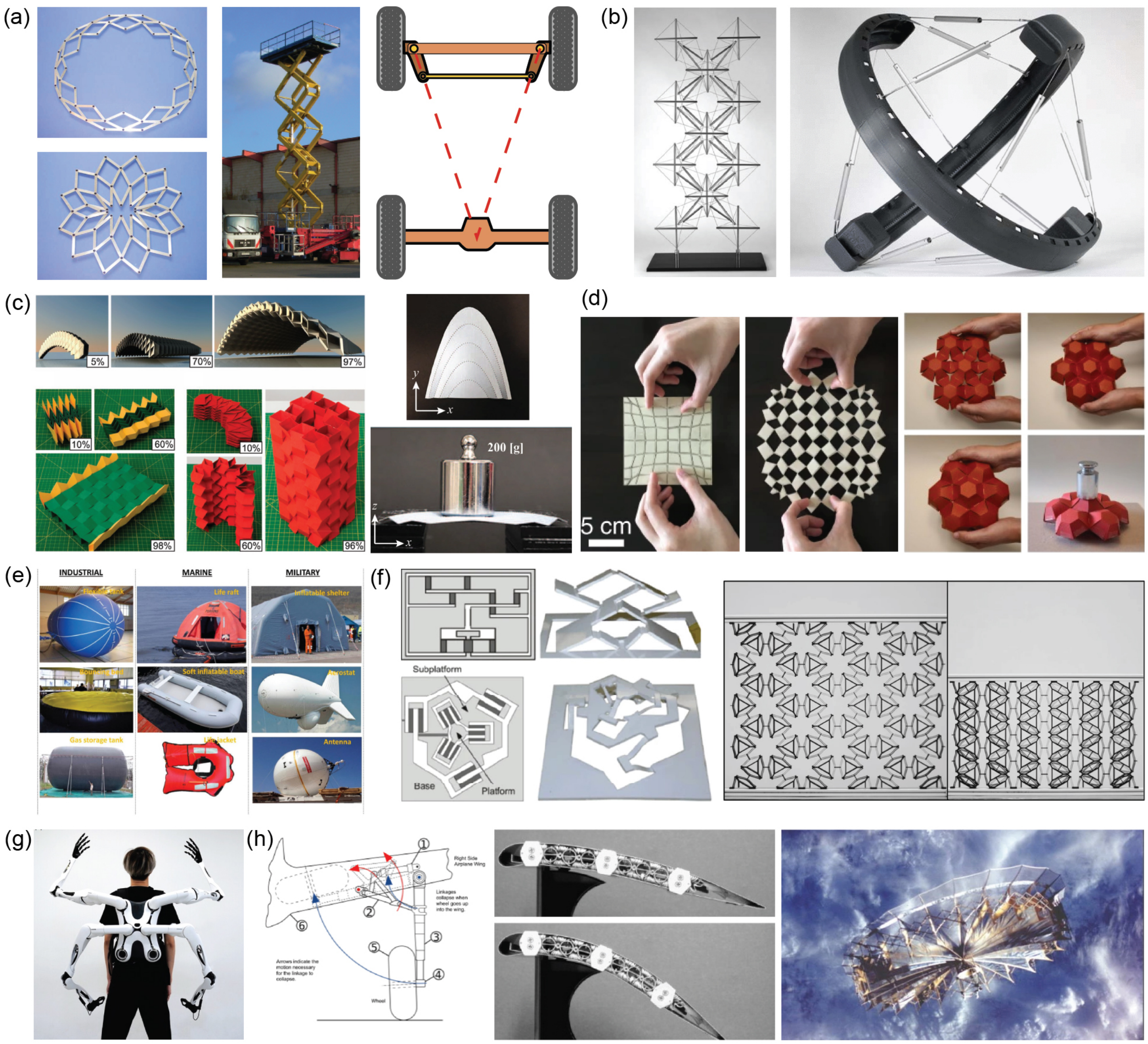


Figure 1.1: Examples of reconfigurable systems across engineering disciplines. (a) Bar-linked mechanisms: Hoberman linkage [13] (left, licensed under CC BY-NC-ND 3.0.), Scissor lift [14] (middle, licensed under CC BY-SA 2.0 Germany) and Ackermann steering linkage [15] (right, licensed under CC BY-SA 3.0); (b) Tensegrity structures: planar tensegrity mast [16] (left, adapted with permission from Sage Publications) and sperical tensegrity robot [17] (Copyright © 2016, IEEE); (c) Origami tubes for stiff columns, bridges and domes [18] (left) and curved-crease origami [19] (right, adapted with permission from Springer Nature); (d) Kirigami systems: planar auxetic kirigami [20] (left, adapted with permission from Springer Nature) and pop-up kirigami domes [21] (right, adapted with permission from Elsevier); (e) Membrane based inflatable systems [22] (licensed under CC BY-NC-ND 4.0); (f) Compliant mechanisms [23] (adapted with permission from Springer Nature) and multistable metamaterials [24] (licensed under CC BY 4.0); (g) Robotic arms [25] (licensed under CC BY 4.0); and (h) Aerospace systems: aircraft landing gear [26] (left), shape morphing wings [27] (middle, adapted with permission from Elsevier) and mesh reflector antenna deployed in orbit [28] (right, adapted with permission from Springer Nature).

rotational joints to produce prescribed motion, expansion, and controlled deployment. Tensegrity assemblies consist of isolated compressive struts within a network of prestressed tensile tendons. The balance between these compressive and tensile members enables the creation of lightweight, deployable space masts and towers, as shown in Fig. 1.1 (b). Origami-inspired systems use folding of flat sheets along prescribed crease patterns to generate complex, three-dimensional structures. Origami principles help create structures with intricate surface topologies, architected mechanical responses, and enable compact storage, deployment, and reconfigurability, as shown in Fig. 1.1 (c). Kirigami systems extend the idea of origami further by incorporating strategically placed cuts or slits into thin sheets of material. These cuts help generate auxetic behavior, enhanced flexibility, tunable mechanical properties, and large shape transformations that are difficult to achieve solely through folding, as shown in Fig. 1.1 (d). Pneumatic and membrane-based structures use thin flexible sheets that gain shape and stiffness through internal pressure, prestress, or supporting boundary elements, as shown in Fig. 1.1 (e). Compliant, multistable, and metamaterial systems use elastic instabilities and geometric nonlinearities to program motion, dissipate energy from external loads, or maintain multiple stable states, as shown in Fig. 1.1 (f). Robotic and aerospace systems combine structural components with actuation, sensing, and control to interact with their environment, as shown in Fig. 1.1 (g) and (h).

Despite significant progress, incorporating deployable and reconfigurable systems into conventional large-scale load-bearing structures remains challenging. Civil structures must satisfy strict requirements for stiffness, strength, stability, constructability, reliable connections, and predictable load transfer under prescribed loading conditions. Many reconfigurable systems provide effective mechanisms for shape change, but their changing geometry, complex joints, and material assumptions pose challenges during large-scale implementation. For instance, origami-inspired systems offer powerful ways to transform flat sheets into complex three-dimensional forms. However, at large scales, the thin-sheet mechanics of origami must be adapted to incorporate finite-thickness panels to ensure robust load transfer between components. Even with thickness-accommodation strategies, these systems can develop unintended moments, localized strains, and complex hinge behavior along creases. Compliant mechanisms, multistable systems, and metamaterials provide useful strategies for programmed deformation and energy absorption from external loads. However, many remain tied to small-scale architectures, specialized materials, or unit-cell-level behavior, which do not directly translate to large structural assemblies with conventional members, joints, and supports. Thus, existing systems offer many principles for creating deployable and reconfigurable systems, but far fewer methods for incorporating those principles into established structural topologies that engineers already understand and trust. The next section uses this distinction to define the specific research problem addressed in this dissertation.

1.3 Problem Statement: Integrating Reconfigurability into Existing Structural Forms

The preceding section shows that reconfigurable systems have been studied across many fields. However, many of these systems are designed with a bottom-up philosophy, in which the unit cell is repeated to create a larger system. This dissertation considers the inverse problem of introducing reconfigurability into an existing structural topology without losing the structural function that made the original topology useful.

The central gap is the lack of generalized frameworks for transforming existing structural topologies into reconfigurable systems. Conventional structures such as trusses, frames, and hulls are typically designed to withstand prescribed loads in a fixed configuration. In contrast, reconfigurable systems must satisfy two coupled requirements: they must move between useful configurations and retain predictable mechanical behavior during and after reconfiguration. Existing studies often address one of these requirements in isolation. Mechanism-based studies emphasize motion, while conventional structural design emphasizes load resistance. The unresolved problem is how to combine these requirements in a systematic way.

This gap has three parts. First, there is a need for design methods that begin with conventional structural topologies, rather than with specialized reconfigurable mechanisms. Second, there is a need for analysis tools that can capture the kinematics, load demands, and stability of the resulting systems through large geometric changes. Third, there is a need to understand how added mechanical features, such as rotational stiffness and nonlinear hinge behavior, can turn reconfiguration from a geometric capability into a functional structural response.

We address this broad problem in the context of two structural systems in this dissertation. The primary focus is on truss systems due to their simple and well-defined geometry, connectivity, and load paths. This makes trusses an appropriate platform for studying how static structural topologies can be transformed into reconfigurable systems. The dissertation then extends this broader philosophy to ship hulls, where curved origami principles are used to introduce compact storage, rapid deployment, and on-demand shape-morphing behavior within an established hull form. These two contexts differ in geometry and application. However, they share a common objective: to move from isolated reconfigurable mechanisms to functional reconfigurable structural systems.

1.4 Objectives of the dissertation

This dissertation addresses the central gap identified in the previous section through four objectives.

Objective 1: Develop a framework for transforming static trusses into reconfigurable systems.

We develop a framework for transforming static truss designs into reconfigurable systems using principles of planar quadrilateral linkages. The framework introduces kinematic degrees of freedom into selected truss units, enabling the transformed systems to change shape while preserving the primary load paths, stiffness, and peak load capacity of the original designs. We also develop fabrication strategies that allow the transformed systems to deploy as intended without member interference or loss of strength due to out-of-plane moments.

Objective 2: Develop computational tools for kinematic and mechanical analysis.
We develop computational tools to evaluate the kinematic and structural response of the reconfigurable trusses. The kinematic tool uses a sequential actuation framework to simulate deployment paths and quantify the packing efficiency of the transformed systems. The structural tool uses a geometric nonlinear analysis framework with axial bar and torsional spring elements to evaluate actuation load demands, behavior during load reversal, and structural response during large deformations.

Objective 3: Incorporate programmed hinge behavior into reconfigurable trusses for functional applications. We develop programmed rotational hinges that introduce prescribed nonlinear moment-rotation behavior into reconfigurable truss systems. This objective combines a physical hinge design based on magnetic interactions between alternating-polarity magnets with nonlinear torsional spring models that capture bistable and multistable rotational responses. We show that these programmed hinges can assign multiple stable states to independent degrees of freedom in reconfigurable trusses, enabling functional applications such as reusable impact absorption infrastructure.

Objective 4: Broaden the framework beyond civil truss systems through curved-crease origami hulls.
We extend the dissertation's philosophy beyond civil structures by using curved-crease origami principles to obtain shape-morphing ship hulls. This objective develops flat-foldable and rapidly deployable hull systems that match conventional planing hull geometries and morph shape for on-demand tunability of hydrodynamic performance. Through this extension, we show that reconfigurability can also be used to rethink established structural forms in naval architecture to achieve

functional benefits, such as adaptable hydrodynamic performance.

1.5 Organization of the Dissertation

The dissertation is organized as follows to address the objectives described in the previous section:

Chapter 2 introduces a method for transforming static trusses into reconfigurable systems. This method uses Grahsof's flat-foldability criterion for planar quadrilateral linkages to strategically introduce a new node in every triangular unit of a given truss. These added nodes convert triangular units into flat-foldable quadrilateral linkages, thereby introducing kinematic degrees of freedom that enable global reconfigurability. A general workflow is presented for applying these transformations to most single-chorded truss designs, and it is supported by a sequential kinematic analysis framework to study the deployment behavior of the resulting systems. This workflow is first demonstrated on traditional truss designs and then extended to topology-optimized trusses with arbitrary geometries, to highlight its broader applicability. The chapter also presents experimental structural analysis of proof-of-concept prototypes to demonstrate fabrication feasibility and to show that the transformed systems retain load-bearing capacity and stiffness of the original designs.

Chapter 3 develops a Three-node Torsional Spring (3NTS) element to introduce rotational stiffness at the folding joints of the reconfigurable trusses developed in Chapter 2. The chapter presents a first-principles mathematical formulation of a torsional coil spring element and its implementation within a nonlinear structural solver. The formulation is validated against analytical solutions for two benchmark examples, demonstrating its ability to capture structural instabilities associated with elastic buckling. The chapter then shows how torsional springs can be integrated into reconfigurable trusses to provide functional benefits such as resistance to load reversal, controlled path-dependent deployment, and energy absorption under external loading.

Chapter 4 extends the 3NTS formulation from Chapter 3 by introducing hinge-level nonlinear constitutive models for bounded rotation, magnetic multistability, and idealized bistability. These models enable the simulation of programmable mechanical response in reconfigurable trusses, including sequential deployment and path-dependent multistability. This capability establishes a foundation for future functional applications of reconfigurable trusses such as reusable infrastructure for impact-energy management.

Chapter 5 broadens the perspective of the dissertation by extending the transformation of static structural topology beyond civil engineering and into naval architecture. In this chapter, the focus shifts from reconfigurable truss systems to shape-morphing hull structures inspired by curved-crease origami. The chapter develops curved-crease-origami-inspired planing hulls that can be deployed rapidly from flat sheets, match the shape and therefore, hydrodynamics of conventional planing hulls, and utilize active folding to change shape and tune hydrodynamic performance of the hull on demand. A proof-of-concept prototype is also presented, together with a discussion of the steps required for real-world implementation.

Chapter 6 summarizes the main contributions of the dissertation, discusses its limitations, and outlines directions for future research.

Three appendices are included to provide additional background for the analytical tools and methods used throughout the dissertation.

Appendix A presents the mathematical formulation for the positional analysis of planar quadrilateral linkages, which serves as the basis of the sequential kinematic analysis framework used to study deployment of reconfigurable trusses. This appendix also includes the positional analysis of the special six-bar linkage case discussed in Chapter 2, together with a note on interpreting the percentage of actuation of each kinematic degree of freedom in the reconfigurable truss systems.

Appendix B contains the supplementary videos of fabricated reconfigurable truss prototypes and their load-tests discussed in Chapter 2, to demonstrate failure type and load-carrying capacities.

Appendix C describes the theoretical basis of the Powersea software used for the hydrodynamic analyses of planing hulls in Chapter 5. This appendix briefly outlines the *low-aspect-ratio strip theory* used in the software, states the key assumptions, and explains why this approach is suitable for comparative hydrodynamic studies of curved-crease origami and conventional planing hulls.

CHAPTER 2

Transforming Static Trusses into Shape-morphing Systems using Principles of Quadrilateral Linkages

Trusses are valued for their simple design principles and efficient load-bearing capability. However, their slow assembly process and static topology restrict rapid deployment and reconfiguration for functional use. Mechanical linkages, in contrast, offer rapid deployability and reconfigurability but are challenging to apply as large-scale civil structures. These challenges raise the question: Can the design versatility of trusses be combined with the kinematic advantages of mechanical linkages to create structurally efficient, deployable, and reconfigurable large-scale systems? To that end, we present a method inspired by flat-foldable quadrilateral linkages to transform static trusses into compactly stowable, reconfigurable systems. An additional node is introduced on the tensile members of triangular units based on Grashof linkage principles. This node converts triangles into flat-foldable quadrilateral linkages, enabling system-level reconfigurability while preserving the load capacity, stiffness and stability of the structure. We show that the Fink, Scissor, and Warren trusses can be transformed into reconfigurable systems, achieving up to 93% and 60% reduction in convex hull area and maximum length, respectively, upon actuation of all degrees of freedom. Our method also extends to topology-optimized trusses, enabling the design of functional, shape-morphing trusses for arbitrary geometries, loads, and support conditions. Proof-of-concept prototypes, including a reconfigurable cantilever and a three-meter Warren truss bridge, validate feasibility while demonstrating load capacities and stiffness comparable to their static counterparts. We believe the proposed method will advance the design, analysis and fabrication of sophisticated bar-linked reconfigurable structures with potential applications in deployable infrastructure, aerospace systems, robotic components, consumer devices, metamaterials, and more. This chapter has been adapted from **Patil, H. Y.**, & Filipov, E. T. (2026). Transforming static trusses into shape morphing systems using principles of quadrilateral linkages. International Journal of Solids and Structures, 113913 [29].

2.1 Introduction

Trusses embody a simple yet efficient engineering concept of arranging material in a network of triangles to create stable, load-bearing structures. This simplicity has made trusses a preferred solution for supporting large loads for around 5000 years. The earliest occurrence of trusses dates back to the third-millennium BCE [30] when timber trusses were employed to support grain storage and other agricultural activities. Over centuries, truss designs have evolved to find new forms and applications with the adoption of new materials, analysis methods and construction techniques. The transition from timber to iron and steel trusses [31, 32], coupled with advancements in analytical and finite element methods [33–35], has enabled precise sizing of members to make trusses more efficient at carrying loads. Furthermore, modern computational tools such as topology optimization [36] have enabled engineers to design and construct increasingly complex truss structures to accommodate any geometry, load, or support conditions. Despite these advances, modern trusses lack reconfigurability. The slow member-by-member assembly process of trusses prevents rapid deployment, while their static topology restricts shape-morphing for functional use. Once constructed, trusses serve a single purpose and require complete dismantling for relocation, limiting adaptability and compact storage for transport. These limitations underscore the need for innovative designs that introduce reconfigurability in truss structures without compromising structural integrity.

The need for reconfigurable systems for functional use has motivated extensive research into various bar-linked mechanical linkages. One well-studied example is the Scissor linkage [37]. This linkage utilizes straight or angulated rods to form tessellations of parallelograms, enabling transformations along linear [38], radial [39] and spherical [40] paths. Scissor linkages have been employed in a variety of applications, including deployable bridges [41], retractable arches [42, 43], retractable roofs [44], and support structures for space antennas [45]. Beyond Scissor mechanisms, overconstrained linkages such as the Bricard [46], Myard [47] and Bennett [48] linkages have been utilized to design large mobile structures by tessellating individual unit cells. These linkages have been proposed for large-scale applications, including a deployable mesh-reflector antenna [49]. Another class of reconfigurable structures is tensegrity [50], which utilizes the equilibrium of compressive bars and tensile cables to create lightweight, deployable space masts [51], robots [52] and more. Bar linkages have also been explored to design mechanisms capable of polyhedral scaling [53]. Furthermore, optimization techniques have helped achieve desired kinematic behavior and deployed shapes while ensuring adequate structural performance [54]. Some studies have also investigated topology optimization of bar-linkages [55] and the feasibility of generating various deployed shapes using single element type [56]. A wide array of bar-linked mechanisms now enable the design of reconfigurable systems for diverse applications. However, most design approaches begin by developing a deployable unit or component, which is then assembled into a larger system. While this bottom-up approach has proven effective, the overall system behavior is often constrained by the characteristics and limitations of its fundamental unit.

In contrast, starting with a desired static bar-linked structure and modifying its members to incorporate a mechanism for reconfigurability offers broader design possibilities for designing deployable systems. One such approach involves replacing rigid truss members with telescoping beams. This approach has enabled the one-dimensional deployable truss to achieve linear transformation [57] and the shape-morphing hinged truss to exhibit bending and twisting [58]. However, the transformation limits of telescopic beams often constrain overall reconfigurability. In cases where large motion is achieved, the manual release of locks to allow telescoping reduces deployment efficiency. Others have explored substituting truss bars with scissor linkages [59] and tensile members with cables [60] to enable structural transformations. While promising, these approaches often introduce complex gear-driven joints and actuators, increasing fabrication challenges. [61] presents an example of the decomposition of triangular truss units into linkages by integrating semi-rigid joints that permit reconfigurability. A similar concept is seen in the Heatherwick Rolling Bridge [62, 63], which incorporates hydraulic actuator supported hinges for structural reconfiguration. However, these existing approaches to decompose truss units represent isolated cases and lack a generalized workflow that can be applied to a broad range of truss structures. This gap highlights the need for a systematic method to transform static trusses into deployable, scalable and efficient reconfigurable systems.

To that end, this chapter aims to introduce a general workflow that converts static trusses with a series of triangles into foldable, reconfigurable systems. The proposed workflow leverages the concepts of quadrilateral linkages [64] to strategically introduce an additional node within each triangular unit of a given truss. The placement of these nodes is guided by the flat-foldability criterion for planar Grashof quadrilateral linkages, the geometry of the triangular unit, and the force characteristics of the truss members. Using this workflow, we demonstrate the transformation of different conventional truss designs into reconfigurable systems, showcasing their theoretical capacity for flat-folding and analyzing their kinematic behavior. Additionally, we illustrate how this approach is generalizable and can be applied to topology-optimized trusses, producing reconfigurable trusses with arbitrary geometries. Lastly, we show that the proposed reconfigurable trusses not only enable shape morphing but also ensure that the structure maintains its load capacity and structural stiffness by fabricating proof-of-concept prototypes. The method proposed in this work is available as an open-source code linked in Section 2.7.

The remainder of the chapter is organized as follows. Section 2.2 presents the methodology for transforming a static truss into a reconfigurable system. This section outlines the criteria for node placement in triangular units of a truss and a workflow for achieving system-level reconfigurability. Next, Section 2.3 applies the proposed method to three traditional truss designs. Here, we demonstrate the ability to transform conventional trusses into reconfigurable systems and assess the kinematic behavior of the trusses upon the sequential actuation of their kinematic degrees of freedom. The kinematic behavior of reconfigurable trusses is simulated using a sequential kinematic analysis program outlined in Appendix A. Section 2.4 extends the proposed method to topology-optimized trusses with arbitrary geometries. The subsections explain post-processing steps for topology-optimized trusses and kinematic analysis of three reconfigurable topology-optimized trusses. Next, Section 2.5 describes fabrication strategies for efficient shape-morphing behavior and demonstrates the structural behavior of proof-of-concept reconfigurable truss prototypes. Finally, concluding remarks are presented in Section 2.6.

2.2 Transforming Static Triangulated Trusses into Reconfigurable Systems

In this section, we detail the process of transforming triangular elements into reconfigurable quadrilateral linkages by introducing an additional node. We also outline a workflow based on the geometric and force-based criteria for the node placement to achieve system-level reconfigurability.

2.2.1 From Triangles to Reconfigurable Quadrilateral Linkages

A triangle is the only polygonal shape that cannot be deformed without changing the length of its sides. This unique property endows triangles with exceptional structural rigidity, making them a fundamental element in the design of truss structures. However, traditional trusses are static, and there is an ever-increasing demand for reconfigurable structures that morph their shape to serve diverse purposes. To transform a static truss into a reconfigurable system, we propose a method of transmuting the triangular units of a truss into quadrilateral linkages [64] by introducing an additional node. This additional node is introduced such that the resulting quadrilateral linkage satisfies the Grashof condition. According to the Grashof condition [65], the linkage will permit full rotation of the shortest link with respect to the adjacent links if the sum of the shortest (S) and the longest (L) links is less than or equal to the sum of the other two links (P & Q). Furthermore, the linkage can be reconfigured to fold flat and achieve a maximum reduction in its convex hull area when the node placement fulfills the *flat-foldability criterion*, expressed as:

$$S + L = P + Q\,. \tag{2.1}$$

Figure 2.1 (a) demonstrates the proposed method of transmuting a triangle into a flat-foldable quadrilateral linkage, also known as a Grashof linkage that fulfills the flat-foldability criterion.

2.2.2 Rules for Determining Node Location in an Individual Triangle

The precise location of the newly introduced node is determined based on the lengths of the members forming the triangular unit. In our method, the member of the triangular unit that remains fixed relative to the moving links in the resulting quadrilateral linkage is called the Ground link, while the member decomposed by adding a node is called the Candidate link. We introduce a new node that splits the Candidate link into two links. The link adjacent to the Ground link is made the shortest link S. The length S is derived such that the four links formed after the introduction of the node comply with the flat-foldability criterion discussed in Section 2.2.1, such that the linkage can fold flat with respect to the Ground link. It is important to note that the orientation of the S, L, P and Q links in the formed quadrilateral linkage is interchangeable and does not affect flat-foldability as long as the condition $S + L = P + Q$ is met. However, to keep things simple and consistent, we ensure that the smallest link S is always positioned next to the Ground link. The formulae to determine length S based on the side lengths of the triangle and the combination of Candidate and Ground links are shown in Fig. 2.1 (b-n).

Triangles can be categorized as Equilateral, Isosceles, or Scalene based on the lengths of their sides. Depending on the side lengths and the possible assignment of the Candidate and Ground links, there are 13 ways to introduce a node in a triangle and transform it into a flat-foldable linkage. In our analysis, we assume that the side lengths of the triangles satisfy the inequality $L_1 < L_2 < L_3$. All sides have the same length for an *Equilateral* triangle, and any valid combination of the Candidate and Ground links results in the same configuration. As a result, there is only one way to introduce a node in an equilateral triangle—at the midpoint of the Candidate link. The formula to find length S in this case is illustrated in Fig. 2.1 (b).

Isosceles triangles can be of two types. First, the sides of equal length are longer than the third, and second, the sides of equal length are shorter than the third. When the two equal sides of a triangle are longer than the third side, there are three possible orientations for the Candidate and Ground links: (Fig. 2.1c) the shorter side serves as the Ground link, and one of the longer sides is the Candidate link, (Fig. 2.1d) one of the longer sides serves as the Ground link, with the other longer side as the Candidate link; and (Fig. 2.1e) one of the longer sides serves as the Ground link, with the shorter side as the Candidate link. The formulae to find length S in these three cases when the sides equal in length are longer than the third side are shown in the respective figures. Similarly, when the two equal sides of a triangle are shorter than the third side, we have three more possible orientations for the Candidate and Ground links, as shown in Fig. 2.1 (f-h). The formulae to find length S in these three cases when the sides equal in length are shorter than the third are shown in respective figures. Thus, when we encounter an Isosceles triangle in a truss, there are six different ways to introduce a node.

In the case of a *Scalene* triangle, where all sides are unequal in length, there are six possible combinations of the Candidate and Ground links. For each of these six possible combinations, the

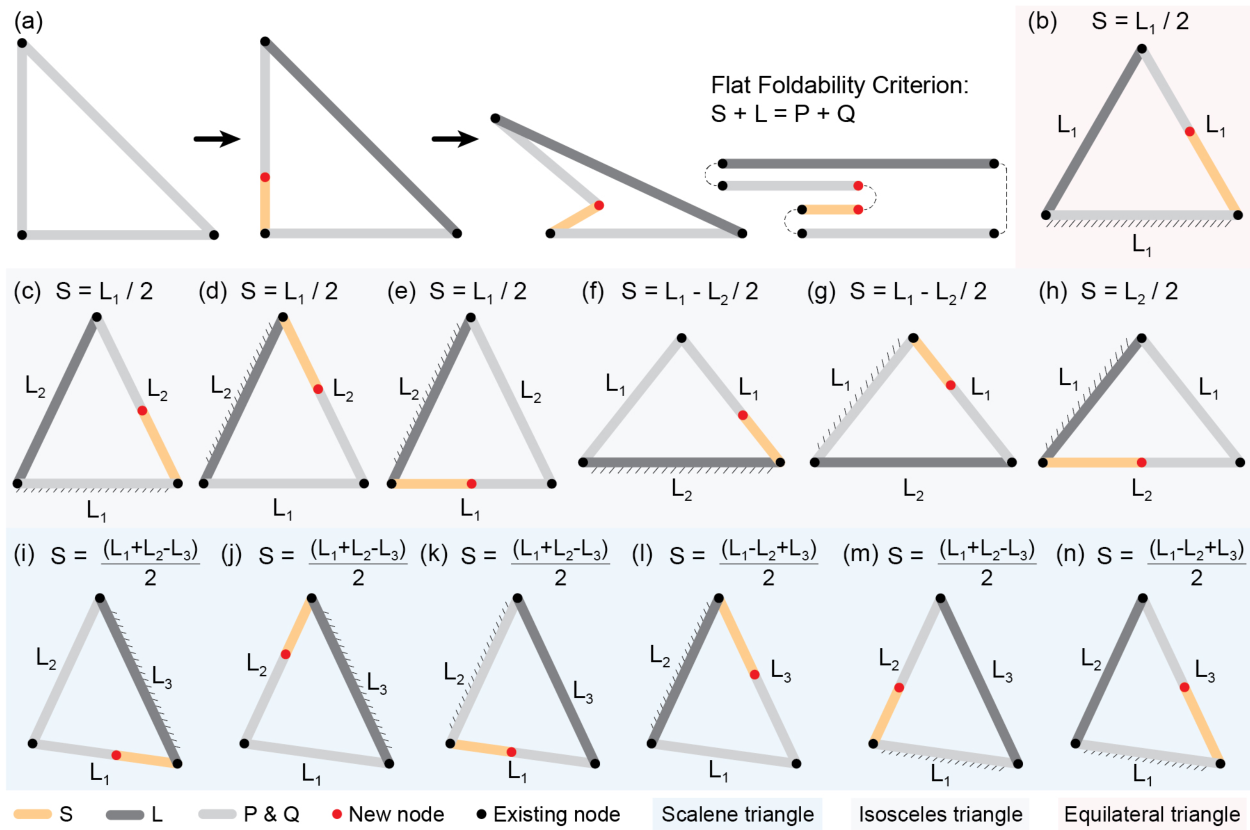


Figure 2.1: (a) Decomposing a triangle into a Grashof quadrilateral linkage, following the *flat-foldability* criterion; The location of the node to be introduced in a triangular unit is determined by calculating the length of the shortest link S when the triangle is: (b) Equilateral; (c - e) Isosceles with the common sides larger than the third side; (f - h) Isosceles with the common sides smaller than the third side; (i - n) Scalene. For all triangles, $L_1 < L_2 < L_3$. The Ground link is shown with a hatched side.

formulae to determine the length S at which the node must be placed along the Candidate link are shown in Fig. 2.1 (i-n).

2.2.3 Workflow for System Level Node Placement

In this subsection, we outline a workflow for identifying a Candidate link in each triangular unit of a static truss by considering both member forces and the configuration of adjacent triangles. We then apply the unit-level rules to introduce a new node on the Candidate links, enabling system-level reconfigurability for the initially static trusses.

Although any truss member within a triangular unit can be split by introducing a node, we focus on tensile members as Candidate links in our work. When subjected to loads, a node introduced on a tensile member remains stable without requiring locking or additional treatment. In contrast, splitting a compressive member requires locking the reconfigurable node in its deployed state to prevent unintended movement, which adds to the complexity of the deployment process. Moreover, introducing a node on a compressive member makes the structure vulnerable to buckling failure, as even minor imperfections that prevent the two links from being collinear would induce bending moments. Theoretically, limiting our focus to tensile members also preserves the structure's stiffness and peak load capacity while enabling reconfigurability. Incorporating compressive members as Candidate links could provide alternative pathways for introducing reconfigurability into truss structures, which will be explored in future work. It is important to note that the distinction between tensile and compressive members is valid for a range of load cases that result in the same overall combination of tensile and compressive members in a given truss. Load cases outside of this range would require re-evaluation based on the new tensile and compressive member assignment.

With the selection criteria in place, we now detail a workflow that is programmed into an algorithm to identify and process Candidate links across an entire truss structure. We work with the following simplifying assumptions. First, we assume the trusses are planar. Second, we restrict the scope of the workflow to trusses composed of a single layer of triangular units arranged in sequence between the top and bottom chords. Third, we assume the truss is statically determinate. Based on these assumptions, the workflow processes each triangle sequentially, tracking the following elements within the truss at any given step: (*i*) Ground link, G; (*ii*) Candidate link, C (not to be confused with compressive link); (*iii*) Avoid link, A; (*iv*) Current triangle, $\triangle_c$; (*v*) Previous triangle, $\triangle_p$; and (*vi*) Next triangle, $\triangle_n$. At each step within the current triangle, the Ground link (G) is typically defined as the side shared with the previous triangle ($\triangle_p$), while the Avoid link (A) refers to the side shared with the next triangle ($\triangle_n$).

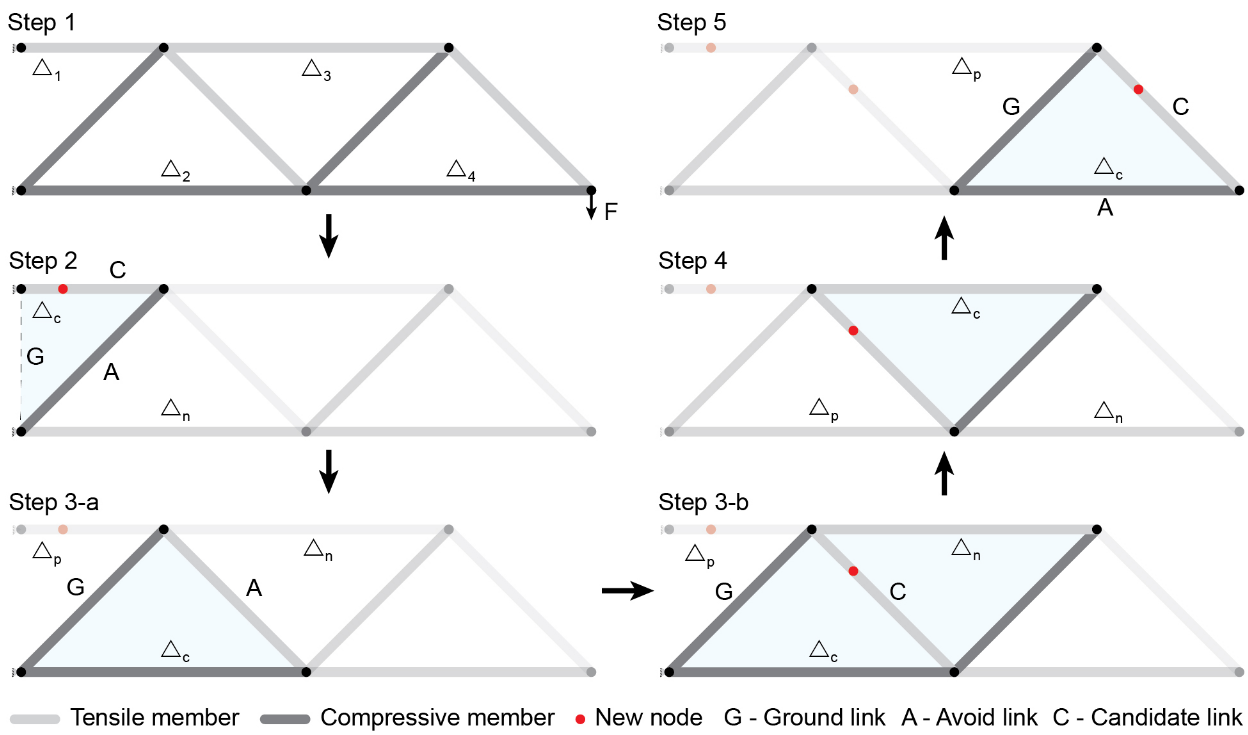


Figure 2.2: A visual representation of how the workflow processes triangles sequentially to introduce nodes.

To demonstrate how the Candidate link is selected, let us consider the cantilever truss example in Fig. 2.2. In Step 1, the triangles are numbered, and the tensile and compressive members are identified. Starting from Step 2, each triangle is analyzed individually. To start in Step 2, $\triangle_1$ is the current triangle and $\triangle_2$ is the next. The member between the two supports acts as the Ground link, while the side shared between $\triangle_c$ and $\triangle_n$ becomes the Avoid link. The remaining side, being a tensile member, is designated as the Candidate link (C), and a node is introduced based on the rules described in Section 2.2.2, which consider the lengths of G, C, and A.

In Step 3-a, $\triangle_2$ becomes the current triangle, with $\triangle_1$ and $\triangle_3$ as the previous and next triangles, respectively. The Ground and Avoid links are assigned as described earlier. Because the third side of $\triangle_c$ is not a tensile member, we check whether the Avoid link is tensile and if $\triangle_c$ and $\triangle_n$ are congruent. If both conditions are satisfied, the avoid link is reclassified as the Candidate link, and a node is introduced based on the side lengths of the current triangle, as shown in Step 3-b.

This approach consolidates two triangular units into a unique 6-bar linkage with a single kinematic degree of freedom. If the two triangles are not congruent, the same operation could be performed, however the system would not be able to fold flat.

The process continues to Step 4, where $\triangle_3$ becomes the current triangle. Throughout the process, we maintain a history of the nodes introduced in the structure. If a node has already been placed on the Ground link of the current triangle in the previous step, the workflow skips that triangle to avoid placing multiple nodes in a triangle. Consequently, in Step 4, no new node is added, and we proceed to the final triangle. This approach ensures that only one new node is introduced for any given triangle.

In Step 5, $\triangle_4$ becomes the current triangle, with $\triangle_3$ as the previous triangle. The common side between the two triangles is assigned as the Ground link. Of the two remaining sides, the compressive member becomes the avoid link while the tensile member is assigned as the candidate link. A node is then introduced on this Candidate link.

While this process applies to most triangles in the truss, three additional rules are embedded in the workflow to handle special cases. First, if the first triangle in the structure does not have two support nodes, the compressive member connected to the support node is classified as the Ground link. Second, the workflow skips the current triangle, leaving it as a static part of the system if: (a) none of the non-ground links are tensile, or (b) the side shared between $\triangle_c$ and $\triangle_n$ is tensile but the triangles are not congruent. Therefore, only the tensile nature of the non-ground links is considered for the selection of the Candidate link. Third, the tensile member with the higher cross-sectional area is selected as the Candidate link if two tensile members exist in the current triangle and no side is shared with the next triangle. A video illustrating the steps of the proposed workflow is provided as Supplementary Video 1.

Each node (pinned joint) introduced by the proposed workflow introduces a degree of freedom, imparting a mobility of one to the truss system. This added mobility, while enabling the structure's reconfigurability, introduces potential movement that renders the system structurally unstable. However, upon the application of an external force, tensile forces are generated along the members where new nodes have been introduced. This tension effectively secures the pinned joints, as the force pulls the tensile members taut, limiting relative movement around each pin. Consequently, the applied force stabilizes the structure under load. Thus, the reconfigurable trusses generated through our approach can be considered force-stable and can exhibit shape-morphing behavior in the absence of an applied external load.

2.3 Transforming Traditional Trusses into Reconfigurable Systems

In this section, we demonstrate the reconfigurable trusses obtained with the proposed method and examine their spatial movement using three representative designs commonly found in practice: (*i*) Scissor truss, (*ii*) Fink truss, and (*iii*) Warren truss.

A sequential kinematic simulator was developed to analyze the kinematic behavior of the modified truss systems. This program utilizes the analytical solution of a planar quadrilateral linkage to simulate the motion of each kinematic degree of freedom (DOF) in the structure, one at a time. Users can specify the extent of actuation for each DOF, with 0% representing the initial state and 100% corresponding to the flat-folded configuration. At 0% actuation, the decomposed links of the Candidate link are collinear, forming an angle of 180° relative to each other. An X% actuation corresponds to X% of the total angular rotation required to rotate the shorter of the decomposed links relative to the Ground link and make them collinear. Depending on the geometry of the quadrilateral linkage, the 100% actuated state results in the shorter of the decomposed links folding to either 0° or 180° relative to the Ground link. Details are provided in Appendix A.

During the actuation of each DOF, we track two parameters to assess the packing density of the

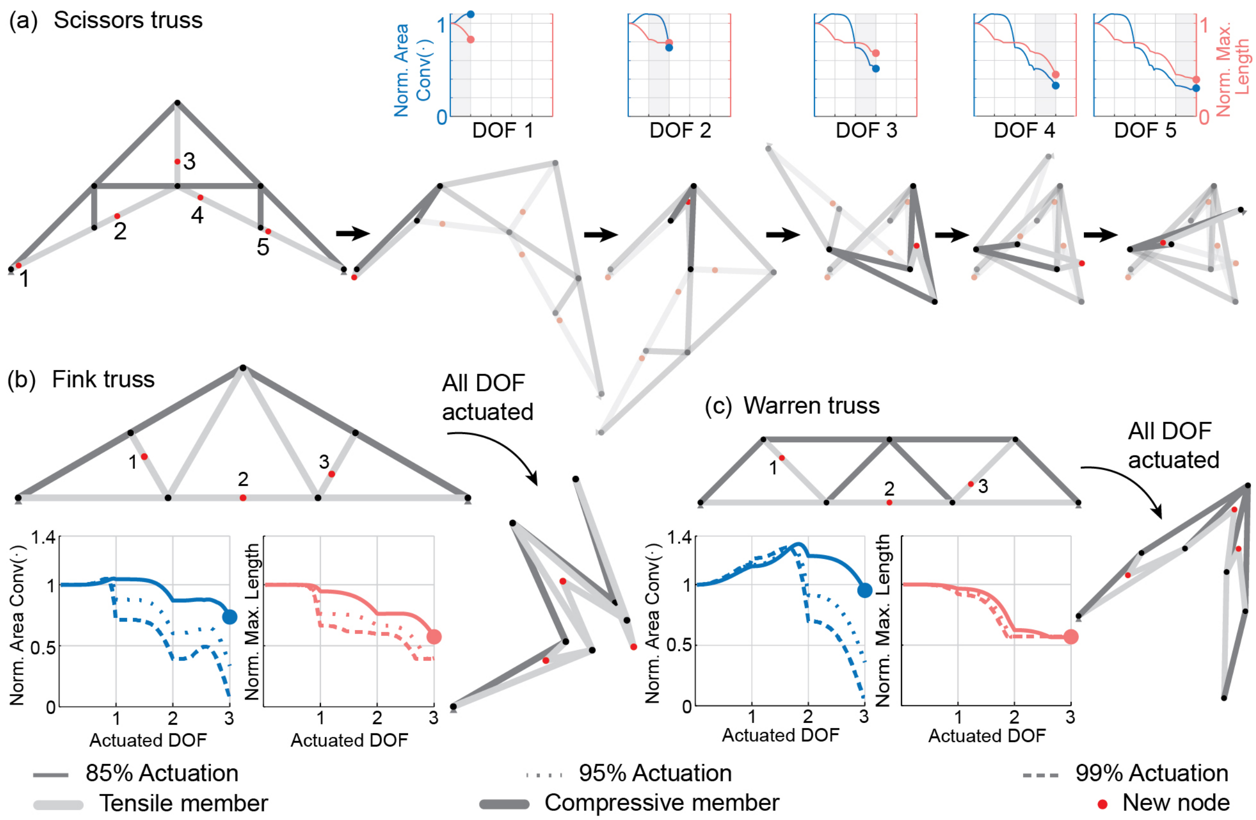


Figure 2.3: (a) Sequential actuation of each kinematic degree of freedom, illustrating the kinematic path of a reconfigurable Scissor truss. The dual Y-axis plot displays changes in the convex hull area normalized by the initial convex hull area of the truss (left axis) and the maximum structural length normalized by the initial maximum length of the truss (right axis) as the configuration evolves. The initial and final configurations after actuating all kinematic degrees of freedom are shown for (b) the Fink truss and (c) the Warren truss, along with the normalized convex hull area and maximum length plots for 85%, 95%, and 99% actuation of each DOF.

structure as it changes shape. The packing density is evaluated using: (*i*) area of the convex hull, $Conv(\cdot)$ of the truss nodes, and (*ii*) the maximum length of the truss structure. The convex hull of a set of points in a plane (or in higher-dimensional space) is the smallest convex shape that contains all the points in that set [66]. As the truss morphs, the convex hull area is normalized by its initial, non-actuated area to measure relative compactness. The maximum length of the truss is defined as the greatest distance between any two points within this convex hull and is normalized by the initial maximum length of the truss to provide a consistent basis for comparison across configurations.

Figure 2.3 (a) demonstrates the step-by-step actuation of each kinematic degree of freedom in a

Scissor truss, which has been modified using our proposed method to enable reconfigurability. Five new nodes (highlighted in red and numbered) have been introduced, resulting in five independent kinematic DOFs numbered from one to five from left to right. Moving sequentially from left to right in Fig. 2.3 (a), each DOF is actuated up to 85% of its maximum limit (approaching a flat state). While each DOF could theoretically reach 100%, we limit the actuation to 85% here to provide a more clear visualization of the kinematic motions. Throughout this actuation process, the area of the convex hull (left axis) and the maximum length of the hull (right axis), are tracked and illustrated in the plots above each actuated DOF in Fig. 2.3 (a). Even with each DOF actuated only to 85%, the structure achieves a compacted area that is approximately 0.3 times its initial footprint. The structure also achieves a reduction in maximum length to about 0.4 times the initial truss length when all DOFs are actuated to 85%.

Figure 2.3 (b) and (c) illustrate the Fink and Warren trusses in their pre-actuation and post-actuation states, respectively. The post-actuation configurations illustrated here are achieved by actuating each kinematic DOF to 85% of its initial state for each truss. For both trusses, we also demonstrate the changes in the structure's convex hull area and the maximum length compared to their initial states. These changes are depicted for three cases, wherein each DOF is actuated to 85% (solid), 95% (dotted), and 99% (dashed) of its initial state.

For the Fink truss, we observe that actuation of 85%, 95%, and 99% across all three DOFs results in the compacted structure achieving a convex hull area of 0.73, 0.34, and 0.07 times its initial state, respectively. Correspondingly, the maximum length of the compacted structure is reduced to 0.57, 0.39, and 0.39 times the initial length. In the case of the Warren truss, actuation of 85%, 95%, and 99% across all three DOFs reduces the compacted structure's convex hull area to 0.95, 0.36, and 0.07 times its initial state, respectively. The reduction in the maximum length of the structure plateaus at 0.56 times its initial state, irrespective of the extent of actuation of each kinematic DOF of the truss.

From the three examples of reconfigurable trusses, we observe varying levels of compactness based on the type of truss and the extent of actuation. For instance, the scissor truss achieves significant reduction of the convex hull area even with 85% actuation of its DOFs (final state that is $0.3\times$ of the initial). In contrast, the Fink and Warren trusses show a minimal reduction in their footprint at the same actuation (final states that are $0.7\times$ and $0.9\times$ of their initial). However, with 99% actuation of all DOFs, all three trusses achieve up to a 93% reduction in convex hull area. In other words, if we can sufficiently actuate all the DOFs of the system, we can achieve a large reduction in the overall area that a stowed truss will take up. It is important to note that this level of compact packing is theoretical and can only be achieved if the overlap of structural members is allowed. In practical applications, strategies like distributing bars in and out of the plane (as discussed in Section 2.5.1) must be implemented to achieve adequate packing without overlap. Furthermore, while a large reduction in the volume of the truss is possible, the maximum length of the truss can be reduced to only a fraction of the initial length, such as $0.4\times$ for the Scissor, $0.4\times$ for the Fink, and $0.5\times$ for the Warren truss. Often the reduction in maximum length of the truss plateaus with the first DOFs, and further actuation of the system does not produce a notable reduction in the system length. For instance, when actuating the Warren truss, the compressive members on the side and the top chord of the truss become collinear, and do not fold further, even as the system becomes flat.

2.4 Reconfigurable Trusses for Arbitrary Geometries

In this section, we show that our approach enables the transformation of arbitrary truss geometries into reconfigurable systems. We apply our node introduction workflow to trusses obtained using topology optimization for arbitrary geometries, loads, and support conditions. We first discuss how our method can be integrated with ground-structure-based topology optimization and then discuss how results from continuum optimization can be adopted into reconfigurable systems.

2.4.1 Arbitrary Truss Geometries from Ground-structure-based Topology Optimization

In this subsection, planar trusses for arbitrary geometries, loads, and support conditions are obtained using the ground-structure-based topology optimization method (GRAND) developed by [67]. GRAND begins by discretizing the design domain into a mesh of nodes connected with a network of potential truss members to form the *ground structure*. The complexity of the network is determined by a parameter called connectivity level. At level 1 connectivity, each node is connected only to its immediate neighbors, resulting in a sparse network. Higher connectivity levels introduce connections to distant nodes, adding potential members and redundancy to the structure. Selecting an appropriate connectivity level is crucial, as it influences the complexity of the resulting design and computational effort.

The goal of the optimization is to identify the lightest truss structure that satisfies the given load and support conditions. GRAND achieves this goal by minimizing the total volume of material of the truss members, $V = l^\top a$, where l and a represent the lengths and cross-sectional areas of the members, respectively. The method redistributes material among truss members by adjusting their cross-sectional areas and removing unnecessary members whose areas fall below a user-defined threshold. The optimization process maintains static equilibrium and keeps member stresses within allowable limits to ensure the final design meets all structural performance criteria while minimizing material use.

Truss structures generated using GRAND require post-processing before being integrated with our node introduction workflow to create reconfigurable trusses for arbitrary geometries. Since GRAND often outputs trusses with unused nodes and bars with zero cross-sectional area, the first step involves removing unused elements, renumbering the nodes, and updating the bar connectivity. The next step identifies nodes connected to exactly two collinear bars. These nodes are removed, and the collinear bars are replaced with a single bar whose cross-sectional area is the maximum of the two original bars. The optimized structure may also lack a direct connection between the support nodes due to the specified boundary conditions. This lack of connection is addressed by introducing zero-force members between the supports with a cross-sectional area equal to the minimum of the bar areas in the optimized structure. Additionally, if intermediate nodes lie along the line connecting the supports, zero-force members are added to connect all collinear nodes. Occasionally, a stray bar may connect the force application point to the rest of the structure. In such cases, the node of force application is connected to the nearest available node on the structure or directly to a support. These post-processing steps ensure that the resulting truss structure is primarily composed of closed triangular units.

As an illustrative example, Fig. 2.4 (a, top left) shows a level-1 ground structure loaded as a cantilever. This structure is first optimized using GRAND and then post-processed and transformed into a reconfigurable system using the proposed method of node introduction. Sequential kinematic simulations are performed on the reconfigurable cantilever truss to examine its kinematic behavior in space. The reconfigurable cantilever truss exhibits a reduction in the convex hull area up to 0.34, 0.12, and 0.07 times its initial state when all three kinematic DOFs are actuated to 85%, 95%, and 99%, respectively. In contrast, the reduction in the maximum length of the structure is minimal and plateaus at 0.97 times its initial length despite the full actuation of all DOFs.

While the topology-optimized cantilever truss effectively demonstrates the applicability of the proposed method for trusses with arbitrary geometries, the transformation process becomes increasingly challenging as the domain shapes become complex. For instance, consider the L-shaped and Serpentine beams illustrated in Fig. 2.4 (b) and (c), respectively. The ground-structure-based topology optimization of these geometries can exhibit highly intricate bar connectivity, including overlapping bars and polygonal shapes. Currently, our workflow is not equipped to handle such complex structures — a limitation we identify as a key direction for future work (see Section 6.2).

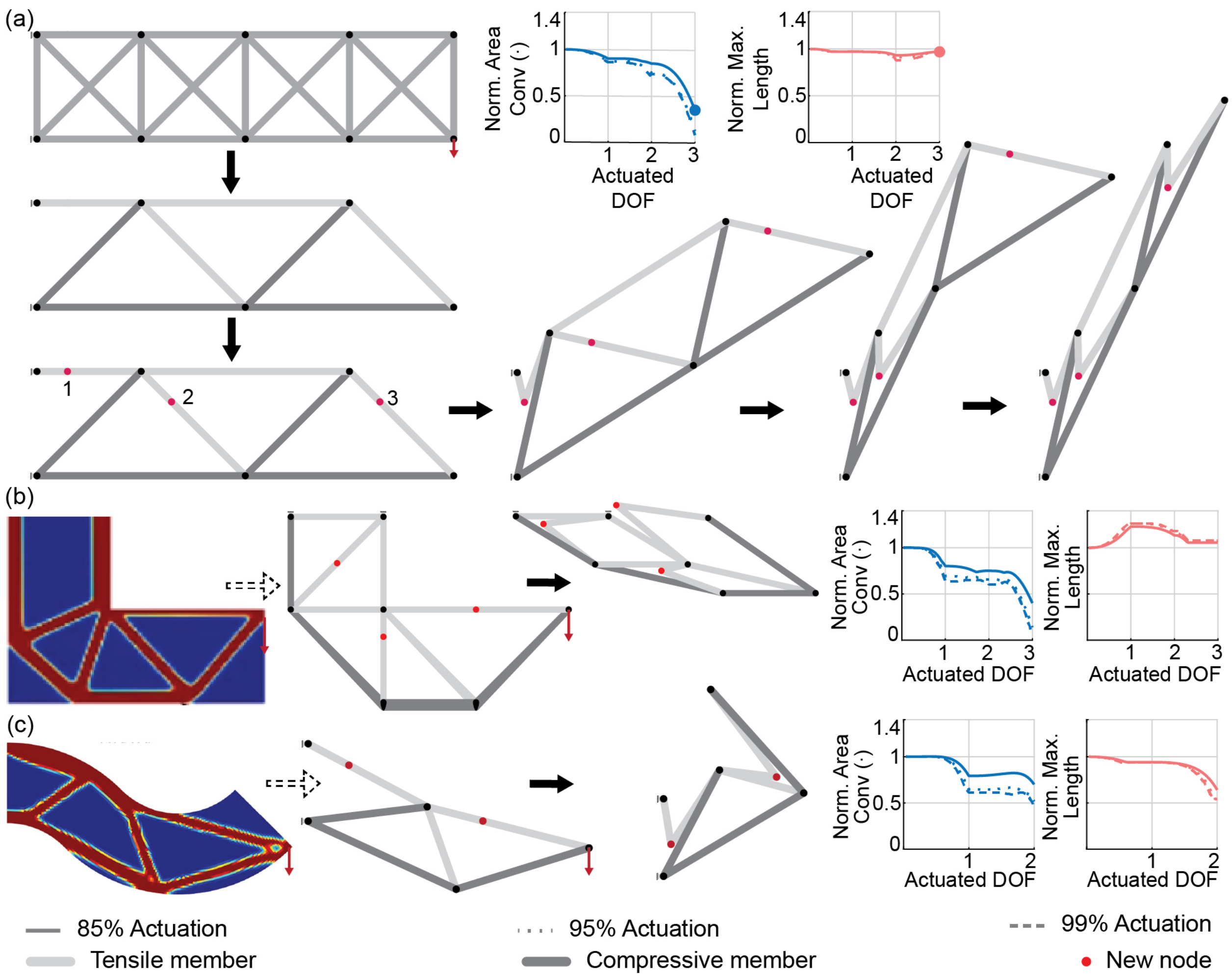


Figure 2.4: (a) Cantilever truss. Anti-clockwise from top left: level-1 ground structure for a cantilever problem, optimized cantilever truss obtained from ground structure optimization, cantilever truss transformed into a reconfigurable structure, and sequential actuation of each kinematic degree of freedom up to 85% of its full extent. Plots depicting the convex hull area and maximum length of the structure, relative to its initial state, are shown for 85%, 95%, and 99% actuation of each DOF as the structure morphs; (b) L-shaped and (c) Serpentine truss. Clockwise from top left: topology optimized structure (adapted from [68] under the Creative Commons Attribution 4.0 International License), approximate simplified truss structure modified to obtain a reconfigurable system, stowed orientation of the structure at 85% actuation of all DOFs, variation in normalized maximum length and normalized convex hull area of the structure at 85%, 95% and 99% actuation of all DOFs.

2.4.2 Arbitrary Truss Geometries from Continuum Topology Optimization

Continuum topology optimization has become a widely adopted tool in structural engineering for identifying the efficient distribution of material within prescribed design constraints. While similar to ground-structure optimization, continuum optimization does not enforce strict stress constraints during the optimization process. Instead, the continuum optimization works by optimizing the stiffness or compliance by adjusting the distribution density of continuous material in a domain. Various continuum optimization methods have been developed that serve as conceptual guides for engineers designing new structures [69]. However, raw results from continuum optimization often require additional filtering or post-processing to ensure structures have fewer, well-defined and practically manufacturable members, particularly in civil engineering applications where direct additive manufacturing is impractical. Engineers frequently use their judgment to simplify intricate continuum optimization results into simpler truss-like designs that retain core load-carrying mechanisms while accommodating design constraints such as domain geometry, loads, and supports [70]. In line with this practice, we use simplified versions of continuum-optimized structures as the starting point for our reconfigurable truss designs.

We use the continuum-optimized results from [68] (adapted under the Creative Commons Attribution 4.0 International License) as a benchmark for the L-shaped truss and serpentine truss designs. The simplified versions of these trusses, which can be transformed into reconfigurable systems, are presented in Figures 2.4 (b) and (c), respectively. These examples are chosen from the literature for simplicity, though other continuum-optimized structures could also be adapted using the same approach. We intentionally select slender geometries that resemble single-chord truss systems and make geometric modifications to ensure compatibility with our node introduction workflow. In cases where a quadrilateral is encountered in the continuum-optimized structures, we either add a diagonal to split it into two triangles or remove the shortest member to reduce it to a single triangle. For the L-shaped domain, the geometry is adjusted to obtain congruent triangles. While these simplified structures differ from the original continuum-optimized designs, they still fit within the initial envelope and carry a load using the same overall mechanisms. The morphed configurations of these two truss structures at 85% actuation of each degree of freedom are shown in Fig. 2.4 (b) and (c).

We observe that the L-shape truss, when actuated to 85%, 95%, and 99% across all three DOFs, reduces its convex hull area to 0.4, 0.14, and 0.06 times its initial state, respectively. Correspondingly, the maximum length of the compacted structure increases to 1.05, 1.07, and 1.07 times the initial length. While the L-shaped truss effectively reduces the area it occupies, the increase in maximum length (about 7%) occurs because the structure flattens along its longest side in the final state. For the Serpentine truss, one triangular unit remains unaffected by the node introduction workflow due to the compressive nature of the bars and the non-congruent next triangle (a constraint established earlier in this chapter). Nonetheless, an actuation of 85%, 95%, and 99% across all two DOFs of the Serpentine truss reduces the convex hull area to 0.7, 0.52, and 0.45 times its initial state, respectively. The maximum length of the compacted structure was reduced to 0.64, 0.54, and 0.54 times the initial maximum length of the truss, respectively. We acknowledge the challenges of rendering these complex truss structures reconfigurable and have identified them as potential areas for future research.

2.5 Physical Design of Reconfigurable Trusses and Proof-of-concept Prototypes

This section outlines the basic design principles, physical fabrication, and mechanical performance of the reconfigurable truss systems introduced in this chapter. First, we introduce the strategies used to distribute truss members in and out of the plane, ensuring proper shape-morphing behavior while avoiding out-of-plane bending moments during loading. We then detail the fabrication of two reconfigurable truss structures — a cantilever truss and a Warren truss, and load test these structures to demonstrate that these systems maintain stiffness and strength, making them suitable for large-scale applications.

2.5.1 Distributing Truss Members In and Out of Plane

Real-world static planar trusses typically consist of members that converge at the nodes or joints along the central plane of the truss. This member placement ensures load transfer along the central plane of the truss without generating out-of-plane bending moments that could induce global deformations and buckling. However, placing members in the same plane at every joint is incompatible with the reconfigurable truss designs presented in this study. Our designs must account for member contact and avoid overlap to ensure the expected shape-morphing behavior. The reconfigurable trusses must, therefore, be fabricated using techniques that strategically distribute truss members across the central plane to ensure desired shape transformations and a densely packed state. This distribution of members must ensure that the applied loads are distributed evenly about the central plane without generating unwanted out-of-plane bending moments.

We utilize a color-coded schematic to illustrate the distribution strategy of truss members across the central plane. This schematic assigns a specific color to each truss member based on its distance from the central plane, organizing them into distinct levels, as illustrated in Fig. 2.5 (a). Members marked in red are placed closest to and immediately on either side of the central plane. Green-marked members are placed at a distance of one member thickness, blue at two, yellow at three, and purple at four member thicknesses. The proof-of-concept prototypes are fabricated with a two-ply configuration incorporating the color-coded schematic to define the separation between member levels. Each truss member in the two-dimensional trusses shown in Fig. 2.5 (b) and (e) has a corresponding member positioned equidistantly on the opposite side of the central plane. This two-ply configuration ensures that loads act along the central plane, preventing moment generation

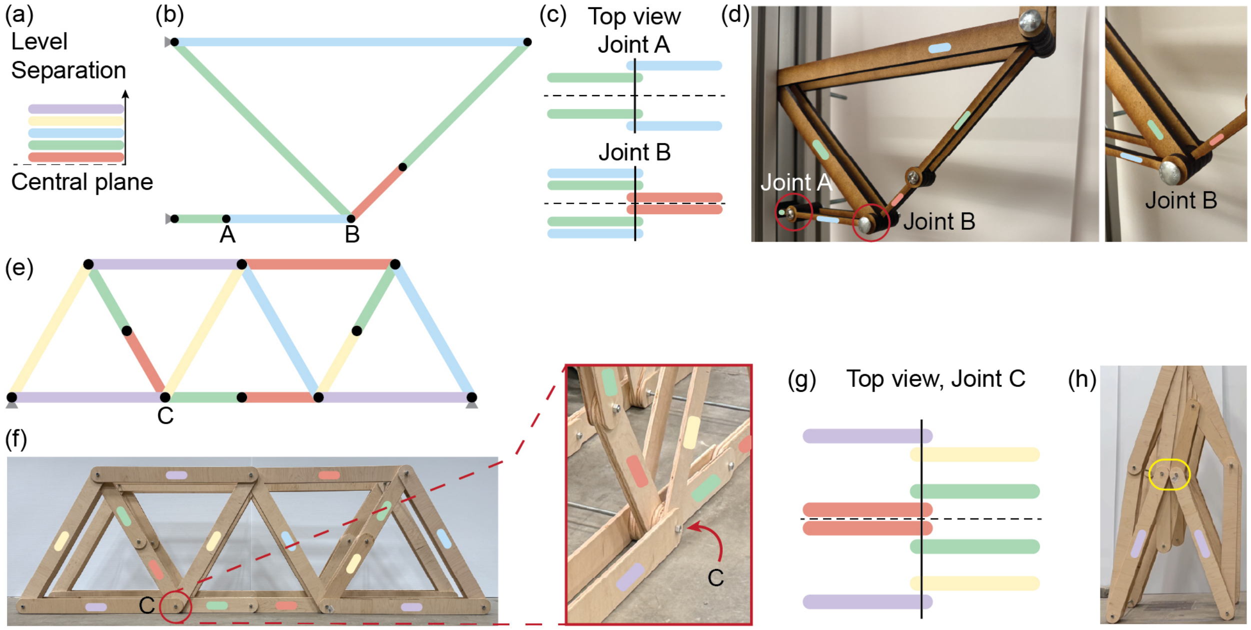


Figure 2.5: (a) Color-coded schematic of the out-of-plane level separation of truss members; (b) Side view of reconfigurable cantilever truss indicating the out-of-plane level separation of its members; (c) Top view of joints A and B illustrating the position of the members across the central plane (dashed); (d) Oblique view of the reconfigurable cantilever truss prototype showing the fabricated joints A and B; (e) Side view of reconfigurable Warren truss indicating out-of-plane level separation of its members; (f) Side view (left) and oblique view of joint C (right) of the fabricated reconfigurable Warren truss prototype indicating out-of-plane level separation of its members; (g) Top view of joint C illustrating the position of the members across the central plane (dashed); (h) Truss members coded with the same color cannot overlap and thus would encounter contact as the structure is folded towards a compact state (highlighted by the lilac rounded rectangle).

that could induce torsional deformations in the structure.

To ensure the truss exhibits the expected shape-morphing behavior and achieves a compact folded state, we aim to nest members as close to the central plane as possible. This arrangement is illustrated for select joints in the fabrication of the reconfigurable cantilever and Warren trusses in Fig. 2.5 (c) and (g), respectively. Nesting the morphing members in this manner allows them to achieve a dense folded state while keeping the unit self-contained and preventing member contact. Since the prototypes follow a two-ply configuration, the width of each joint becomes at least twice the sum of the widths of all connected members in the two-dimensional design. As the complexity of the structure increases, particularly when more members converge at a single joint, members need to be positioned farther away from the central plane. In such cases, achieving the desired shape transformation may require more layers than simply twice the number of members connected to the joint in the two-dimensional design. An example of this case is depicted in Fig. 2.5 (g), where four members meet at joint C, yet the joint's width exceeds twice the sum of the widths of all connected members. This additional width is necessary when considering the entire design of the truss system in Fig. 2.5 (e).

An important observation is that increasing the number of layers alone may not be enough. Proper nesting of joint components is also crucial. Protruding joint edges can prevent the system from folding all the way and lead to an inefficiently packed structure, even with adequate layer spacing at individual joints. For example, the Warren truss cannot fold completely flat because members with the same separation level encounter contact as the structure folds towards a compact state (Fig. 2.5 h). Optimizing for compactness requires careful joint design, either by nesting protruding edges within member bodies or by ensuring joints lie flush with member surfaces. Proper nesting will help achieve a more efficient folded state. Future work can refine these design strategies to improve structural compactness and reconfigurability.

2.5.2 Comparison of Stiffness and Strength Between Reconfigurable and Static Trusses

The static and reconfigurable versions of a cantilever truss, shown in Fig. 2.6 (b) (left and right, respectively), were fabricated using the strategies discussed in Section 2.5.1. Each truss member was laser cut out of a 6.35 mm thick Medium Density Fiberboard (MDF) sheet. Consequently, all members had a cross-sectional width of 6.35 mm when viewed from the top, as illustrated in Fig. 2.6 (a, top). The tensile members featured a square cross-section of 6.35 mm by 6.35 mm, with side profile dimensions as depicted in Fig. 2.6 (a, bottom). The compressive members had a rectangular cross-section, measuring 6.35 mm in width and 19.05 mm in depth, with side profile dimensions as shown in Fig. 2.6 (a, middle). The increased depth of the compressive members was chosen to prevent buckling and ensure structural failure by fracturing the tensile members. Truss components were assembled using 6.35 mm bolts, with connections designed to allow rotational movement rather than friction-based locking. Both truss configurations were tested under a loading rate of 12.7 mm/min using the Mark-10 ESM1500 load testing machine. The trusses were mounted on a test frame with pinned connections using 6.35 mm bolts. The test frame was constructed from 80–20 aluminum extrusions and firmly secured to the Mark-10 machine with machine screws. Displacements of the test frame observed during sample testing were orders of magnitude smaller than those experienced by the tested trusses. Therefore, the frame can be reasonably assumed to act as a rigid body for connecting the trusses.

Three samples of each truss configuration—static and reconfigurable—were load tested to compare structural performance. Both configurations exhibited failure through tensile fracture of the bottom member as depicted in Fig. 2.6 (d). The static trusses exhibited an average peak load capacity of 387 N and a stiffness of 42.32 N/mm, while the reconfigurable truss exhibited an average peak load capacity of 433 N and a stiffness of 32.06 N/mm. The theoretical peak load for the two-ply truss design was calculated as 725 N, with a displacement of approximately 3 mm at the loading point, resulting in a calculated stiffness of 241 N/mm. The yield forces for individual members were determined as 725 N for tensile and 806 N for compressive members. Additionally, the critical buckling load for the longest compressive member was calculated to be 982 N, ensuring that failure does not occur due to member buckling. These theoretical values were derived using a Young's modulus (E) of 4 GPa, tensile yield stress (σ_{Ty}) of 18 MPa, and compressive yield stress (σ_{Cy}) of 10 MPa based on [71]. A video of the load tests for both the static and reconfigurable cantilever trusses is included in Supplementary Video 2.

Two key discrepancies were observed in the test results. First, the static and reconfigurable trusses showed differing peak load and stiffness despite their expected equivalence. Second, the

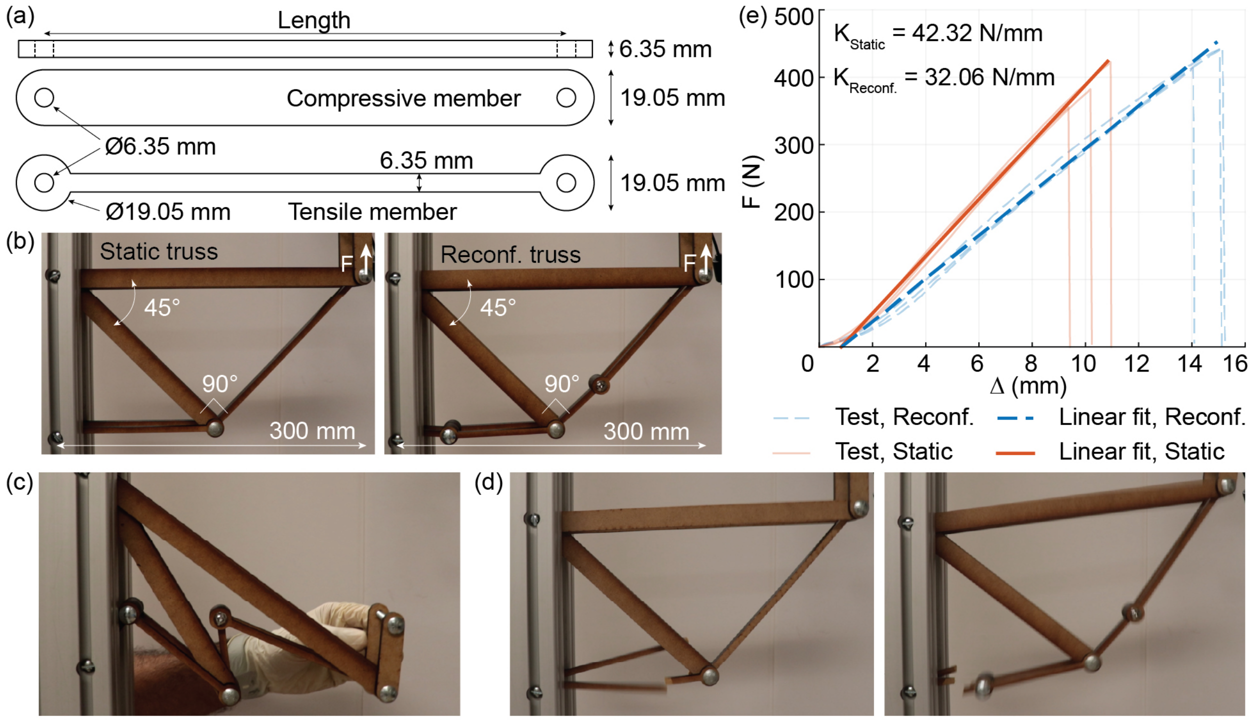


Figure 2.6: (a) Dimensions of the members used for the cantilever truss: top view of all members (top), side view of the compressive (middle) and tensile (bottom) members; (b) Static (left) and reconfigurable (right) versions of the cantilever truss on Mark-10 ESM1500 testing rig; (c) Stowed configuration of the reconfigurable cantilever truss; (d) Failure of the tensile members of the static (left) and reconfigurable (right) cantilever truss; (e) Load-displacement curves for the static and reconfigurable cantilever truss. Faint lines depict the raw data from three separate load tests, while the opaque lines represent the fitted curves used to calculate the stiffness.

experimental peak load and stiffness were significantly lower than theoretical predictions. The lower stiffness of the reconfigurable truss can be attributed to increased joint slip due to the higher number of joints present compared to the static truss. This slipping delays the full engagement of the members and is evident from the higher displacement of the loading point in the load-displacement curves of the reconfigurable truss (indicated by faint blue dashed lines in Fig. 2.6 (e)). The reconfigurable truss also contains more material than the static truss. Because the structure is tested by pulling upward, this additional material requires a greater force to overcome the increased gravitational resistance before reaching the failure load in the members. As a result, the reconfigurable truss exhibits a higher peak load than the static truss.

The difference between experimental and theoretical performance likely stems from a combination of (*a*) weakening of the mechanical properties of MDF due to the degradation of its internal structure caused by extreme heat applied during laser cutting[72, 73]; and (*b*) joint play caused by non-zero clearance between the screw and the hole. Each joint in the truss assembly is far from ideal and exhibits play, which increases structural displacements under loading. This non-ideal joint behavior further contributes to the observed reduction in stiffness compared to theoretical values. We also believe the joint interplay reduces the strength of the system because parallel members, while equidistant from the central plane, may still carry unequal forces, causing one member to fail first. The combined effects of material degradation and joint play are also reflected in the fluctuating peak loads observed across individual tests, providing additional insight into the deviations in peak load capacity between the two trusses.

Despite the observed discrepancies, the static and reconfigurable trusses exhibit reasonably high peak load and structural stiffness that are close to each other. With the improved fabrication of joints and members, we expect the trusses to reach strength and stiffness comparable to the theoretically predicted values.

2.5.3 Structural Behavior of a Three Meter Long Reconfigurable Warren Truss Bridge

A three-meter-long reconfigurable Warren truss bridge, shown in Fig. 2.7 (b), was fabricated to demonstrate the scalability of the proposed method for large-scale reconfigurable trusses. The truss members were cut from 1.27 cm thick Maple plywood sheets, giving them a cross-sectional width of 1.27 cm, as seen in the top view in Fig. 2.7 (a, top). Each member had a 1.27 cm diameter hole at both ends and a cross-sectional depth of 10 cm, as shown in the side view in Fig. 2.7 (a, bottom). The Warren truss featured members of two lengths: 1 m and 0.5 m (center-to-center between joint holes). Member connections were made using 1.27 cm diameter steel threaded rods secured with machine screw nuts. These joints were designed to allow member rotation without acting as a friction joint, enabling the truss to be reconfigured and stowed compactly, as shown in Fig. 2.7 (c) and Supplementary Video 2. The truss also featured a two-ply configuration to ensure that loads were transferred along the central plane of the truss without inducing moments. Two identical Warren trusses were assembled with an out-of-plane offset, as shown in Fig. 2.7 (b, right), resulting in a total structural width of 0.6 m. The truss was laterally braced with 0.16 cm steel cables to prevent global buckling.

Next, a three-point load test was conducted using an Instron machine to evaluate the structural behavior of the reconfigurable Warren truss bridge. The truss was simply supported at the end nodes, with the load applied at the center across its entire width. Figure 2.7 (d) illustrates the schematics of the loading point and support constraints. Loads were applied cyclically in a displacement-controlled manner, with the structure subjected to two loading cycles at each specified displacement, except for the first and last displacements. The applied displacements were 3.6 mm, 7.1 mm, 7.6 mm, 8.6 mm, 10.2 mm, 12.7 mm, 15.3 mm, 17.8 mm, and 20 mm. The load-displacement response of the cyclic three-point load test is presented in Fig. 2.7 (e). The load-displacement curve revealed slight structural degradation between the first and second cycles for every specified displacement, with the peak load being lower by an average of 2.4% in the second cycle. However, there was an average increase of 17% in stiffness during the second cycle compared to the first at the same displacement level. By the final loading stage, the structure had accumulated approximately 7 mm of downward plastic deformation. The loading phase of the first cycle at each specified displacement was plotted against the displacement of the structure after removing accumulated residual deformation to explore stiffness degradation during loading. The resulting load-displacement curve is presented in Fig. 2.7 (f). The truss exhibits initial stiffening after loading up to the first and second specified displacements due to the complete engagement of the joints and members. As shown in Fig. 2.7 (g), the compressive members at the end of the struc-

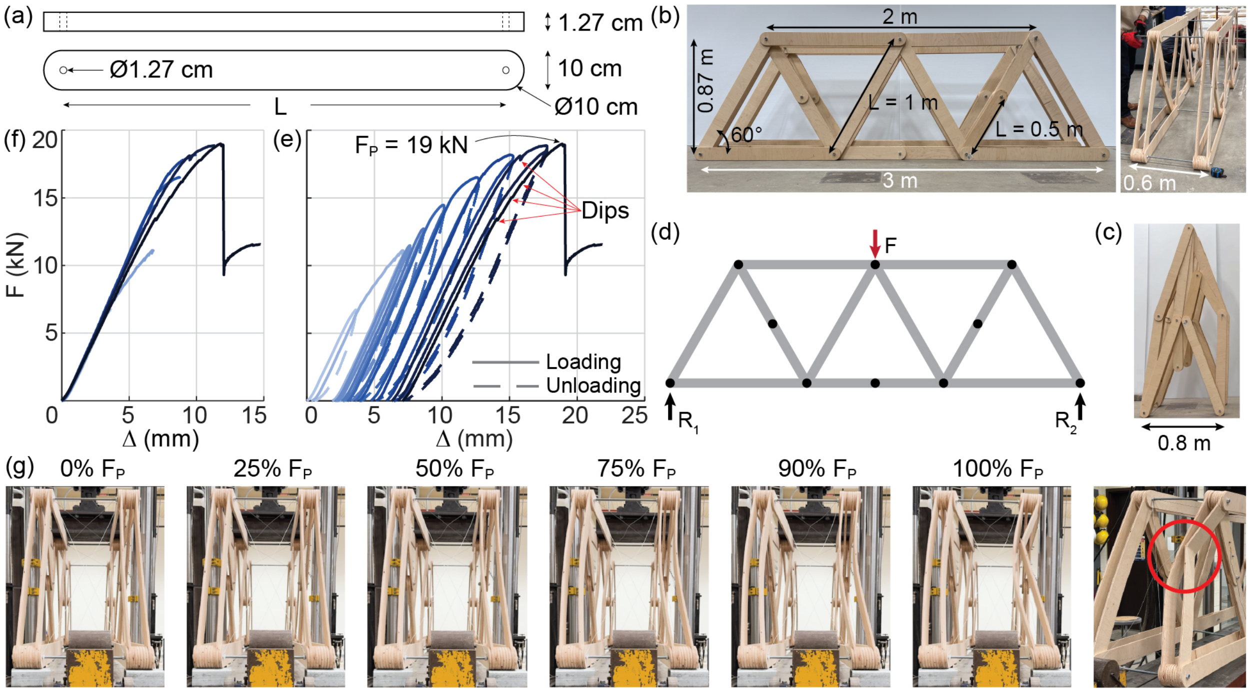


Figure 2.7: Warren truss prototype. (a) Dimensions of the Maplewood members, shown in top (upper) and side (lower) views; (b) Side (left) and oblique (right) views detailing the overall dimensions of the structure; (c) Stowed configuration of the Warren truss bridge; (d) Schematic representation of the three-point-loading test, illustrating the load and supports; (e) Complete Load (F) vs. Displacement (Δ) curve of the cyclic three-point-loading test. The truss weighs 38 kg and can support a load of 19 kN; (f) Load-Displacement curve depicting the initial loading phase of each cycle for all specified displacement limits after removing accumulated structural displacement; (g) End-view of the structure during loading—local elastic buckling of the truss members results in shortening of diagonals, which subsequently leads to larger global and local buckling as the structure approaches the point of failure.

ture exhibit some elastic buckling at loads as low as 50% of the failure load (F_p). However, this buckling does not seem to impact the stiffness and load-displacement behavior of the structure. In subsequent loading phases, stiffness degradation occurs towards the end of each phase, with the most significant degradation observed in the final cycle. This degradation can be attributed to the relaxation of lateral bracing steel cables, marked by dips in the load-displacement curves in Fig. 2.7 (e). The loss of cable tension results in insufficient bracing, leading to larger local buckling of truss members. This local buckling triggers global buckling as the load approaches the failure threshold, ultimately resulting in a critical local buckling with structural failure as depicted in Fig. 2.7 (g).

The theoretical peak tensile and compressive forces for individual Warren truss members were determined as 4.726 kN and 4.286 kN, respectively, based on Maplewood's tensile yield stress (σ_{Ty}) of 4 MPa and the compressive yield stress (σ_{Cy}) of 3.6 MPa. For the given loading condition, a 1 kN load generated a maximum tensile and compressive force of 0.87 kN and 0.58 kN, respectively, in the members of a two-dimensional Warren truss. Since the Warren truss bridge prototype featured a four-ply configuration, its theoretical peak load capacity was calculated as 21.9 kN ($4 \times 4.726/0.87$). At this peak load, the maximum compressive force in any member would be 3.13 kN ($21.9 \times 0.58/4$), which is well below the yield compressive force and the estimated 4.52 kN critical buckling load for a 1 m long member, pinned at both ends. These calculations indicate that buckling should not have been the failure mode. During testing, however, the truss failed due to elastic buckling at a peak load of 19 kN, when the compressive force in the failed member was only 2.76 kN (much lower than all calculated limits). The observed buckling, contrary to the expected tensile failure, can be explained by the following three reasons. First, significant play in the truss connections introduced flexibility and caused unintended lateral movements during loading. Second, steel cables added to brace the structure against lateral movements were not tensioned uniformly. This non-uniform tensioning resulted in insufficient bracing and lateral-torsional buckling of the structure. Lastly, the inherent curvature of plywood members reduced their load capacity and made them susceptible to buckling. Additionally, dry weather conditions during the two-month gap between fabrication and testing likely caused the plywood members to lose moisture and increase their curvature. These reasons explain why we observe the initial elastic buckling of the system and the ultimate failure from buckling rather than the tensile fracture that was initially expected.

Despite fabrication defects and unexpected buckling failure, the reconfigurable Warren truss bridge (weighing 38 kg) sustained cyclic loading until failure at a peak load of 19 kN. The sustained load was just 13.2% lower than the theoretical peak load of 21.9 kN for the four-ply Warren truss. Future refinements in design and fabrication are expected to reduce this gap in load capacity. A video of the load test and failure of the Warren truss bridge is included in Supplementary Video 2.

2.6 Concluding Remarks

This chapter introduces a new method for transforming a static truss structure into a reconfigurable system capable of rapid deployment and functional use. The method draws inspiration from flat-foldable Grashof quadrilateral linkages to introduce an additional node in each triangular unit of a static truss. We establish general rules to guide node placement based on the member forces and geometry of the triangular units. These rules ensure global system reconfigurability while maintaining the load capacity and stiffness of the structure to a reasonable extent. We demonstrate the applicability of the proposed method on three commonly observed truss designs — Scissor, Fink, and Warren trusses. Using a sequential kinematic simulator, made available as open-source software, we show that the modified trusses achieve up to a 93% reduction in convex hull area when actuated to 99% across all degrees of freedom. We further explore the versatility of the method by applying it to trusses that were obtained using topology optimization designed for arbitrary geometries, load, and support conditions. Finally, we validate the practicality of the method by fabricating and load-testing two proof-of-concept reconfigurable trusses: a 30 cm long topology-optimized cantilever truss and a three-meter-long Warren truss.

2.7 Data Accessibility

The sequential kinematic simulator used for the simulations in this article is available as an open-source code at: github.com/hardikyp/reconfigurable-trusses. Further improvements and updates to the code will continue to be made.

CHAPTER 3

Three-node Torsional Spring Element Formulation for the Analysis of Reconfigurable Bar-linked Structures

This chapter presents the derivation, validation, and application of a *Three-node Torsional Spring (3NTS)* element for the analysis of bar-linked, reconfigurable structures. The 3NTS element assigns rotational stiffness to a joint (node) of two axial force members (bars) in truss-like assemblies. This element avoids the use of rotational degrees of freedom in the model by recasting its resisting moment into equivalent nodal forces, which are consistent with global equilibrium, thereby keeping the model size compact and computationally efficient. The 3NTS is integrated into standard nonlinear solvers to simulate large-displacement response and validated against analytical solutions of two benchmark examples: the simplest 3NTS structure and the buckling of a vertical column. We further apply the framework to a reconfigurable truss structure from Chapter 2 to illustrate potential functional use cases and outline its broader applicability to metamaterials, kirigami systems, and biomechanical assemblies. An open-source matrix structural analysis tool implementing the 3NTS and axial force members is made available with this note. This chapter has been adapted from **Patil, H. Y.**, & Filipov, E. T. (2026). Three-node torsional spring element formulation for the analysis of reconfigurable bar-linked structures. Journal of Applied Mechanics, 93(3), 034502 [74].

3.1 Introduction

Over the past century, structural element formulations have evolved to capture nonlinearity and connection flexibility, enabling accurate modeling of practical structures [75]. However, the support for modeling joint rotational stiffness within truss-like structures using a torsional (or coil) spring remains limited, especially in formulations with minimal degrees of freedom (DOF) and high computational efficiency.

Joint rotational stiffness is crucial in applications where controlled joint rotation influences the global structural behavior. Some examples include reconfigurable linkages and robotic arms that rely on programmed hinge stiffnesses [76–79], biomechanics where joint stiffness affects function [80, 81], origami-inspired mechanisms where crease stiffness governs panel folding [82–84], and semi-rigid connections in lattice structures and metamaterials that store and release energy to achieve functional reconfiguration [24, 85, 86].

A typical workaround for joint rotational stiffness in two-dimensional bar-linked structures involves the use of semi-rigid joints. These semi-rigid joints introduce a short beam element (3 DOFs per node) at the joint and impose rotational stiffness by assigning ad-hoc stiffness values to the node's rotational degree of freedom [61, 87–89]. While serviceable, this approach may yield spurious results if the user-chosen stiffness value is not representative of the structural behavior [90]. Furthermore, it increases the computational complexity by inflating the model, as representing a single joint by a beam segment requires six DOFs over the two required for a bar-bar joint in two-dimensional space.

We address this gap by introducing a Three-Node Torsional Spring (3NTS) element to model a joint's rotational resistance within bar-linked structures. The 3NTS element consists of two elastic axial force members (bars) joined at an intermediate node, which hosts a torsional spring of lumped rotational stiffness K_T [Nm/rad] (Fig. 3.1). When the relative angle between the bars (α) changes, the spring develops a resisting moment that is recast as equivalent nodal forces consistent with global equilibrium. Since α is a function of the coordinates of the three nodes, the 3NTS requires only six translational DOFs for its representation in a two-dimensional space. Consequently, a two-dimensional bar-linked assembly using the 3NTS elements has the same global DOF count as an equivalent spring-less truss structure. Thus, by avoiding rotational DOFs, this formulation yields a compact model with improved computational efficiency over ad-hoc joint rotational stiffness workarounds.

The proposed 3NTS element provides a minimal, first-principles representation of a joint with rotational stiffness, which the current literature lacks. Furthermore, it decouples the joint's moment-rotation behavior from axial force-displacement effects by allowing standard bar elements to handle the axial and positional effects. This concept draws on the bar and hinge origami models [91–94] where three-dimensional rotational hinges capture crease folding and sheet bending. Here, we distill that concept and apply it to a two-dimensional case to assess the global structural behavior of bar-linked reconfigurable structures with torsional springs. We believe that this reduced-order formulation will aid in the exploration and efficient simulation of system behaviors in biomechanics, robotics, compliant mechanisms, planar metamaterials, and kirigami lattices. The torsional spring implementation is made available freely in Section 3.6

The remainder of this chapter is organized as follows. Section 3.2 derives the torsional spring stiffness matrix. Section 3.3 validates the simulated response of the torsional spring against analytical solutions of benchmark examples. Section 3.4 illustrates the potential application of the developed analysis program to reconfigurable bar-linked structures. Finally, Section 3.5 summarizes the contributions of this work.

3.2 Stiffness Matrix of a Three-node Torsional Spring Element

The stiffness matrix of a two-dimensional three-node torsional spring is derived from constitutive law by evaluating the Jacobian and Hessian of its strain energy. The torsional spring unit shown in Fig. 3.1 consists of three nodes 1, 2, and 3 with undeformed coordinates (X_1, Y_1), (X_2, Y_2), and (X_3, Y_3), respectively. Bar 1 connects nodes 1 and 2, and bar 2 connects nodes 2 and 3. Each bar behaves as an axial-force member. A torsional spring of stiffness K_T is located at node 2, producing a moment only when the relative angle between bars 1 and 2 changes. The nodal coordinates in the deformed configuration are (x_1, y_1), (x_2, y_2) and (x_3, y_3).

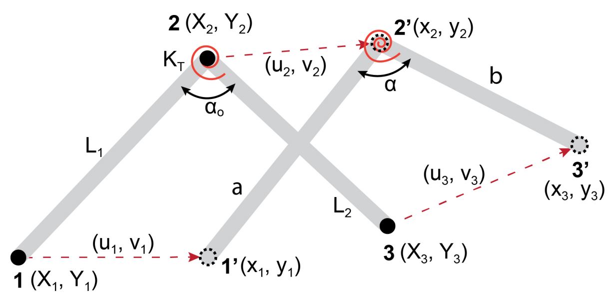


Figure 3.1: A schematic representation of the three-node rotational spring unit depicting the nodal displacements.

Let the position vectors in the undeformed and deformed configurations be $R_1 = [X_1,\ Y_1]^\top$, $R_2 = [X_2,\ Y_2]^\top$, $R_3 = [X_3,\ Y_3]^\top$, and $r_1 = [x_1,\ y_1]^\top$, $r_2 = [x_2,\ y_2]^\top$, $r_3 = [x_3,\ y_3]^\top$, respectively. The global nodal coordinate vectors are expressed as $R = [R_1^\top,\ R_2^\top,\ R_3^\top]^\top$ and $r = [r_1^\top,\ r_2^\top,\ r_3^\top]^\top$, and the corresponding displacement vector is defined as $u = r - R$.

3.2.1 Torsional Spring Constitutive Law

The strain energy of a linearly elastic three-node torsional spring is expressed as

$$U(\alpha) = \frac{1}{2} K_T (\alpha - \alpha_o)^2 \, , \tag{3.1}$$

where K_T is the spring's rotational stiffness, $\alpha = \alpha(r)$ denotes the relative angle between the two connected bars as a function of the global nodal coordinate vector r, and $\alpha_o = \alpha(R)$ represents the undeformed (rest) angle of the spring, defined with respect to the undeformed global nodal coordinate vector R.

The internal nodal force vector is obtained by differentiating the strain energy with respect to the displacement vector:

$$F_{\text{int}} = \frac{\partial U}{\partial u} = \frac{\partial U}{\partial \alpha} \frac{\partial \alpha}{\partial u} \, . \tag{3.2}$$

Recalling that the reference configuration of the spring R is constant, gradients with respect to the displacement vector are equivalent to those with respect to the deformed coordinates, i.e., $\partial u = \partial r$. Definining the gradient of the relative angle as $\nabla \alpha = \partial \alpha / \partial u = \partial \alpha / \partial r$ and the moment generated by the spring as $M = \partial U / \partial \alpha = K_T(\alpha - \alpha_o)$, the internal force can be expressed as

$$F_{\text{int}} = M \nabla \alpha = K_T (\alpha - \alpha_o) \nabla \alpha \, . \tag{3.3}$$

The tangent stiffness matrix is obtained by differentiating the internal force with respect to the deformed global nodal coordinates:

$$K_{\text{spr}} = \frac{\partial F_{\text{int}}}{\partial r} = \frac{\partial}{\partial r} \left(M(r) \nabla \alpha(r) \right) = M(r) \nabla^2 \alpha(r) + \nabla M(r) \nabla \alpha(r)^\top \, . \tag{3.4}$$

Because $M = K_T(\alpha - \alpha_o)$, it follows that $\nabla M = K_T \nabla\alpha$. Substituting these relations into Eq. 3.4 yields the final expression for the spring element stiffness matrix:

$$K_{\text{spr}} = K_T \left[(\nabla\alpha)(\nabla\alpha)^\top + (\alpha - \alpha_o)\nabla^2\alpha \right] \ . \tag{3.5}$$

3.2.2 Relative Angle Between the Bars

Let the vectors along bars 1 and 2 be defined as $a = r_1 - r_2$ and $b = r_3 - r_2$, where $r_i = [x_i, y_i]^\top$ are the nodal position vectors in the deformed configuration. Define the 90° counterclockwise rotation matrix N such that, for any vector v, the product Nv represents a rotation of v by $+90°$. Furthermore, the property $N = -N^\top$ holds true.

$$N = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \tag{3.6}$$

The cosine and sine of the relative angle α between bars 1 and 2 are then expressed as

$$\begin{aligned} C &= a \cdot b = \|a\| \, \|b\| \, \cos(\alpha) \ , \\ S &= a^\top N b = \|a\| \, \|b\| \, \sin(\alpha) \ . \end{aligned} \tag{3.7}$$

Hence, the relative angle α is obtained as

$$\alpha = \tan^{-1}\left(\frac{S}{C}\right) \ . \tag{3.8}$$

It is important to note that this expression when programmed into Matlab does not capture the spring's full range of rotation, as the `atan2` function returns values in the interval $[-\pi,\, \pi]$. To obtain a continuous measure of rotation in the range $[0,\, 2\pi)$, the computed angle in Matlab is modified as

$$\alpha = \mod (\alpha, 2\pi) \ , \tag{3.9}$$

where `mod` function returns the remainder after dividing α by 2π. This treatment assumes that the torsional spring does not undergo self-overlapping deformations.

3.2.3 Derivatives of Bar Vectors and Auxiliary Quantities

The gradient of α with respect to the global nodal coordinates can be derived by differentiating Eq. 3.8. Using the chain rule, we have

$$\nabla\alpha = \frac{1}{1+(S/C)^2}\left(\frac{C\nabla S - S\nabla C}{C^2}\right) = \frac{C\nabla S - S\nabla C}{C^2+S^2} = \frac{C\nabla S - S\nabla C}{\|a\|^2\|b\|^2} \ . \tag{3.10}$$

To evaluate Eq. 3.10, the derivatives of vectors a, b, and the scalar quantities C and S with respect to the nodal coordinates are required. These are given by

$$\begin{array}{lll}
\dfrac{\partial a}{\partial r_1} = \mathbf{I} & \dfrac{\partial a}{\partial r_2} = -\mathbf{I} & \dfrac{\partial a}{\partial r_3} = \mathbf{0} \\
\dfrac{\partial b}{\partial r_1} = \mathbf{0} & \dfrac{\partial b}{\partial r_2} = -\mathbf{I} & \dfrac{\partial b}{\partial r_3} = \mathbf{I} \\
\dfrac{\partial C}{\partial r_1} = b & \dfrac{\partial C}{\partial r_2} = -(a+b) & \dfrac{\partial C}{\partial r_3} = a \\
\dfrac{\partial S}{\partial r_1} = Nb & \dfrac{\partial S}{\partial r_2} = N(a-b) & \dfrac{\partial S}{\partial r_3} = -Na
\end{array} \tag{3.11}$$

3.2.4 Closed-form Expression for Gradient of Relative Angle

Substituting Eq. 3.11 into Eq. 3.10 and applying the vector identity $(x \cdot y)Ny - (x^\top Ny)y = \|y\|^2 Nx$ the gradient components of α can be expressed in compact form as:

$$\begin{aligned}
\nabla\alpha_{11} &= \frac{\partial\alpha}{\partial r_1} = \frac{C(\partial S/\partial r_1) - S(\partial C/\partial r_1)}{\|a\|^2\|b\|^2} = \frac{(a \cdot b)Nb - (a^\top Nb)b}{\|a\|^2\|b\|^2} = \frac{Na}{\|a\|^2} \\
\nabla\alpha_{21} &= \frac{\partial\alpha}{\partial r_2} = \frac{C(\partial S/\partial r_2) - S(\partial C/\partial r_2)}{\|a\|^2\|b\|^2} = \frac{(a \cdot b)N(a-b) + (a^\top Nb)(a+b)}{\|a\|^2\|b\|^2} = \frac{Nb}{\|b\|^2} - \frac{Na}{\|a\|^2} \\
\nabla\alpha_{31} &= \frac{\partial\alpha}{\partial r_3} = \frac{C(\partial S/\partial r_3) - S(\partial C/\partial r_3)}{\|a\|^2\|b\|^2} = \frac{(a \cdot b)(-Na) - (a^\top Nb)a}{\|a\|^2\|b\|^2} = -\frac{Nb}{\|b\|^2} \qquad (3.12)
\end{aligned}$$

$$(3.13)$$

These three 2×1 vectors together form the full 6×1 gradient vector $\nabla\alpha$.

3.2.5 Hessian of the Relative Angle

The Hessian matrix $\nabla^2\alpha$ is of size 6×6 and consists of nine 2×2 sub-blocks corresponding to the second derivatives of α with respect to the nodal coordinates. The sub-blocks are expressed as

$$\begin{aligned}
\mathbf{H}_{11} &= \frac{1}{\|a\|^2}N - \frac{2}{\|a\|^4}(Na)a^\top \\
\mathbf{H}_{22} &= \frac{1}{\|a\|^2}N - \frac{2}{\|a\|^4}(Na)a^\top + \frac{2}{\|b\|^4}(Nb)b^\top - \frac{1}{\|b\|^2}N \\
\mathbf{H}_{33} &= \frac{2}{\|b\|^4}(Nb)b^\top - \frac{1}{\|b\|^2}N \\
\mathbf{H}_{12} &= \mathbf{H}_{21}^\top = \frac{2}{\|a\|^4}(Na)a^\top - \frac{1}{\|a\|^2}N \\
\mathbf{H}_{13} &= \mathbf{H}_{31}^\top = \mathbf{0} \\
\mathbf{H}_{23} &= \mathbf{H}_{32}^\top = \frac{1}{\|b\|^2}N - \frac{2}{\|b\|^4}(Nb)b^\top \qquad (3.14)
\end{aligned}$$

Thus, we have all the necessary information to form the stiffness matrix for a three-node torsional spring and use it in matrix structural analysis solvers.

3.3 Validation of the Three-node Torsional Spring Element

In this section, we validate the derived stiffness matrix of the torsional spring element by comparing the simulated behavior of two torsional-spring fitted bar-linked structures with their respective analytical solutions. The torsional spring element is implemented alongside linearly elastic axial bar elements within non-linear structural analysis solvers [95] to facilitate large-deformation simulations of torsional-spring fitted bar-linked structures. The two examples presented herein demonstrate the element's capability to simulate large-deformation responses, evaluate strain energies stored in each structural element, and reproduce buckling behavior in collinear bar systems under compression.

3.3.1 Comparing Analytical and Simulated Behavior of the Simplest Three-node Torsional Spring Structure

The simplest bar-linked structure with a three-node torsional spring is the element itself, as shown in Fig. 3.2 (a). Nodes A and B are fixed supports, while node C is free. Bars connecting nodes A-B and B-C have a Young's modulus $E = 3 \times 10^6$ Pa, cross-sectional area $A = 0.5$ m^2, and length $L = 1$ m. A torsional spring with stiffness $K_T = 50$ Nm/rad is located at node B. A gradually increasing external load P is applied at node C in the positive Y-direction.

The analytical response of this structure can be obtained using potential energy analysis. When subjected to the applied load P, the bar connecting nodes B and C rotates by an angle θ and elongates by ΔL. The elongation is induced by the component of the applied load along the bar's axis and is expressed as $\Delta L = (PL/EA)\sin(\theta)$. In this deformed state, the total potential energy of the structure is

$$U = \frac{1}{2}K_T\theta^2 - PL\sin(\theta) - \frac{P^2L}{2EA}\sin^2(\theta) + \text{constant}\,. \tag{3.15}$$

The equilibrium condition is obtained by setting $\partial U/\partial\theta = 0$, which yields the following expression for the applied load P:

$$P = \frac{-AEL\cos(\theta) \pm AE\sqrt{\frac{L\cos(\theta)}{EA}\left(4K_T\theta\sin(\theta) + AEL\cos(\theta)\right)}}{2L\sin\theta\cos\theta}\,. \tag{3.16}$$

The corresponding vertical displacement of node C due to the applied load becomes $\Delta = (L + \Delta L)\sin(\theta)$.

Figure 3.2 (b) compares the analytical and simulated load-displacement responses of the simple three-node torsional spring structure. The two curves show excellent agreement over the entire range of deformation. The load increases from zero and appears to approach infinity asymptotically as the bar rotates from 0° to 90°. In reality, the load does not become infinite. Throughout the range of deformation, the applied load is primarily resisted by the torsional spring. As the angular rotation of the bar approaches 90°, however, the load begins to increase approximately in proportion to the axial stiffness of the bar, EA/L, as both the bar and the spring resist the external load. Because this axial stiffness is several orders of magnitude greater than the torsional spring stiffness, the overall response visually appears to asymptote toward infinity in the load-displacement plot. Furthermore, as shown in Fig. 3.2 (c), there is excellent alignment between the work done by the external load and the cumulative strain energy stored in each element type, which confirms energy balance and validates the correct implementation of the torsional spring element within the non-linear solver framework.

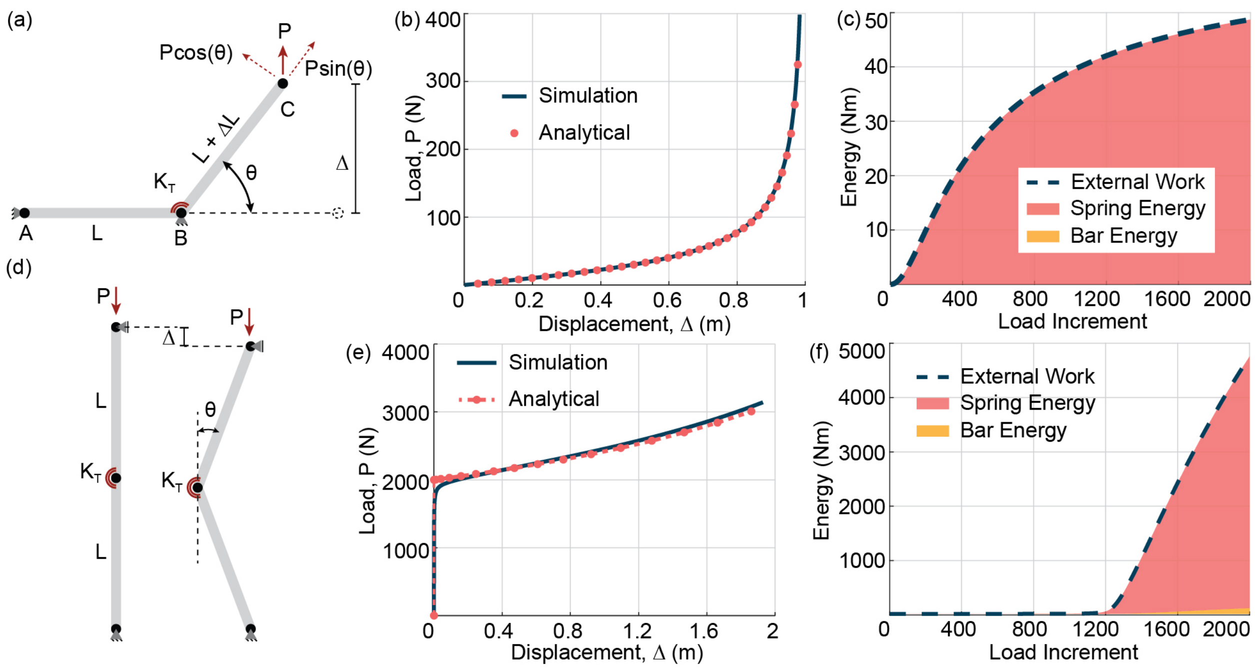


Figure 3.2: Validation of the simulated behavior of bar-linked structures with torsional spring elements against their respective analytical solutions. (a) Simplest structure with a torsional spring element; (b) Comparison of analytical and simulated load–displacement responses for the structure in (a); (c) Simulated external work and stacked internal strain energies (spring and bars) over load increments for the structure in (a); (d) Buckling of a bar-linked vertical column comprising of two bars and a torsional spring element at the central joint; (e) Comparison of analytical and simulated load–displacement responses for the structure in (d); (f) Simulated external work and stacked internal strain energies (spring and bars) over load increments for the structure in (d).

3.3.2 Comparing Analytical and Simulated Buckling Behaviour of a Vertical Column Structure

The vertical column structure shown in Fig. 3.2 (d) consists of two bars with a torsional spring element at the central joint. The bottom end of the column is fixed, while the top end is a roller support that restrains horizontal motion but allows vertical movement. The analytical relationship between the load, P and bar rotation θ can be obtained using potential energy analysis as

$$P = \frac{2K\theta}{L\sin(\theta)} \,. \tag{3.17}$$

The structure becomes unstable when the applied load reaches the critical value $P = 2K/L$. Thereafter, the column undergoes progressive deformation until each bar rotates by $\theta = \pi/2$, at which stage the load reaches $P = \pi K/L$, representing a fully folded configuration. The vertical displacement of the roller end beyond instability can be expressed as $\Delta = 2L(1 - \cos(\theta))$. It should be noted that the potential energy analysis assumes rigid bars, while the non-linear solver models the bars as linearly elastic. Consequently, slight differences are expected in the analytical and simulated buckling behavior of the vertical column.

In the simulation, the bars have a Young's modulus $E = 3 \times 10^9$ Pa, cross-sectional area $A = 1$ m^2, bar lengths $L = 1$ m, and torsional spring stiffness $K_T = 1000$ Nm/rad. A minor geometric imperfection is introduced at the free node to trigger the buckling behaviour. As shown in Fig. 3.2 (e), the simulated structure exhibits instability at a load slightly lower than predicted analytically, due to elastic bar deformations and non-ideal buckling triggered with a geometric imperfection. After the onset of buckling, the load continues to increase till the bars rotate by $\theta = \pi/2$. Overall, the simulated load-displacement curve agrees well with the analytical prediction, capturing the same deformation trend and load magnitude, with minor deviations attributed to elastic bar effects and accumulated numerical drift at large displacements. Moreover, the close agreement between the work done by external loads and internal strain energies shown in Fig. 3.2 (f) validates the torsional spring element behaviour within the non-linear solvers.

3.4 Application to Reconfigurable Bar-linked Structures

With the torsional spring element validated, the developed analysis program can now be extended to study load requirements for actuation and functional applications of reconfigurable bar-linked structures. Previous work by the authors introduced a systematic method to generate reconfigurable trusses from existing static truss designs and demonstrated their stability under design loading conditions [29]. These reconfigurable trusses can be fitted with torsional springs at their reconfigurable joints, enabling them to serve as functional systems capable of storing and releasing energy during load reversals.

The torsional spring implementation within the non-linear solver facilitates the simulation of load-displacement responses in such reconfigurable bar-linked systems. Incorporating torsional springs into these structures allows for the study of impact energy storage through spring strain energy, provided that the springs are locked after reaching their maximum deformation. This configuration also enables the evaluation of the stiffness associated with rotational joints and the estimation of actuation forces required to deploy or reconfigure the structure.

As a preliminary demonstration, a reconfigurable truss structure from our earlier work in Chapter 2 is analyzed here to illustrate the potential application of the developed analysis program. The reconfigurable cantilever truss shown in Fig. 3.3 consists of bars with Young's modulus $E = 3 \times 10^6$ Pa, cross-sectional area $A = 0.5$ m^2, and torsional springs of stiffness $K_T = 50$ Nm/rad. The bar lengths are indicated in Fig. 3.3, and the reader is referred to Chapter 2 for the exact locations of the torsional springs.

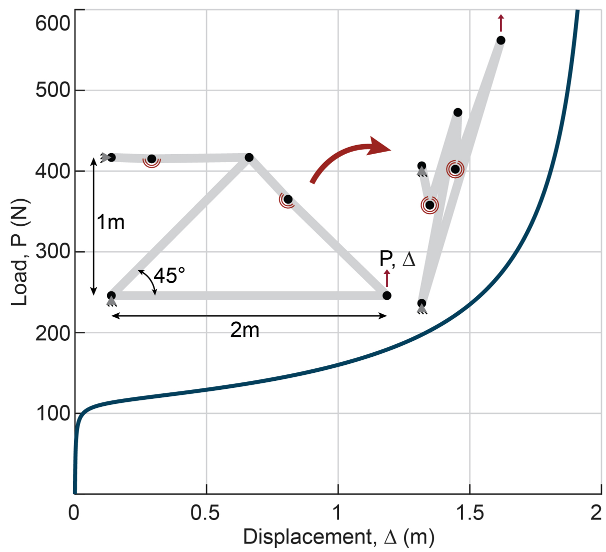


Figure 3.3: Load analysis of the actuation of a reconfigurable cantilever truss

This truss remains force-stable when loaded in the negative-Y direction. When the direction of loading is reversed to induce reconfiguration, the analysis program can be used to evaluate the load-displacement response of the structure for various spring stiffnesses. Between the two orientations shown in Fig. 3.3, an actuation force of approximately 600 N was required to achieve reconfiguration. The resulting load-displacement curve reflects characteristics similar to those observed in the validation examples, exhibiting both spring deformation and buckling behavior, and also illustrates the amount of energy stored within the system due to the applied load. This stored energy can be harnessed for functional purposes when the structure is constrained from returning to its original configuration. With further development, the analysis framework may also enable sequential release of individual torsional springs, providing a mechanism for controlled or rapid deployment of reconfigurable structures.

3.5 Concluding Remarks

This chapter presented the formulation of a three-node torsional spring element stiffness matrix and validated its accuracy through comparison with analytical solutions of two benchmark examples. The torsional spring element successfully captured large-deformation responses and the onset of buckling in spring-fitted bar-linked structures. Finally, an example was included to demonstrate the capability of the developed analysis program to simulate actuation in reconfigurable bar-linked structures equipped with torsional springs. This example previews the potential of the model for the exploration of functional use cases of reconfigurable bar-linked structures. The solver implementation is made available as a matrix structural analysis program accompanying this note.

3.6 Data Accessibility

The data and information that support this chapter are freely available at github.com/hardikyp/bar-linked-torsional-springs. Further improvements and updates to the code will continue to be made.

CHAPTER 4

Programmable Hinge Mechanics for Functional Reconfigurable Trusses

Reconfigurable structures can provide functional mechanical behavior beyond shape change through intentional design of deformation paths, stable states, and load-transfer mechanisms. This capability is especially relevant for applications that require controlled deployment, recoverable deformation, or reusable energy management. However, many existing systems that exploit elastic instabilities, multistability, or architected deformation are studied at material, unit-cell, or component scales. Their translation to larger load-bearing structures remains challenging because practical structural systems must also satisfy requirements for member strength, joint detailing, fabrication, stability, and predictable load transfer. We address this gap by developing hinge-level modeling tools for programming the mechanical response of reconfigurable truss systems. Building on the Three-Node Torsional Spring (3NTS) element introduced in Chapter 3, three nonlinear torsional spring formulations are developed. First, an enhanced torsional spring law is introduced to bound the admissible angular range of the 3NTS element while preserving the linear response over the intended operating range. This formulation enables the simulation of sequential folding, locking, and load redistribution among different kinematic degrees of freedom. Second, a magnetic multistable hinge is introduced using alternating magnetic interactions between permanent magnets. A reduced-order torque-rotation model is incorporated into the 3NTS element to represent repeated energy barriers and multiple stable rotational states. Third, an idealized bistable torsional spring law is formulated to study snap-through transitions and stable-state switching in reconfigurable trusses. Together, these contributions extend the analysis framework from kinematic reconfiguration toward programmable mechanical response in reusable and load-bearing reconfigurable structures.

4.1 Introduction

Reconfigurable structures derive their utility from the ability to change shape and from the opportunity to tailor their mechanical response through geometry, joints, and deformation path. In these systems, reconfiguration can provide a mechanism for controlling stiffness, stability, load transfer, and energy storage. A structure can be designed to move between prescribed equilibrium configurations, resist deformation until a target load threshold is reached, or recover its initial configuration after unloading. This coupling between shape change and mechanical response defines the functional reconfigurable systems considered in this chapter.

One example of functional application arises in infrastructure systems designed to manage energy from impact loads. Many conventional structures used to dissipate mechanical energy, such as protective barriers or vehicle frames, are designed to absorb energy through plastic deformation, fracture, crushing, or other irreversible mechanisms [96]. These systems are effective in reducing transmitted loads, but their deformation often produces permanent damage and limits reuse. Recent work on mechanical metamaterials has shown that external work can also be managed through reversible deformation, elastic instabilities, snap-through transitions, auxetic mechanisms, and multistable energy landscapes [24, 85, 97–102]. These systems demonstrate the potential of reusable energy-management mechanisms. However, many such architectures are studied at material, unit-cell, or component scales. Their translation to large load-bearing structures remains nontrivial because structural systems must also satisfy requirements for member strength, joint detailing, fabrication, stability, and predictable load transfer [103, 104].

The reconfigurable trusses developed in this dissertation provide a structural platform for exploring these ideas at a larger scale. Previous chapters introduced a method for transforming static trusses into reconfigurable mechanisms and a Three-Node Torsional Spring (3NTS) element for modeling rotational stiffness at folding joints. This chapter extends that framework from kinematic reconfiguration to programmable hinge mechanics. The central objective is to develop nonlinear moment-rotation laws at the hinge level and evaluate how these local laws affect the global response of reconfigurable trusses. This approach will enable the analysis of sequential folding, locking, load redistribution, snap-through transitions, and multiple stable configurations.

To this end, this chapter develops three hinge-level modeling capabilities for simulating programmable mechanical behavior in reconfigurable trusses. First, an enhanced torsional spring law is introduced to bound the admissible rotation of the 3NTS element. This law preserves the linear torsional response over the intended operating range and introduces nonlinear stiffening near prescribed angular deformation limits. The enhanced spring law prevents physically inadmissible rotations and enables the simulation of sequential folding, locking, and load redistribution among different kinematic degrees of freedom. Second, a magnetic multistable hinge is introduced to create multiple stable rotational states through alternating magnetic interactions. A reduced-order model is used to represent the periodic torque-rotation response of the hinge, and the corresponding constitutive law is incorporated into the 3NTS element to model multistable response in reconfigurable trusses. In addition to enabling programmable multistable configurations, the magnetic interactions provide a mechanism for dissipating energy under external loading through reconfigurable deformations, enabling the reuse of the structure since no material deformation occurs. Third, an idealized bistable torsional spring law is introduced to define two energetically stable rotational configurations separated by an unstable equilibrium. The constitutive law is incorporated into the 3NTS element to investigate how bistable hinges introduce multiple stable configurations in reconfigurable trusses with one or more kinematic degrees of freedom.

The remainder of this chapter is organized as follows. Section 4.2 introduces an enhanced formulation of the torsional spring that limits the admissible deformation range and enables the analysis of sequential folding, locking, and load redistribution in reconfigurable trusses. Section 4.3 introduces a magnetic multistable torsional hinge that uses polarity-dependent magnetic interactions to create multiple stable rotational states. This section also develops a computational model of the magnetic hinge response within the 3NTS framework. Section 4.4 introduces an idealized bistable torsional spring law and demonstrates how reconfigurable trusses with multiple kinematic degrees of freedom can exhibit multiple stable states. Section 4.5 discusses the implications of programmable hinge mechanics for controlled reconfiguration, reusable structures for impact energy management, and engineered deployment paths. Finally, Section 4.6 summarizes the main contributions of the chapter.

4.2 Modeling Bounded Rotation in Torsional Spring Hinges

This section builds on the Three-Node Torsional Spring (3NTS) element from Chapter 3 to enforce limits on the angular rotation of the joints. The enhanced constitutive law preserves the linear spring response within the intended operating range and introduces nonlinear stiffening near the deformation limits. This bounded response prevents inadmissible hinge rotations and enables the analysis of sequential folding, locking, and load redistribution in reconfigurable trusses.

4.2.1 Enhanced Torsional Spring Constitutive Law

The Three-Node Torsional Spring (3NTS) element was originally formulated using a linear torsional spring constitutive law. In this formulation, the spring deformation is described by the relative angle α defined over the interval $[0,\ 2\pi)$. The spring's restoring moment is proportional to the difference between the current relative angle α and the rest angle α_0. While this formulation works as long as $0 \leq \alpha < 2\pi$, numerical issues arise when the spring undergoes large rotations associated with self-overlap of the connected bars. Since α is evaluated over $[0,\ 2\pi)$, the computed angle wraps from 2π to 0 when the rotation exceeds this range. This behavior produces a sudden discontinuity in the computed spring deformation, leading to inconsistent restoring moments and tangent stiffness with the actual deformation path.

This numerical issue is also inconsistent with the observed physical behavior of the reconfigurable truss systems discussed in this dissertation. In these systems, the bars connected to a torsional spring do not overlap each other during reconfiguration. The fabrication techniques and kinematics of the truss physically limit the allowable rotation at each folding joint, as discussed in Chapter 2 (see Fig. 2.7 (c)). Furthermore, relative angles near 0 or 2π correspond to a fully actuated degree of freedom, where the truss has reached its most compact configuration. Rotation of the spring joint beyond this state becomes meaningless. Therefore, rotations beyond 0 or 2π do not represent the intended deformation range of the torsional springs in the physical structure. This motivates restricting the admissible angular range of the torsional spring.

To address the numerical issue and ensure that the spring model captures a realistic range of deformation for the folding joints, the constitutive law of the 3NTS element is modified using the concepts inspired by the bar-and-hinge model [92]. The modified formulation restricts the admissible deformation range of the spring by introducing nonlinear stiffening near the extreme deformation limits. This stiffening acts as a smooth mechanical stop, providing increasing resistance as the joint approaches its allowable rotation limit. The modified formulation, referred to as the *Enhanced Spring Law*, preserves the original linear response near the rest configuration while creating high-energy barriers at the lower and upper deformation limits.

The enhanced spring law introduces three additional parameters to define the admissible deformation limits of the spring. The parameter, δ, defines the minimum allowable relative angle between the two connected members. Therefore, the lower admissible deformation limit of the spring becomes δ. By symmetry, the upper admissible limit becomes $2\pi - \delta$. The parameters α_1 and α_2 define the lower and upper threshold angles, respectively, at which the spring response transitions from the linear to the nonlinear stiffening regime. These parameters satisfy the following relationship: $0 \leq \delta < \alpha_1 < \alpha_0 < \alpha_2 < 2\pi - \delta \leq 2\pi$. Thus, the interval $[\alpha_1,\ \alpha_2]$ represents the nominal operating range of the spring. Within this range, the enhanced law is identical to the original linear torsional spring law. The intervals $(\delta,\ \alpha_1)$ and $(\alpha_2,\ 2\pi - \delta)$, on the other hand, represent the nonlinear stiffening regions near the lower and upper deformation limits, respectively.

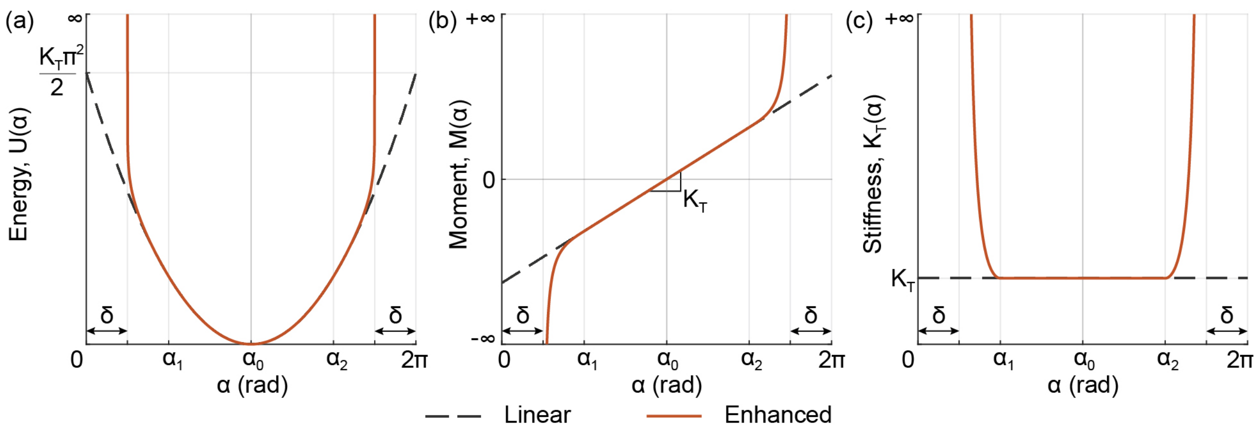


Figure 4.1: Comparison between the original linear torsional spring law (dashed line) and the enhanced spring law (solid line): (a) strain energy; (b) restoring moment; and (c) tangent stiffness as functions of the relative angle α. Here, α_0 is the rest angle, α_1 and α_2 are the lower and upper threshold angles where the spring response changes from linear to nonlinear stiffening, respectively, and δ and $2\pi - \delta$ are the lower and upper admissible deformation limits, respectively.

The strain energy, U(α), of the enhanced spring law is defined as

$$U(\alpha) = \begin{cases} \frac{1}{2}K_T(\alpha_0 - \alpha_1)^2 + K_T(\alpha_0 - \alpha_1)(\alpha_1 - \alpha) \\ -\frac{4K_T(\alpha_1-\delta)^2}{\pi^2} \ln\left|\cos\left(\frac{\pi(\alpha_1-\alpha)}{2(\alpha_1-\delta)}\right)\right|, & \alpha < \alpha_1 \\ \\ \frac{1}{2}K_T(\alpha - \alpha_0)^2, & \alpha_1 \leq \alpha \leq \alpha_2 \\ \\ \frac{1}{2}K_T(\alpha_2 - \alpha_0)^2 + K_T(\alpha_2 - \alpha_0)(\alpha - \alpha_2) \\ -\frac{4K_T((2\pi-\delta)-\alpha_2)^2}{\pi^2} \ln\left|\cos\left(\frac{\pi(\alpha-\alpha_2)}{2((2\pi-\delta)-\alpha_2)}\right)\right|, & \alpha > \alpha_2 \end{cases} \tag{4.1}$$

The restoring moment of the enhanced spring is obtained as the first derivative of the strain energy with respect to the relative angle α as

$$M(\alpha) = \begin{cases} K_T(\alpha_1 - \alpha_0) + \dfrac{2K_T(\alpha_1 - \delta)}{\pi} \tan\left(\frac{\pi(\alpha-\alpha_1)}{2(\alpha_1-\delta)}\right), & \alpha < \alpha_1 \\ \\ K_T(\alpha - \alpha_0), & \alpha_1 \leq \alpha \leq \alpha_2 \\ \\ K_T(\alpha_2 - \alpha_0) + \dfrac{2K_T(2\pi - \delta - \alpha_2)}{\pi} \tan\left(\frac{\pi(\alpha-\alpha_2)}{2(2\pi-\delta-\alpha_2)}\right), & \alpha > \alpha_2 \end{cases} \tag{4.2}$$

The corresponding tangent rotational stiffness of the enhanced spring is obtained as the derivative of the restoring moment with respect to α as

$$K_T(\alpha) = \begin{cases} K_T \sec^2\left(\frac{\pi(\alpha-\alpha_1)}{2(\alpha_1-\delta)}\right), & \alpha < \alpha_1 \\ \\ K_T, & \alpha_1 \leq \alpha \leq \alpha_2 \\ \\ K_T \sec^2\left(\frac{\pi(\alpha-\alpha_2)}{2((2\pi-\delta)-\alpha_2)}\right), & \alpha > \alpha_2 \end{cases} \tag{4.3}$$

Figure 4.1 (a-c) compares the original linear torsional spring law (dashed line) with the enhanced spring law (solid line) in terms of strain energy, restoring moment, and tangent rotational stiffness. It is important to note that the values of δ, α_1, and α_2 used in Fig. 4.1 are chosen for visual representation only and are not up to scale.

The spring response can be divided into three regions. Firstly, when $\alpha_1 \leq \alpha \leq \alpha_2$, the enhanced law reduces to the original 3NTS constitutive law. The tangent stiffness is constant and equal to K_{T}, the restoring moment varies linearly with $\alpha - \alpha_0$, and the strain energy varies quadratically with $\alpha - \alpha_0$. The energy minimum occurs at $\alpha = \alpha_0$, where the restoring moment is zero. Thus, the enhanced law does not alter the spring response in the intended operating range. In the lower stiffening region, where $\delta < \alpha < \alpha_1$, the spring stiffens as the relative angle decreases below α_1. The tangent stiffness increases according to a secant-squared function and tends to infinity as $\alpha \to \delta^+$. The restoring moment becomes increasingly negative. This moment resists further closure of the joint. The strain energy also grows rapidly as the lower admissible limit is approached. In the upper stiffening region, where $\alpha_2 < \alpha < 2\pi - \delta$, the spring stiffens as the relative angle increases beyond α_2. The tangent stiffness again tends to infinity, now as $\alpha \to (2\pi - \delta)^-$. The restoring moment becomes increasingly positive. This moment resists further deformation of the joint. The strain energy also grows rapidly near the upper admissible limit.

4.2.2 Path-dependent loading and Sequential Folding Enabled by Enhanced Torsional Spring

The enhanced spring law formulated in the previous subsection allows the 3NTS element to model large rotations while enforcing prescribed angular limits. This capability is important for reconfigurable trusses with multiple independent kinematic degrees of freedom. In such systems, individual units may fold, lock, or remain active at different stages of deformation. The enhanced spring law allows the simulation of these transitions without numerical failure caused by angle wrapping, spring overlap, or physically inadmissible rotations. This subsection demonstrates these capabilities using two examples.

The first example is the stacked square-linkage truss shown in Fig. 4.2 (a). The structure consists of two vertically stacked square units. Each unit is a special case of a six-bar linkage, formed by a square frame and two diagonal bars with a rotational joint at their connection. The joint at the midpoint of each diagonal is modeled using the enhanced torsional spring. Each unit, therefore, contributes one independent kinematic degree of freedom to the structure.

The structure is approximately 2 m tall. Each non-diagonal member has a length of 1 m. The bars have an elastic modulus $E = 3 \times 10^9$ Pa and a cross-sectional area $A = 0.5$ m^2. Each torsional spring has a linear-regime stiffness $K_T = 10$ Nm/rad. A geometric imperfection of $10°$ is assigned to each torsional spring joint to simulate buckling deformation of the diagonal bars. The admissible rotation range is bounded between $15°$ and $345°$ and the spring response remains linear between $30°$ and $330°$. Outside this interval, the tangent stiffness increases asymptotically as the spring angle approaches the prescribed maximum angular deformation limits. Using the notation from the previous subsection, these enhanced spring law parameters are set to $\alpha_1 = 30°$, $\alpha_2 = 330°$, and $\delta_1 = 15°$.

The load-displacement response of the stacked square-linkage truss is shown in Fig. 4.2 (b). At the beginning of loading, both units resist the applied vertical load. The structure carries a load of approximately 180 N before instability develops in the lower unit. After this instability, the load drops as the lower unit folds. The lower unit continues to deform until the relative angle of its torsional spring approaches the lower angular limit of $15°$. At this stage, the enhanced spring law makes the torsional spring increasingly stiff, preventing any further deformation of the lower unit.

After the lower unit locks, additional load on the system is redistributed and transferred to the upper unit. The load then increases again, leading to a second instability in the upper unit. The upper unit folds like the lower unit until its torsional spring also approaches the lower angular limit. Once both the units are locked, any further increase of load on the structure is resisted by a combination of the axial deformation of the bars and the extremely stiff torsional springs. This produces the final steep increase in the load-displacement curve near $\Delta = 1.6$ m, as shown in Fig. 4.2 (b).

This example illustrates the main advantage of the enhanced spring law. The formulation can capture sequential folding, locking, and load redistribution in a reconfigurable truss with multiple independent kinematic degrees of freedom. A conventional linear torsional spring cannot represent this behavior reliably because it does not impose a physical angular range. The enhanced law therefore provides a way to study collapse sequences, deployment sequences, and actuation paths in systems in which different units activate at different stages of deformation.

The second example considers the reconfigurable Warren truss and its various configurations shown in Fig. 4.2 (c). The truss is 6 m long and 1 m high. The left support is pinned, and the right support is modeled as a roller. Three enhanced torsional springs are placed at the locations marked with red arcs, as shown in Fig. 4.2 (c). Since the structure has three independent kinematic degrees of freedom, the structure can occupy several intermediate configurations depending on which units remain folded or open. Excluding symmetric cases, the truss has five unique starting configurations

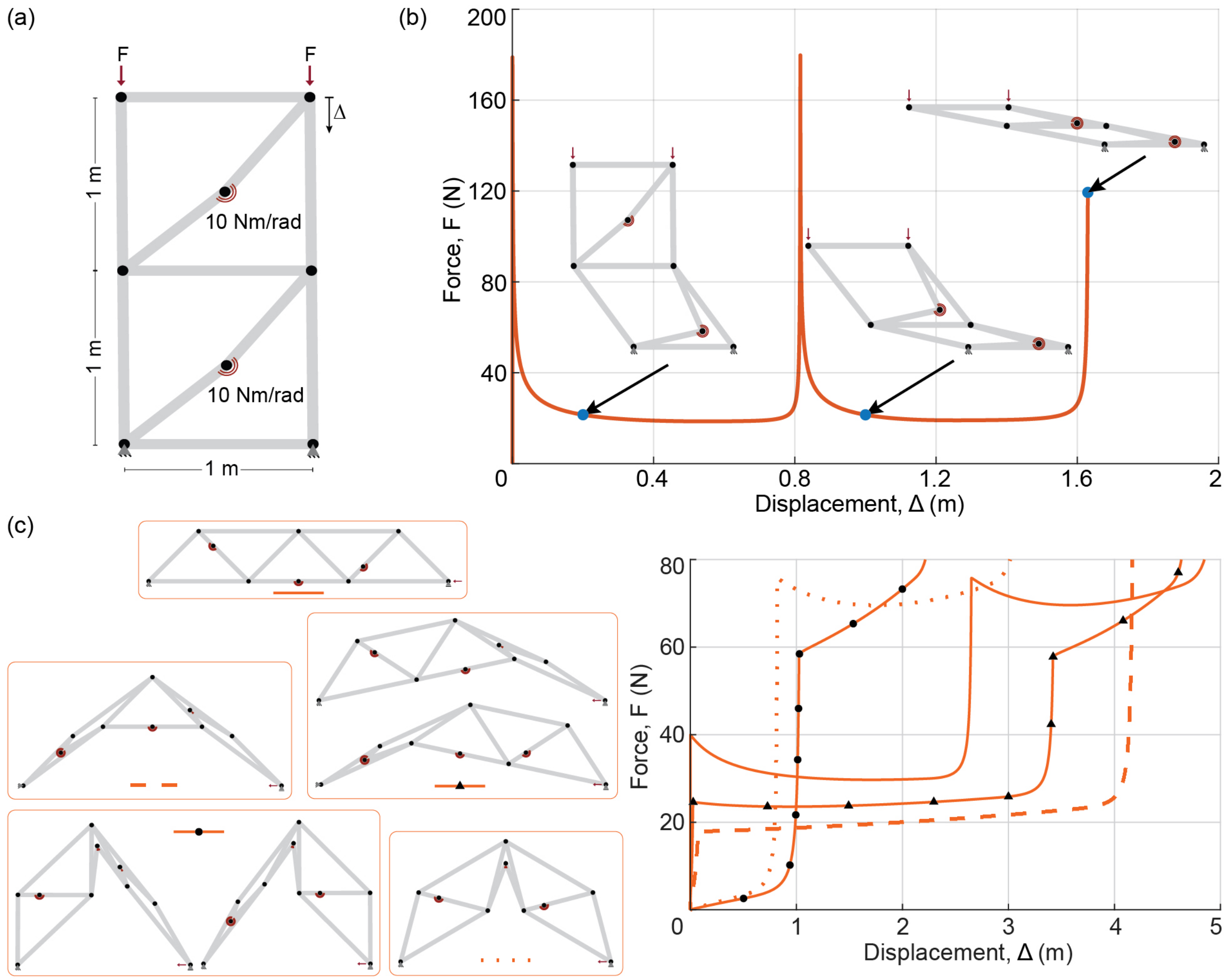


Figure 4.2: (a) Vertically stacked square-linkage truss with two independent folding units under compression. Each unit contains an enhanced torsional spring at the midpoint of the diagonal member; (b) Load-displacement response showing sequential development of instability, folding, and locking of the structure's kinematic units; (c) Reconfigurable Warren truss configurations produced by folding different combinations of its independent kinematic units; (d) Load-displacement responses for folding various Warren truss configurations to the same compact state. The intermediate force demand depends on the folding path, even when the maximum force required to fold the entire structure is the same.

shown in Fig. 4.2 (c).

Each configuration is loaded by applying a vertical displacement at the roller support in the negative y-direction. The corresponding load-displacement curves are shown in Fig. 4.2 (d). Each torsional spring has a geometric imperfection of $0.1°$, and the linear-regime torsional stiffness is set at $K_T = 20$ Nm/rad.

The results show that all loading paths eventually require approximately 80 N to reach the most compact final state. However, the force demand varies substantially during the intermediate stages of folding. This variation occurs because each starting configuration activates a different set of kinematic degrees of freedom. Some configurations deform through a gradual folding sequence. Others require one or more units to snap through or lock before the remaining units can continue to fold. As individual units reach their angular limits, the enhanced springs restrict further rotation and redirect deformation to the remaining active units.

Therefore, the final compact configuration alone does not define the actuation demand. The required force also depends on the folding path used to reach that configuration. The enhanced torsional spring law makes this path dependence accessible in simulation. It allows the force demand to be evaluated for different deployment or collapse sequences while maintaining physically admissible joint rotations. This capability is useful for designing reconfigurable trusses that must fold in a prescribed order, avoid unintended locking, or minimize actuation forces along a selected kinematic path.

4.3 Magnetic Multi-stable Torsional Hinges

This section presents a magnetic multistable torsional hinge that uses polarity-dependent interactions between permanent magnets to create multiple stable rotational states. The physical design, fabrication, and reduced-order torque–rotation model are first introduced to characterize the hinge behavior. The multistable hinge response is then incorporated into the 3NTS element formulation, enabling programmable mechanical response in the reconfigurable trusses.

4.3.1 Prototype and Reduced-order Simulated Response of a Magnetic Multistable Hinge

The magnetic torsional hinge prototype uses polarity-dependent interactions between permanent magnets to create preferred rotational states. The hinge is formed at the joint between two axial members whose ends remain adjacent during rotation. Each member terminates in a circular hub, which provides the mating surface for the magnetic connection, as shown in Fig. 4.3 (a). Each hub face contains eight identical N35-grade cylindrical neodymium magnets with a diameter of 6 mm and a thickness of 2 mm. The magnets are seated in individual holes and are equally spaced at 45° intervals around the hinge axis. The center of each magnet lies on a pitch circle with a radius of 13.75 mm. The opposing hub uses the same geometric layout, so corresponding magnets align face-to-face when the two hubs are in selected relative orientations. The exposed poles alternate between north and south around each hub face, and the pole sequence on the opposing hub is arranged so that aligned magnet pairs have opposite-facing poles in the preferred configurations. As the hubs rotate relative to each other, the facing magnet pairs move between attractive and repulsive alignments, producing a periodic magnetic torque and multiple stable rotational states. The hub dimensions and magnet layout are shown in Fig. 4.3 (a).

As one hub rotates relative to the other, the facing magnet pairs periodically move between attracting and repelling configurations. Stable states occur at relative rotations where the two magnet arrays form an energetically favorable pole arrangement. For the eight-magnet alternating-polarity pattern used in this prototype, the magnetic layout repeats every $\pi/2$ radians, resulting in four stable states over one full rotation. Figure 4.3 (b, left) shows the 3D-printed hinge components with embedded magnets, and Fig. 4.3 (b, right) shows the four stable configurations of the magnetic multistable hinge integrated into a four-bar linkage unit.

A reduced-order magnetic interaction model is used to estimate the torque required to drive

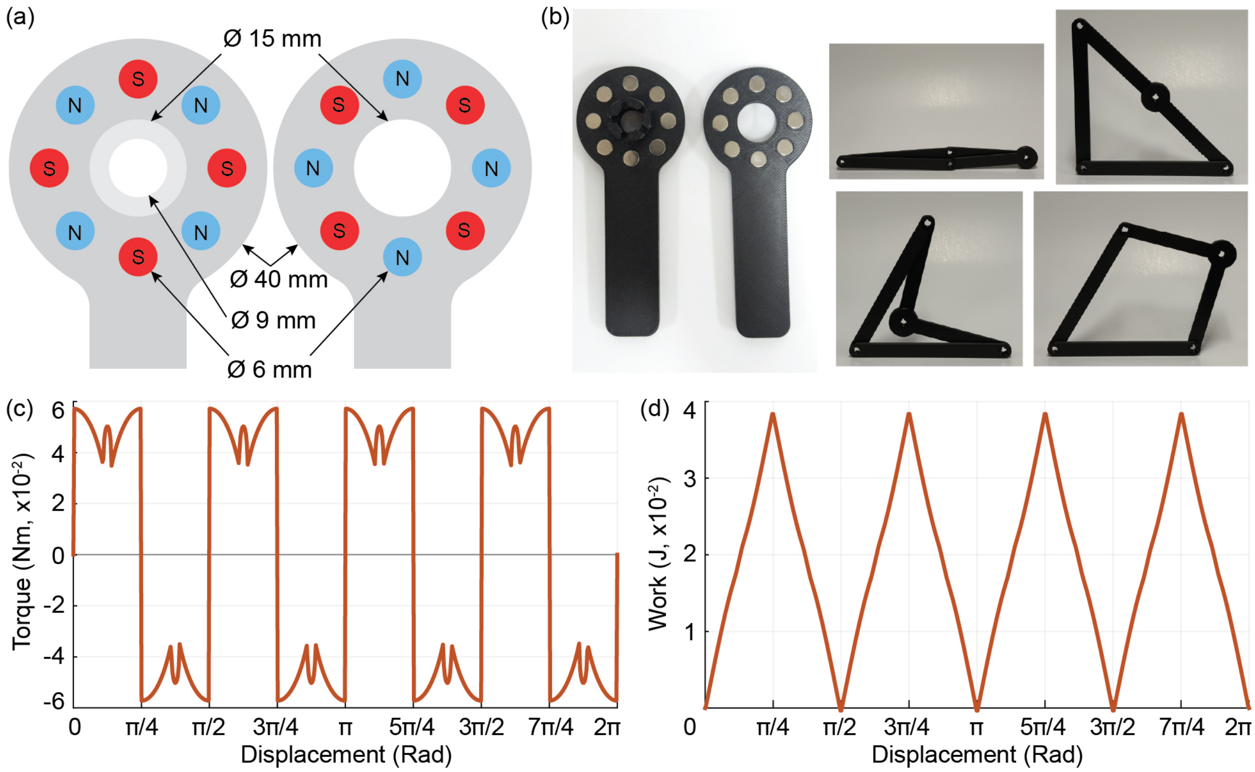


Figure 4.3: Magnetic multistable torsional hinge prototype: (a) Magnet placement and dimensions of the torsional hinge hub; (b) 3D printed and assembled prototype of a magnetic multistable hinge in a 4-bar linkage unit; Simulated behavior of the magnetic multistable torsional hinge using a reduced-order magnetic interaction model: (c) Torque-displacement, and (d) Work-displacement response over one full rotation.

the hinge through multiple stable states and the corresponding work associated with this rotation. The model provides a simple way to interpret the periodic peaks and troughs in the energy landscape produced by the alternating magnet-polarity pattern. The resulting torque-displacement and work–displacement relationships are later used to model multistable hinge behavior in the 3NTS element for multistable reconfigurable truss simulations.

Each magnet is modeled as a cylindrical disc with diameter D, radius $r = D/2$, thickness t, remanence B_r, and axial gap g between the magnets. The axial magnetic field at the opposing face is approximated as:

$$B = \frac{B_r}{2}\left[\frac{z_1}{\sqrt{z_1^2 + r^2}} - \frac{z_0}{\sqrt{z_0^2 + r^2}}\right] \tag{4.4}$$

where, $z_0 = g$ and $z_1 = g + t$. The corresponding magnetic pressure is estimated using

$$P = \frac{B^2}{2\mu_0} \tag{4.5}$$

where $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{N/A^2}$ is the permeability of the free space. The units of P are N/m^2, which are equivalent to J/m^3. Therefore, the magnetic pressure can be interpreted as an approximate magnetic energy density. In this reduced-order model, the magnetic energy is estimated by multiplying this energy density by the effective interaction volume between the magnets. The force associated with a change in magnet position is then obtained by differentiating the magnetic energy with respect to the corresponding displacement direction.

For the rotating assembly, one hub face is held fixed while the opposing hub face rotates over it. The effective overlap area between the two magnet arrays is denoted by $A(\theta)$, where θ is the angular position of the rotating face of the hub. The total magnetic interaction energy of the system at any given instance then becomes

$$U(\theta) \approx PV(\theta) \tag{4.6}$$

where $V(\theta) = \ell A(\theta)$ is the effective interaction volume between the two hub faces and $\ell = t + g$ is the interaction height, taken as the sum of the magnet thickness and axial gap. The torque associated with the magnetic interaction is approximated from the rate of change of the overlap area with respect to the angular displacement θ:

$$\tau(\theta) \approx -P\ell\frac{dA}{d\theta} \tag{4.7}$$

The simulated torque required to rotate the magnetic multistable hinge and the corresponding work are shown in Fig. 4.3 (c) and (d), respectively. The alternating magnet polarity produces a periodic response with a period of $\pi/2$ radians, resulting in four repeating regions over one full rotation. In the torque-rotation response, sharp transitions occur from a positive to negative torque as magnet pairs on either face switch between attractive and repulsive configurations. The corresponding work-rotation response forms a periodic energy landscape with four repeated energy wells. The local minima correspond to stable configurations, while the local maxima represent energy barriers between adjacent stable states. These four energy minima are consistent with the stable states observed in the fabricated four-bar linkage prototype.

4.3.2 Constitutive Modeling of the Multistable Magnetic Torsional Hinge

The magnetic multistable hinge response introduced in the previous section is incorporated into the 3NTS element through a scalar nonlinear constitutive law. The simulated torque-rotation response of the hinge contains an alternating sequence of positive and negative torque plateaus. This behavior is idealized using a smoothed square-wave moment-rotation relationship. In this idealization, the restoring moment alternates between prescribed positive and negative values, and hyperbolic tangent functions provide smooth transitions between adjacent plateaus.

The strain energy associated with the hinge is obtained by integrating the moment-rotation relationship. The tangent stiffness is obtained by differentiating the same moment-rotation relationship with respect to the relative rotation. Because the moment law is represented as a smoothed square wave, the corresponding strain energy takes the form of a smoothed triangular wave. This energy landscape captures the repeated energy barriers produced by the magnetic hinge. Local minima of the energy define stable angular configurations, whereas local maxima define unstable transition states between neighboring stable configurations.

The present subsection focuses on the magnetic response within the angular interval $\alpha_1 < \alpha < \alpha_2$. The response outside this interval is modeled using the nonlinear stiffening formulation described in Section 4.2.1. Within the interior interval, the hinge response is expressed in terms of the shifted rotation $\theta = \alpha - \alpha_0$, where α_0 is the rest angle of the hinge. The interior response is defined as the superposition of a weak linear torsional spring and a nonlinear magnetic contribution. The weak linear contribution regularizes the constitutive response by providing a nonzero baseline stiffness. Without this term, the tangent stiffness can approach zero over the smoothed moment plateaus, which can make the global tangent stiffness matrix singular or poorly conditioned during nonlinear solution. Here, K_1 is defined as the weak linear spring stiffness and M_0 is defined as the magnitude of the resisting moment offered my the magnetic joint.

The periodic magnetic interactions are represented using a phase variable $\phi(\theta) = \mathrm{mod}\,(|\theta|, p)$. Here p is the angular period of the repeating magnetic features and is defined as $p = (\alpha_2 - \alpha_0)/N$. N is the number of periods and $\alpha_3 = p/2$ is the half-period. The smoothed transition coordinates are defined as

$$\xi_0 = \frac{\phi}{\varepsilon}, \qquad \xi_1 = \frac{\phi - \alpha_3}{\varepsilon}, \qquad \xi_2 = \frac{\phi - p}{\varepsilon}. \tag{4.8}$$

The parameter ε controls the sharpness of the transition between neighboring magnetic states. Smaller values of ε produce sharper transitions, while larger values produce smoother transitions.

The magnetic contribution to the restoring moment is defined as

$$M_{\mathrm{mag}}(\theta) = \mathrm{sgn}(\theta) M_0 \left[\tanh(\xi_0) - \tanh(\xi_1) + \tanh(\xi_2)\right]. \tag{4.9}$$

This expression provides a smooth approximation of the alternating moment response produced by the magnetic hinge. The corresponding magnetic tangent stiffness is

$$K_{\mathrm{mag}}(\theta) = \frac{M_0}{\varepsilon}\left[\mathrm{sech}^2(\xi_0) - \mathrm{sech}^2(\xi_1) + \mathrm{sech}^2(\xi_2)\right]. \tag{4.10}$$

The tangent stiffness is concentrated near the transition regions where the magnetic moment changes rapidly between adjacent plateaus.

The magnetic energy is obtained by integrating the magnetic moment contribution. The energy is written as

$$U_{\mathrm{mag}}(\theta) = M_0\varepsilon\left[\mathrm{logCosh}(\xi_0) - \mathrm{logCosh}(\xi_1) + \mathrm{logCosh}(\xi_2) - c_0\right], \tag{4.11}$$

where c_0 sets the reference value of the magnetic energy. The function $\mathrm{logCosh}(\cdot)$ denotes $\log(\cosh(\cdot))$, which is the antiderivative of $\tanh(\cdot)$.

The complete interior constitutive law is obtained by adding the weak linear elastic contribution to the magnetic contribution. The strain energy is represented as

$$U(\alpha) = \frac{1}{2}K_1\theta^2 + U_{\mathrm{mag}}(\theta). \tag{4.12}$$

The restoring moment becomes

$$M(\alpha) = K_1\theta + M_{\mathrm{mag}}(\theta). \tag{4.13}$$

The tangent stiffness is

$$K_T(\alpha) = K_1 + K_{\mathrm{mag}}(\theta). \tag{4.14}$$

The tangent stiffness, restoring moment and strain energy with respect to the angular deformation

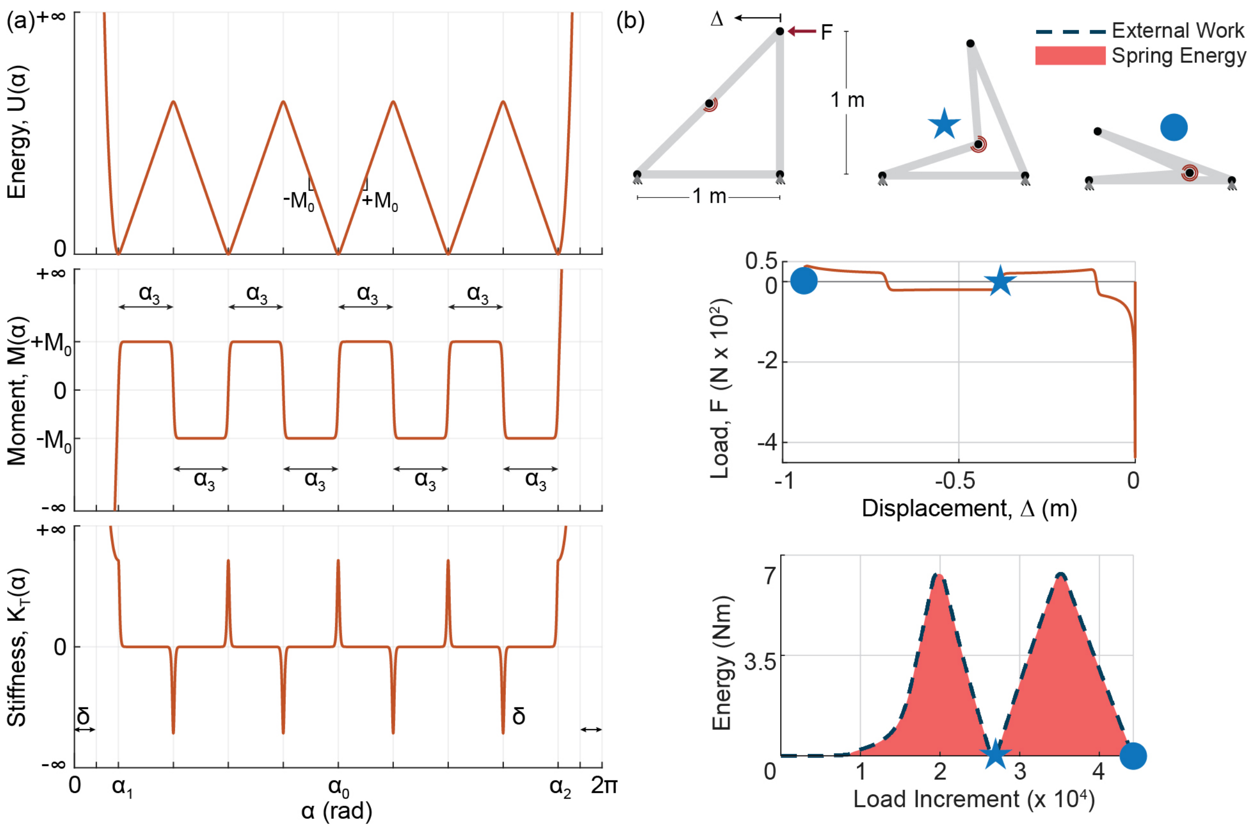


Figure 4.4: Modeling magnetic multistable torsional hinge: (a) Schematic representation of the strain energy (top), restoring moment (middle), and tangent stiffness (bottom) curves for a multistable magnetic torsional hinge with respect to the relative angle α of the hinge; (b) Load vs. Displacement (middle) and System energy vs. Displacement (bottom) curves for the buckling of colinear bars in compression fitted with a bistable torsional hinge.

of a magnetic multistable hinge obtained using the equations above is shown in Fig. 4.4 (a).

4.3.3 Structural Analysis of a 4-bar Linkage Fitted with Magnetic Multistable Torsional Hinge

This subsection demonstrates the nonlinear response of a four-bar linkage unit equipped with a magnetic multistable hinge, shown by the red arc in Fig. 4.4 (b, top). The unit has pinned supports at the base nodes. The horizontal and vertical members are each 1 m long, and the two diagonal segments are each $1/\sqrt{2}$ m long. All bars are modeled with elastic modulus $E = 3 \times 10^9$ N/m^2 and cross-sectional area $A = 0.5$ m^2. The multistable hinge is placed at the joint along the diagonal member. The weak linear stiffness of the torsional spring is set to $K_1 = 1 \times 10^{-4}$ Nm/rad, and the alternating magnetic moment amplitude is set to $M_0 = 1$ Nm. These values are selected only to demonstrate the modeling capability. They are not intended to represent a specific magnet design. In practice, the parameters can be calibrated from the measured or computed moment-rotation response of a physical hinge.

A horizontal load is applied to the top node in the negative x-direction. The resulting load-displacement curve is shown in Fig. 4.4 (b, middle). At small displacements, the structure resists deformation, and the applied load required to further deform the structure increases to approximately 400 N. Beyond this point, an instability develops, and the diagonal member begins to fold about the magnetic hinge. As the hinge rotates away from its initial stable configuration, the magnetic interaction transitions from an attracting configuration to a repelling configuration. This transition produces a reversal in the applied load, shown by the negative-load branch in the load-displacement response.

The corresponding energy response is shown in Fig. 4.4 (b, bottom). The spring energy increases as the hinge moves away from its initial stable configuration and reaches a local maximum near the energy barrier. After this barrier is crossed, the hinge snaps into the next stable state with the magnets in attracting configuration. This first stable state is marked by the blue star in the load-displacement and energy plots and is also illustrated in the deformed configuration in Fig. 4.4 (b, top). Continued displacement produces a second cycle of energy storage and release. The hinge again passes through a repelling configuration and then settles into another attracting configuration, marked by the blue circle.

This example shows that the proposed torsional spring formulation can reproduce the essential mechanics of a magnetic multistable hinge within the nonlinear truss analysis tools. The model captures the load reversals, snap-through transitions, and repeated energy barriers associated with successive stable configurations.

4.4 Bi-Stable Torsional Springs

The previous sections introduced piecewise nonlinear behavior into the 3NTS element through the enhanced spring law and the magnetic multistable hinge law. This section extends that idea to an idealized bistable torsional spring. Although a physical realization of this specific spring law is not considered here, the formulation serves as a demonstration of the modeling capabilities of the tools developed in this work. In particular, it shows how the 3NTS element can be used to simulate programmed mechanical response in reconfigurable trusses with multiple stable configurations.

An ideal bistable torsional spring is a useful model problem because it contains two stable rotational states separated by an unstable transition region. When such springs are embedded in reconfigurable trusses, the resulting structural response can include multiple equilibrium configurations and sequential snap-through events during deployment or loading. This makes the model suitable for studying how nonlinear hinge laws can be used to program stable states in systems with one or more kinematic degrees of freedom.

The bistable response is defined by prescribing the tangent stiffness as a function of the relative hinge rotation. This stiffness-first approach is appropriate because the desired bistable behavior can be easily specified through the sign and variation of the tangent stiffness. The tangent stiffness is positive near the first stable configuration, which corresponds to the undeformed or rest state. It then becomes negative over an intermediate range of rotation, creating an unstable transition region and an energy barrier between stable states. After this transition, the tangent stiffness returns to a positive value, forming the second stable region. The bistable law is therefore constructed as a smooth stiffness landscape with an initial positive-stiffness regime, a negative-stiffness regime, and a post-transition positive-stiffness regime.

This approach avoids the need to first assume a strain-energy function and then differentiate it to obtain the restoring moment and tangent stiffness. That traditional formulation is effective when a suitable smooth energy function is known in advance. For the engineered bistable response considered here, it is easier to prescribe the tangent stiffness and integrate it to obtain the restoring moment. The spring energy remains mechanically consistent because it is obtained from the work-conjugate pair of restoring moment and relative rotation. In the structural examples, the spring energy is evaluated from the internal moment and hinge rotation during the nonlinear simulation. The stiffness-first formulation also provides direct control over the bistable response, since the locations and depths of the energy wells can be adjusted by changing the positive- and negative-stiffness regimes over the prescribed range of rotation.

The following subsection presents the constitutive equations used to define the bistable response of the 3NTS element.

4.4.1 Constitutive Modeling of a Bistable Torsional Spring Law

The constitutive law of the bistable torsional spring is defined over the angular interval $\alpha_1 < \alpha < \alpha_2$. Outside this interval, the spring follows the nonlinear stiffening response introduced in the enhanced spring law from Section 4.2.1. The rest angle α_0 denotes the first stable equilibrium configuration, and the signed relative rotation is defined as $\theta = \alpha - \alpha_0$. This law is assumed to be symmetric about α_0. Therefore, the tangent stiffness is written in terms of the rotation magnitude $|\theta|$, while the sign of the restoring moment is recovered using $\chi = \mathrm{sgn}(\theta)$.

Let K_T denote the initial positive stiffness, K_n denote the negative stiffness, and K_p denote the post-snap positive stiffness. The angular values for the transition of stiffness are denoted by θ_1, θ_2, and θ_3, where $\theta_1 < \theta_2 < \theta_3$. These transition rotations define the locations at which the stiffness changes between regimes. For $0 \leq |\theta| \leq \theta_1$, the stiffness remains constant at K_T. For

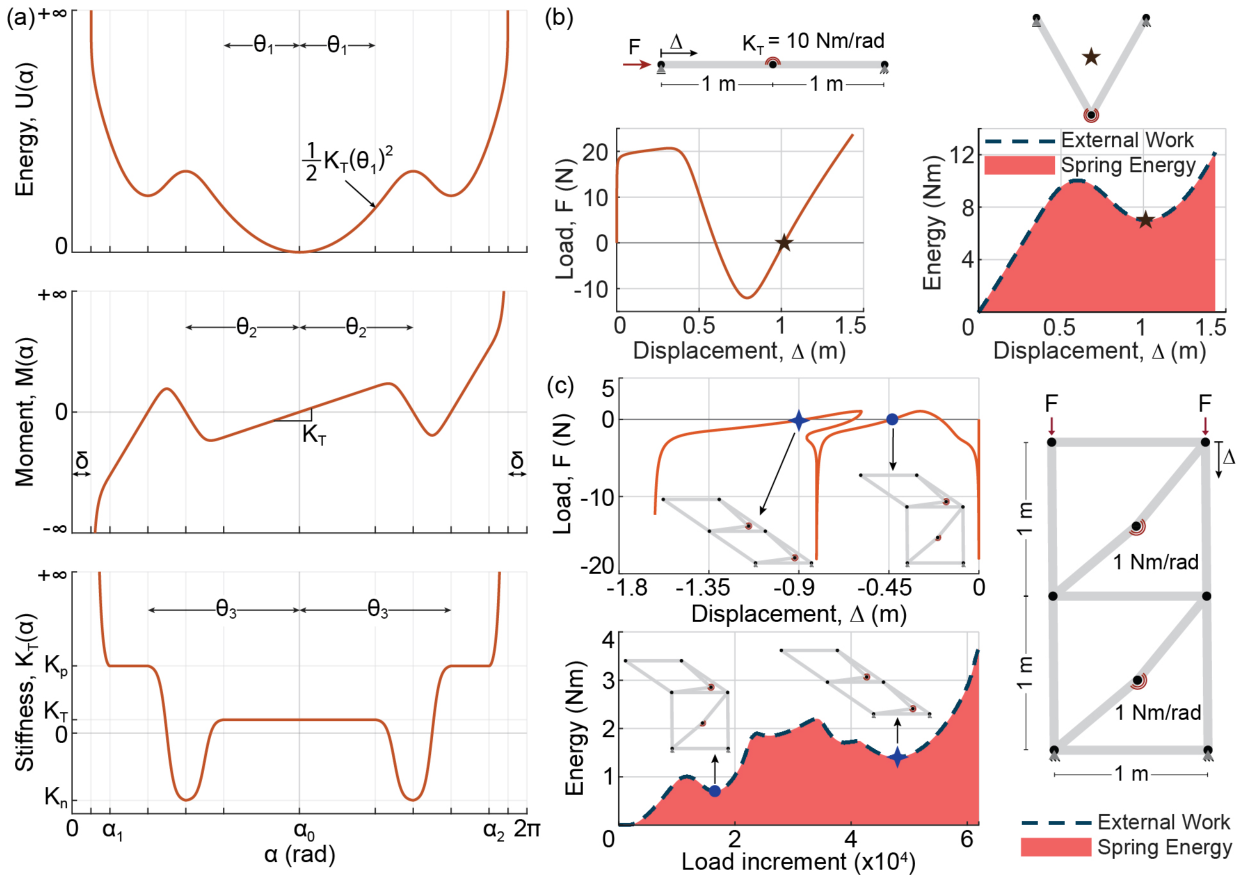


Figure 4.5: Bistable torsional hinge: (a) Schematic representation of the strain energy (top), restoring moment (middle), and tangent stiffness (bottom) curves for a bistable torsional hinge with respect to the relative angle of the hinge; (b) Load vs. Displacement (left) and Strain energy vs. Displacement (right) curves for the buckling of colinear bars in compression fitted with a bistable torsional hinge.

$\theta_1 \le |\theta| \le \theta_2$, the stiffness transitions from K_T to K_n, introducing the negative-stiffness region. $\theta_2 \le |\theta| \le \theta_3$, the stiffness transitions from K_n to K_p. The stiffness remains constant at K_p for rotations beyond θ_3.

To avoid discontinuities in the tangent stiffness, the transitions are smoothed using the normalized function

$$q(s) = \frac{\tanh\left[\beta\left(s-\frac{1}{2}\right)\right] + \tanh\left(\frac{\beta}{2}\right)}{2\tanh\left(\frac{\beta}{2}\right)}, \qquad 0 \le s \le 1. \tag{4.15}$$

Here, the parameter β controls the sharpness of the transition. Larger values of β produce sharper transitions between stiffness regimes, while smaller values produce more gradual transitions. The function satisfies $q(0) = 0$ and $q(1) = 1$, so it provides a smooth interpolation between two prescribed stiffness values. The tangent stiffness in the bistable region is then defined as

$$K_T(\theta) = \begin{cases} K_T, & 0 \leq |\theta| \leq \theta_1, \\ K_T + (K_n - K_T) q\left(\dfrac{|\theta| - \theta_1}{\theta_2 - \theta_1}\right), & \theta_1 < |\theta| \leq \theta_2, \\ K_n + (K_p - K_n) q\left(\dfrac{|\theta| - \theta_2}{\theta_3 - \theta_2}\right), & \theta_2 < |\theta| \leq \theta_3, \\ K_p, & |\theta| > \theta_3. \end{cases} \tag{4.16}$$

This expression directly defines the desired stiffness landscape as shown in Fig. 4.5 (a, bottom).

The restoring moment is obtained by integrating the tangent stiffness with respect to the relative rotation. To write the moment in closed form, the primitive of $q(s)$ is defined as

$$Q(s) = \int_0^s q(\eta)\, d\eta \,. \tag{4.17}$$

Using the definition of $q(s)$, this primitive can be written as

$$Q(s) = \frac{s}{2} + \frac{\ln\cosh\left[\beta\left(s - \frac{1}{2}\right)\right] - \ln\cosh\left(-\frac{\beta}{2}\right)}{2\beta\tanh\left(\frac{\beta}{2}\right)} \,. \tag{4.18}$$

The normalized transition variables are

$$s_n = \frac{|\theta| - \theta_1}{\theta_2 - \theta_1} \,, \qquad s_p = \frac{|\theta| - \theta_2}{\theta_3 - \theta_2} \,. \tag{4.19}$$

The moment magnitude can then be expressed as

$$M_{\text{mag}}(\theta) = \begin{cases} K_T|\theta|, & 0 \leq |\theta| \leq \theta_1, \\ \\ K_T\theta_1 + (\theta_2 - \theta_1)\left[K_T s_n + (K_n - K_T)Q(s_n)\right], & \theta_1 < |\theta| \leq \theta_2, \\ \\ M_{\text{peak}} + (\theta_3 - \theta_2)\left[K_n s_p + (K_p - K_n)Q(s_p)\right], & \theta_2 < |\theta| \leq \theta_3, \\ \\ M_{\text{well}} + K_p(|\theta| - \theta_3), & |\theta| > \theta_3. \end{cases} \tag{4.20}$$

The constants M_{peak} and M_{well} ensure continuity of the moment between adjacent branches. These constants are defined as

$$M_{\text{peak}} = K_T\theta_1 + (\theta_2 - \theta_1)\left[K_T + (K_n - K_T)Q(1)\right] \ , \tag{4.21}$$

and

$$M_{\text{well}} = M_{\text{peak}} + (\theta_3 - \theta_2)\left[K_n + (K_p - K_n)Q(1)\right] \ . \tag{4.22}$$

The signed restoring moment is obtained by restoring the sign of the relative rotation:

$$M(\alpha) = \chi M_{\text{mag}}(\theta) \ . \tag{4.23}$$

This definition makes the moment antisymmetric about α_0, while the tangent stiffness remains symmetric about α_0, as shown in Fig. 4.5 (a, middle).

The strain energy is obtained by integrating the moment from the reference configuration to the current hinge angle:

$$U(\alpha) = \int_{\alpha_0}^{\alpha} M(\tilde{\alpha})\, d\tilde{\alpha} \ . \tag{4.24}$$

and is shown in Fig. 4.5 (a, top). This strain energy curve shows the first stable configuration at $\alpha = \alpha_0$, the unstable energy barrier at $|\alpha - \alpha_0| = \theta_2$ characterized by the energy local maximum and the second stable configuration of the spring at $|\alpha - \alpha_0| = \theta_3$ characterized by the energy local minimum.

4.4.2 Structural Response of Reconfigurable Trusses with Bistable Torsional Hinges

This subsection presents the analysis of two example structures fitted with bistable torsional hinges. A physical prototype of these idealized bistable torsional springs has not yet been developed. Therefore, the examples are intended as computational demonstrations of the modeling capability. They show how bistable hinge laws can be used to study snap-through, post-transition stiffening, and transitions between multiple stable states in the reconfigurable trusses.

Both examples use the bistable torsional spring law described in the previous subsection. The bistable hinge's transition points are set to $\theta_1 = 60°$, $\theta_2 = 90°$, and $\theta_3 = 120°$. The negative stiffness value is defined using $K_n = -5K_T$, and the post-transition positive stiffness branch is defined using $K_p = 5K_T$. The admissible hinge rotation is bounded using the same asymptotic stiffening strategy introduced in the enhanced spring law from Section 4.2.1. The corresponding angular-limit parameters are $\delta = 15°$, $\alpha_1 = 30°$, and $\alpha_2 = 330°$. These limits restrict the deformation of the hinge in the interval (15°, 345°).

The first example consists of two collinear bars connected by a bistable torsional hinge and loaded in compression, as shown in Fig. 4.5 (b). The left end is modeled as a roller support, the right end is modeled as a pinned support, and the bistable torsional spring is placed at the connection between the two bars. Each bar has a length of 1 m, elastic modulus $E = 3 \times 10^9$ N/m^2, and cross-sectional area $A = 0.5$ m^2. The baseline torsional spring stiffness is $K_T = 10$ Nm/rad.

As the imposed displacement (Δ) increases in the positive x-direction, the applied load initially rises to approximately 20 N. An elastic instability then develops at the central hinge, and the structure snaps away from the initial collinear configuration. During this transition, the load required to continue deformation decreases and becomes negative. This response indicates that the structure tends to move toward the next stable state once the instability is triggered. The second stable state occurs near $\Delta = 1$ m. This configuration is marked with a star on the load-displacement curve, the energy plot, and the corresponding deformed configuration in Fig. 4.5 (b). At this point, the spring-energy curve exhibits a local minimum, consistent with a stable equilibrium configuration. Further displacement beyond this point drives the hinge into the post-transition positive-stiffness regime. As a result, the applied load increases again.

The second example considers a reconfigurable truss with two independent kinematic degrees of freedom. The structure, shown in Fig. 4.5 (c), is the same stacked square-linkage truss introduced in Fig. 4.2 (a), but each spring is now modeled as a bistable torsional hinge. In this example, the baseline torsional stiffness is set to $K_T = 1$ Nm/rad for each hinge.

As vertical compression is applied to the structure, the load magnitude first increases rapidly with little global displacement. An elastic instability then develops in the upper square-linkage unit. The upper unit snaps into its next stable state, producing a sharp drop in the applied load. This first stable state is marked with a blue circle in Fig. 4.5 (c). After this transition, the upper hinge enters its post-transition positive-stiffness branch, and additional deformation is redirected to the lower unit.

With continued compression, the load rises again until a second instability develops in the lower square-linkage unit. The lower unit then snaps into its next stable state. This second stable state is marked with a blue star in Fig. 4.5 (c). The blue circle and blue star are also identified on the spring-energy plot, where they correspond to local minima associated with stable equilibrium configurations.

These examples show that bistable torsional hinges can introduce multiple stable configurations into reconfigurable trusses. In the single-hinge example, the structure transitions between two stable states. In the stacked truss, the two independent kinematic degrees of freedom produce a sequence of stable-state transitions. The analysis, therefore, demonstrates how nonlinear hinge laws can be used to program the deployment paths and stable configurations of reconfigurable truss systems.

4.5 Discussion

This chapter developed modeling tools to introduce programmed rotational-hinge behavior into reconfigurable truss systems. First, the enhanced torsional spring law enabled control of the admissible range of hinge rotation and simulation of sequential folding in multi-degree-of-freedom reconfigurable trusses. Second, a magnetic hinge concept was introduced to realize multiple stable rotational states through alternating magnetic interactions. A magnetic multistable constitutive law was developed and incorporated into the three-node torsional spring (3NTS) element to facilitate the nonlinear analysis of reconfigurable trusses with multistable joints. Third, an idealized bistable spring constitutive law was introduced to design bistable rotational joints, enabling reconfigurable trusses with multiple kinematic degrees of freedom to exhibit multistable structural behavior. Together, these contributions extend the analysis framework from purely kinematic reconfiguration toward reconfigurable structures with programmable mechanical response.

Programmable Structural Response for controlled reconfiguration. A primary implication of this work is the ability to tune the response of trusses with multiple kinematic degrees of freedom. The magnetic hinge response depends on the magnet arrangement, polarity sequence, spacing, and magnetic strength. These design variables can be used to adjust the rotational resistance, the angular spacing between stable states, the height of the energy barriers, and the number of accessible stable configurations. As a result, the global structural response can be programmed through local joint behavior. This capability is especially useful for reconfigurable trusses, where each independent mechanism can be assigned a different stiffness, stability threshold, or preferred configuration. The same ideas can also be extended to actively controlled hinges. For example, replacing permanent magnets with electromagnets would allow the resistance and stability landscape to be modified during operation. This would enable increasing or decreasing deployment resistance, activating selected joints, or guiding the structure through prescribed configurations in real time.

Towards reusable infrastructure for impact energy dissipation. This chapter suggests a route toward reusable structures that manage impact energy through controlled reconfiguration rather than permanent material damage. Conventional impact-absorbing systems typically dissipate energy through plastic deformation, which makes them effective but often leaves them permanently damaged. In contrast, a reconfigurable truss fitted with magnetic multistable hinges can deform under external loading by moving from one stable configuration to another without yielding the primary members. During this transition, the external load performs work to move the magnetic assembly from an attracting state to a repelling state and then to the next attracting state. The energy from the external disturbance is therefore used to overcome the magnetic energy barrier and reconfigure the structure. This mechanism allows the structure to absorb impact through a controlled change in geometry, rather than through material damage. Because the magnetic bonds are broken and reformed without plastic deformation, the structure can be restored to its original configuration after loading. This behavior provides a basis for reusable energy-management systems that accommodate external loads through recoverable deformation, controlled collapse, and repeated transitions between stable states.

Engineered deployment paths for reconfigurable structural systems. The combined ability to model sequential folding and multistability creates opportunities for controlling how reconfigurable structures move through complex environments. A reconfigurable truss can be designed to pass through a prescribed sequence of stable or metastable configurations rather than deform along an unconstrained path. Each stable state can serve as a temporary equilibrium configuration during deployment, while the energy barriers between states can regulate when and how the structure transitions to the next configuration. This capability is relevant for systems that must avoid obstacles, pass through confined spaces, or maintain intermediate deployed states during operation. Potential applications include robotic structures, deployable space systems, and support frames for membrane or origami-based systems. In these cases, the truss can be designed to remain kinematically compatible with an external surface while also providing stiffness, load transfer, and stable intermediate configurations during deployment and stowage. More broadly, this approach allows the deployment path itself to become a design variable, enabling structures that not only change shape, but do so through controlled, mechanically programmed states.

Considerations for future work.

Several limitations remain before the proposed multistable hinge concepts can be used as practical load-bearing reconfigurable systems. The examples in this chapter are intended to demonstrate the 3NTS element's nonlinear multistable modeling capability, rather than to optimize a specific physical design. The magnetic hinge parameters used in the numerical studies are therefore idealized. In a physical prototype, these parameters must be calibrated from experimental moment-rotation measurements or from higher-fidelity magnetic-field simulations that account for magnet size, spacing, polarity arrangement, eccentricity, edge effects, and housing geometry. The current formulation also emphasizes quasi-static nonlinear response. This is appropriate for studying equilibrium paths, snap-through events, and stable configurations, but it does not fully capture inertia, loading rate effects, damping, contact, friction, or impact dynamics. These effects are especially important if the hinges are used for energy-management or protective applications, where the loading rate and the post-snap dynamic response may strongly influence the transmitted forces. Practical implementation also requires attention to fabrication tolerances, joint backlash, frictional losses, connection details, magnet alignment, durability under repeated cycling, and the interaction between local hinge behavior and global truss stability. Scalability is another important issue. Magnetic forces are sensitive to separation distance and may not scale directly to large structural loads without larger magnet arrays, mechanical amplification, or hybrid actuation strategies. Finally, the placement and properties of multistable hinges were not optimized in this chapter. Future work should develop design methods that select hinge locations, stable-state spacing, and energy-barrier heights to produce prescribed deployment paths, target load thresholds, and reliable transitions between configurations. These limitations do not reduce the value of the presented work, but they define the next steps required to move from demonstrative numerical models toward validated, reusable, and load-bearing reconfigurable structures.

4.6 Concluding Remarks

This chapter presented a set of hinge-level modeling tools for programming the mechanical response of reconfigurable truss systems. The enhanced torsional spring law introduced admissible rotation limits and enabled the analysis of sequential folding, locking, and load redistribution. The magnetic multistable hinge demonstrated that alternating magnetic interactions can create multiple stable rotational states, and its constitutive response was incorporated into the 3NTS element to achieve programmable multistable behavior in reconfigurable trusses. The idealized bistable spring law further demonstrated how nonlinear hinge behavior can produce multiple stable configurations in trusses with one or more kinematic degrees of freedom. Together, this chapter shows that reconfigurable trusses can achieve programmed shape change through prescribed equilibrium paths and stable states. These capabilities provide a foundation for reusable impact-management systems, controlled deployment in robotic and space structures, and mechanically compatible support frames for origami- and membrane-based deployable systems.

CHAPTER 5

Rapidly Deployable Hulls and On-demand Tunable Hydrodynamics with Shape Morphing Curved Crease Origami

This chapter extends the central philosophy of this dissertation beyond conventional civil engineering structures and into naval architecture. The preceding chapters focused on reconfigurable truss systems, where static structural topologies were transformed into systems capable of controlled shape change and programmable mechanical response. This chapter applies the same principle to hull structures, where geometry directly influences function. In this context, reconfigurability is used not only for compact storage and rapid deployment, but also for tuning hydrodynamic performance. Although the application shifts from structural trusses to naval hulls, the underlying objective remains the same: to use geometric transformation as a mechanism for functional reconfiguration.

Traditional hull fabrication relies on labour- and time-intensive methods to generate smooth, curved surfaces. These conventional methods often lead to hull surface topologies that are static in design with hydrodynamics aimed at handling a broad range of sea conditions but not optimized for any specific scenario. In this chapter, we introduce a method of rapidly fabricating planing hulls using the principles of curved-crease origami. Starting from a flat-folded state, the curved-crease origami hulls can be deployed to match traditional planing hull shapes like the VPS (deep-V, Planing hull with Straight face) and the GPPH (General Purpose Planing Hull). By extension of the ability to conform to a desired shape, we show that the curved-crease origami hulls can emulate desired hydrodynamic characteristics in still as well as wavy water conditions. Furthermore, we demonstrate the shape-morphing ability of curved-crease origami hulls, enabling them to switch between low and high deadrise configurations. This ability allows for on-demand tuning of the hull hydrodynamic performance. We present proof-of-concept origami hulls to demonstrate the practical feasibility of our method. Hulls fabricated using the curved-crease origami principles can adapt to different sea states, and their flat foldability and deployment facilitate easy transport and deployment for rapid response naval operations such as rescue missions and the launch of crewless aquatic vehicles. This chapter has been adapted from **Patil, H. Y.**, Maki, K. J., & Filipov, E. T. (2024). Rapidly deployable hulls and on-demand tunable hydrodynamics with shape morphing curved crease origami. Journal of Fluids and Structures, 130, 104176 [105].

5.1 Introduction

Advanced boat-building and sailing technologies have existed since 6000 BCE [106]. Despite the impressive technological strides in naval engineering over centuries, hull fabrication continues to be an enduring and intricate task. Hull fabrication relies on labor-intensive methods that involve connecting numerous flat sheets through bolting, riveting, or welding to form smooth, curved surfaces [107]. Achieving the necessary smooth curved surfaces also requires complex casting, milling, and molding procedures. Another challenge is that a significant portion of hull fabrication occurs in landlocked regions, necessitating the transportation of ship parts to coastal docks for assembly [108]. This transportation further extends the construction timeline and associated costs. These current hull fabrication techniques lead to designs with static surface topologies, resulting in hydrodynamics that are a compromise to accommodate a broad spectrum of sea conditions. Consequently, these hulls are not optimized for specific conditions, often leading to inefficient power consumption, poor maneuverability, and reduced passenger comfort and safety. For instance, crew members experience musculoskeletal pain and mental fatigue due to repeated exposure to vertical accelerations caused by the porpoising motion of vessels [109]. These multifaceted challenges underscore the need to adopt innovative approaches in the shipbuilding industry, focusing on more versatile fabrication processes and improved adaptability in hull performance.

Origami, the age-old art of paper folding, presents a promising extension to the existing fabrication techniques used by the shipbuilding industry. In recent years, origami has transcended its traditional boundaries and found remarkable applications in engineering structures [110]. One key characteristic of origami-inspired structures is their ability to transform flat sheets into complex, three-dimensional structures. This geometrically transformative process dynamically changes the properties of structures and the surrounding fluid media to meet multiple design objectives by varying the extent of folding, as shown by [111] and [112]. Origami-inspired designs embody several advantageous characteristics, each with unique applications. For instance, structures designed using origami principles can be rapidly deployed from compact configurations and easily reconfigured for various functions [113]. This reconfigurability has potential applications, such as creating stents for minimally invasive surgeries in biomedical engineering [114] and improving the storage and deployment of satellites and solar arrays in space structures [115]. Origami principles are also scale-independent [116], making them applicable to structures ranging from deployable stadium canopies to micro-scale sensors. Additionally, the active folding capability enables these structures to change form on demand, increasing their adaptability to various loads and conditions — a strategy often employed by biological organisms to survive in strong winds or water currents [117]. Thus, incorporating origami principles into ship hull design offers benefits such as flat foldability, efficient storage and transport, and rapid deployment.

While traditional origami surfaces such as Miura-ori [118] are characterized by rough and angular profiles, curved-creasing — a subset of origami, provides a groundbreaking approach to fabricating hull-type structures in naval engineering. Research by [119] demonstrates that folding a sheet of material along arbitrary non-intersecting curved creases enables rapid and easy fabrication of smooth curved surfaces. Curved-crease origami preserves the inherent merits of conventional origami techniques while offering numerous benefits suited for hull fabrication. For instance, curved creasing grants precise control over surface topologies by tuning the crease geometry and extent of actuation. Furthermore, it enables the fabrication of complex 3-D shapes that embody benefits such as reduced joint complexity, increased global stiffness, and improved resistance to buckling [19].

Motivated by the potential of curved-crease origami principles, this chapter presents an approach for fabricating planing hulls. Planing hulls commonly feature a V-shaped bottom and a transom stern design, which generates hydrodynamic lift to support the hull weight and enable high-speed travel. As a result, these hulls are widely used for applications such as Coast Guard boats, rescue vessels, and ocean racers [120]. [121] provides a detailed review of the topology and hydrodynamics of planing hulls. The inherent geometry of planing hulls (see Section 5.2) makes them an ideal candidate for incorporating curved-crease origami-based fabrication techniques. Thus, as a first step towards curved-crease origami-based hulls, we limit the scope of our exploration to idealized planing hulls. While we agree that extending origami principles to general hull forms poses several challenges at this stage, it still holds significant potential for curved surface fabrication and will be explored separately in the future.

By leveraging the principles of origami, we demonstrate the capability to reproduce a predetermined target planing hull shape with precision. This capability enables the emulation of a desired hydrodynamic behavior and embodies the fabrication efficiency inherent to curved crease origami. Furthermore, through the strategic utilization of active folding in curved-crease origami hulls, we illustrate their ability to undergo a characteristic shape change in real-time. This shape morphing behavior enables on-demand tunability of hydrodynamic properties and renders them applicable across a wide variety of sea conditions. Hulls fabricated using the proposed methods can adapt to any sea state, and their flat foldability and rapid deployment will facilitate easy transport and deployment for naval operations such as covert rescue missions and the launch of crewless decoy marine vehicles. These vehicles will also improve sea-keeping performance, power efficiency, maneuverability, and passenger comfort, making them suitable for civilian recreational use.

The remainder of the chapter is organized as follows. Section 5.2 reviews standard planing hull geometries and curved folding procedures. The subsections discuss the planing behavior and typical geometric features of a planing hull, the development of a curved-origami crease pattern for the planing hull, an overview of the numerical model for curved-origami folding, and geometric parameter optimization to match curved-origami hulls to their target shape. Next, Section 5.3 explores the hydrodynamic analyses and performance evaluation of curved origami planing hulls against target hull shapes. The core findings of this chapter, including the on-demand shape morphing capabilities of curved-origami hulls and tunable hydrodynamic characteristics for a wide range of sea conditions, are presented in Section 5.4. A discussion on the scalability of origami hulls with an emphasis on curvature analysis for various construction materials is presented in Section 5.5. Finally, concluding remarks summarizing the key contributions are included in Section 5.6.

5.2 Geometry and Curved Folding

5.2.1 Planing Hull Geometry

In a state of rest, buoyant forces are responsible for bearing the weight of a floating hull. Buoyancy is also crucial at lower speed since the hull stays afloat by displacing a volume of water. However, as the speed increases, hydrodynamic lift becomes dominant over the buoyant force and supports the weight of the hull. This hydrodynamic lift allows a planing hull to rise and glide on the water surface.

A typical planing hull has distinct geometric features that enable it to generate hydrodynamic lift. These geometric features of a planing hull (some illustrated in Fig. 5.1 (a)) can be defined as follows: (*i*) *Bow*: the foremost part of the hull; (*ii*) *Chine*: line along which a sharp change in the angle of cross-section occurs; (*iii*) *Deadrise*: the angle between the bottom of the hull and horizontal surface; (*iv*) *Draft*: the distance between the waterline and the deepest part of the boat in steady state; (*v*) *Hull Bottom*: bottom surface of the hull (also called the planing surface); (*vi*) *Keel*: bottom-most longitudinal structural element of a hull; (*vii*) *Sheer line*: the uppermost edge of the hull; (*viii*) *Stern*: the aftmost part of the hull; (*ix*) *Transom*: the flat board extending from the keel to the deck at the stern; (*x*) *Trim*: the angle between the horizontal surface and the line connecting the forward and the aft draft; and (*xi*) *Waterline*: the level water reaches on the hull's side. In some cases, the hull has a parallel section and can be divided into two parts longitudinally, namely the prismatic and non-prismatic sections. The prismatic section often forms the middle to aft half of the hull where the keel and chine lines are parallel. On the other hand, the non-prismatic section forms the front portion of the hull where the keel and chine lines are no longer parallel.

It is important to note that the planing hulls discussed in this chapter are hard-chined vessels, which means that the hull exhibits a distinct change of angle along the chine, where the bottom surface meets the sides of the boat. Additionally, these planing hulls feature a typical V-shaped bow with straight surfaces instead of an axe, bulbous, parabolic, or flared bow.

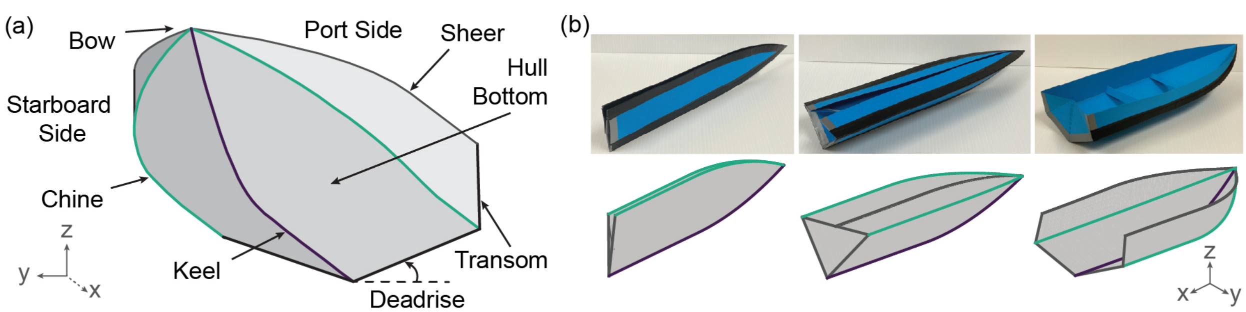


Figure 5.1: (a) Geometric features of a typical planing hull: specifically a deep V-shape, Planing hull with a Straight planing surface (VPS); (b) Sequential deployment of curved-crease origami hull designed to match the shape of a VPS hull: paper model (top) compared with Bar and Hinge simulation (bottom). A transom is not included in the simulation.

The curved-crease origami-inspired planing hulls offer a method to change some of the aforementioned geometric features that directly affect the hull's planing ability, stability, ride comfort, maneuverability, and power efficiency. Throughout its deployment, the curved crease origami hulls can vary the deadrise, hull length, topology of the curved planing surface, and beam length. The deployment process from a flat-folded to a deployed-hull state is shown in Fig. 5.1 (b) for both a 20-inch long paper prototype and for a corresponding Bar and Hinge simulation (discussed in the following section). The planing hull design, obtained using the bar and hinge model, is presented here without the transom to reveal distinctive features of the hull, including the chine, keel, and profile of the planing surface. However, the transom can be incorporated into the model, akin to the paper prototype shown in Fig. 5.1 (b).

5.2.2 Crease Geometry and Bar and Hinge Model for Curved Folding

A curved crease pattern presented in Fig. 5.2 (a) is developed to match the VPS hull designed by [122]. VPS stands for deep V-shape, Planing hull with a Straight planing surface, and resembles the shape illustrated in Fig. 5.1 (a). This crease pattern unfolds into a 3D shape with all the geometric features of a typical planing hull. The geometric parameters used to define this crease pattern are: (*i*) width of the hull's bottom surface between the chine and the keel, H; (*ii*) height of the tip, h_{tip}; (*iii*) overall length of the crease pattern, L_{OA}; (*iv*) length of the prismatic part of the chine, L_P^c; (*v*) length of the non-prismatic part of the chine, L_N^c; (*vi*) length of the prismatic part of the keel, L_P^k; and (*vii*) length of the non-prismatic part of the keel, L_N^k. The curved portions (non-prismatic section) of the chine and the keel curves are parabolic and denoted by y_c in Eq. 5.4 and y_k in Eq. 5.3, respectively. For a given overall length of the crease pattern, the length of the non-prismatic section for the chine and keel curves are denoted by L_N^c in Eq. 5.1 and L_N^k in Eq. 5.2, respectively. These quantities can be computed as:

$$L_N^c = L_{OA} - L_P^c \; , \tag{5.1}$$

$$L_N^k = L_{OA} - L_P^k \; , \tag{5.2}$$

$$y_k = \left[\frac{h_{\text{tip}}}{{L_N^k}^2}\right] \times \left(x - L_P^k\right)^2 \; , \tag{5.3}$$

$$y_c = \left[\frac{h_{\text{tip}} - H}{{L_N^c}^2}\right] \times (x - L_P^c)^2 + H \; . \tag{5.4}$$

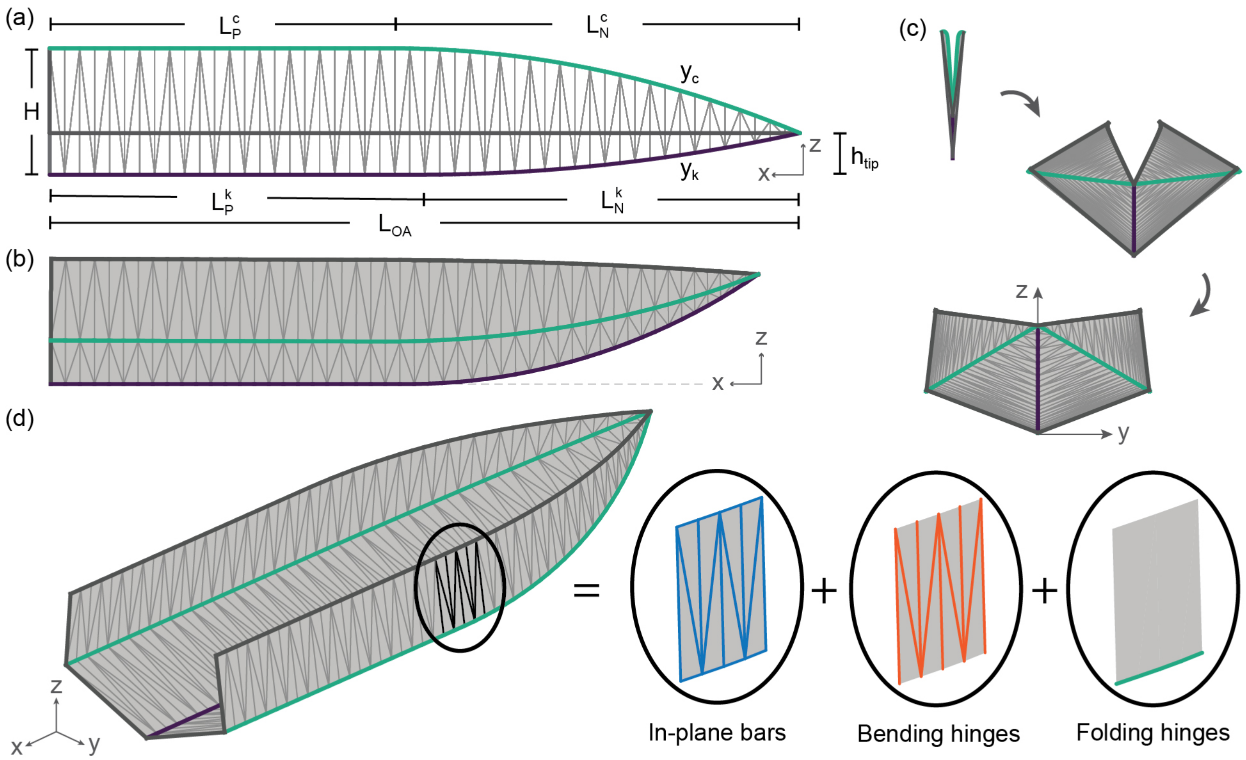


Figure 5.2: (a) Geometric parameters defining the curved crease pattern for a planing hull shape; (b) Folded curved-crease origami hull shape viewed from starboard side; (c) Flat, intermediate, and fully folded (clockwise from top) curved-crease origami hull shape viewed from the Transom (also known as Stern) side; and (d) Decomposition of the Bar and Hinge model into its components: in-plane bars, bending hinges, and folding hinges.

The curved folding simulation of the origami crease pattern into a planing hull shape is accomplished through the Bar and Hinge model developed by [94]. This model offers a simplified numerical approach that effectively simulates the fundamental behaviours of curved origami folding by employing three distinct elements, as illustrated in Fig. 5.2. These elements can be described as follows: (*i*) three-dimensional truss elements, or bars, that capture in-plane stretching and shearing of the sheet; (*ii*) rotational springs, or hinges, that capture the out-of-plane bending of the sheet by consolidating bending into discrete rotations along the bar lengths; and (*iii*) another set of hinges that capture the rotations at the creases or fold lines.

The crease pattern is meshed using a combination of the three elements, where the stiffness of each element is determined based on material properties, sheet thickness, and mesh size. Despite the inherent geometric nonlinearity involved in the folding process, the model achieves convergence within a short time frame (less than 10 seconds). This rapid convergence is due to the relatively low number of elements in the Bar and Hinge model compared to finite element methods. Consequently, this rapid simulation enables efficient analysis of curved origami structures, facilitating subsequent shape-matching and optimization processes.

5.2.3 Shape Matching and Optimization

The geometric design parameters presented in Fig. 5.2 (a) can be varied and optimized such that the curved crease origami (CCO) hull shape will match the desired target shape. The matching of CCO hull with target hull shapes is significant for two reasons. Firstly, it enables the reproduction of specific hydrodynamic characteristics by replicating the target shape. Secondly, it facilitates the integration of origami-based design principles in conventional hull design, thereby unlocking the potential to design on-demand shape morphing hulls with tunable hydrodynamics. This section will explore two design variations of the CCO hulls, $\mathrm{CCO}^{\mathrm{VPS}}$ and $\mathrm{CCO}^{\mathrm{GPPH}}$, and compare their shapes with those of the VPS and the General Purpose Planing Hull (GPPH) shapes, respectively.

The quantification of dissimilarity between a CCO hull shape and a target shape is performed using a parameter called the Hausdorff distance, d_H. By definition, the Hausdorff distance rep-

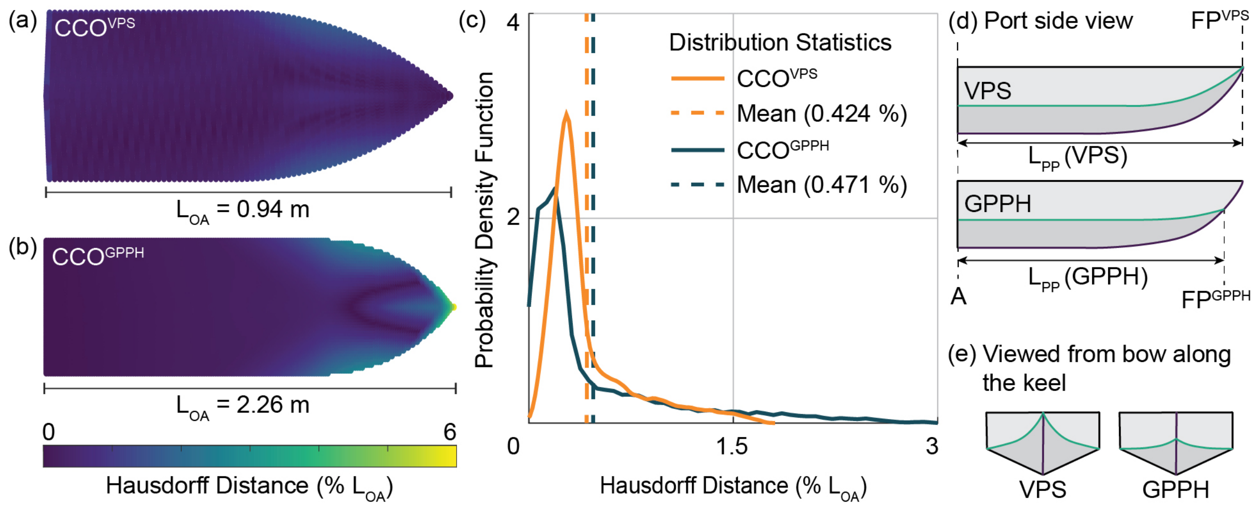


Figure 5.3: Color heat map showing the Hausdorff distance as a percentage of the overall hull length (L_{OA}) for the planing surface of two different CCO hull designs compared with: (a) 0.94 m long VPS hull and (b) 2.26 m long GPPH, respectively; (c) Hausdorff distance distribution statistics for CCO hull designs' planing surface compared with corresponding VPS and GPPH designs; (d) Port side view depicting the location of forward perpendicular (FP), aft perpendicular (AP) and the length between the two perpendiculars (L_{PP}); and (e) Bow view when looked along the keel of the VPS and GPPH showing the difference in the chine and keel configuration.

resents the maximum distance between any point in one set and the closest point in the other set in three-dimensional space, indicating the largest discrepancy between the two point sets. Hence, smaller Hausdorff distances between two sets (two shapes in this case) indicate a closer match. To ensure consistency in comparing the two different design variants of the CCO hulls with their target shapes, we will represent the Hausdorff distance as the percentage of the overall length, L_{OA} of the hull's planing surface. We specifically emphasize comparing the planing surface because its interaction with water is the predominant factor that influences the hydrodynamic behavior of the hull.

The planing surfaces of CCO^{VPS} and VPS hulls are populated with approximately 5000 points, and the Hausdorff distance for each point on the CCO^{VPS} hull is computed with respect to the VPS hull (Fig. 5.3 (a)). As the folded shape of the CCO^{VPS} hull is influenced by the geometric parameters of the curved-crease pattern, a simple optimization problem is formulated to find optimal values that minimize the Hausdorff distance between the two hulls. The objective of this problem is to minimize the mean Hausdorff distance between the two hulls, considering the following design variables: (*i*) overall length of crease pattern (L_{OA}), (*ii*) width of the planing panel (H), (*iii*) tip height (h_{tip}), (*iv*) length of the prismatic part of chine (L_P^c), (*v*) length of the prismatic part of the keel (L_P^k), and (*vi*) deadrise of the hull. The optimizer successfully achieves a mean Hausdorff distance of 0.424% and a maximum Hausdorff distance of under 2% of the overall length of the hull, indicating an excellent match between the two shapes. This method enables on-demand replication of hull shapes using curved-crease origami principles for innovative hull design and hydrodynamic optimization.

A similar shape-matching procedure is performed for the CCO^{GPPH} and the GPPH hulls with approximately 7500 points. In this case, the mean Hausdorff distance is observed to be 0.471%, and the maximum Hausdorff distance is under 6% of the overall GPPH length. The larger Hausdorff distances compared to the previous case, particularly towards the bow of the hull (see Fig. 5.3 (b)), can be explained by the difference in the keel extension of the hull designs. As illustrated in Fig. 5.3 (d) and (e), the keel line of the GPPH extends beyond the chine. Meanwhile, the chine and keel lines of the VPS and both CCO hull designs terminate at the same point on the hull's bow. This difference in keel extension leads to a dissimilar bow shape in the GPPH design compared to the VPS and both CCO hull designs, thus explaining the higher Hausdorff distances.

Despite the minor dissimilarities observed near the bow of the hull in the cases of the VPS and GPPH, the CCO hulls achieve a close match over the prismatic section of the planing surface. The mean Hausdorff distance for the prismatic section of VPS and GPPH equivalent CCO hulls is observed to be 0.258% and 0.281% of the overall length, respectively. Furthermore, the distribution of Hausdorff distances shown in Fig. 5.3 (c) indicates that the mean Hausdorff distance is less than 0.5% of the overall length for most parts of the planing surface that would interact with water. This similarity in shape and deadrise ensures that the CCO hull can potentially exhibit hydrodynamic characteristics comparable to those of the conventional VPS and GPPH hull designs. Detailed findings related to the hydrodynamic properties and the comparison between the CCO hulls, VPS, and GPPH are presented and discussed in the subsequent sections.

5.3 Hydrodynamic Performance Evaluation of Curved-crease Origami Hulls

The ability of curved-crease origami (CCO) hulls to align congruently with target hull shapes motivates a comparative investigation to evaluate whether their hydrodynamic characteristics also match those of the conventional hulls. This section compares the hydrodynamic performance of the two distinct CCO hull variants, CCO^{VPS} and CCO^{GPPH} as introduced in Section 5.2.3, with their corresponding target hull forms, namely the VPS and the GPPH. The hydrodynamic evaluation for all four hull geometries is facilitated by the Powersea tool and encompasses the simulation of hull motion in both still and wavy water (head sea) conditions. Powersea's underlying theory, assumptions and limitations can be found in Appendix C. The geometric and physical characteristics of the hulls passed as input to Powersea can be found in Table 5.1.

Table 5.1: Geometric and physical characteristics of the hulls for hydrodynamic simulations

Hull geometry	**L$_{OA}$** (m)	**L$_{PP}$** (m)	**LCG** (% L$_{OA}$)	**VCG** (m)	**K$_{yy}$** (% L$_{OA}$)	**Weight** (N)	**B** (m)
CCO^{VPS}	0.94	0.92	62	0.056	25	65	0.312
VPS	0.94	0.92	62	0.056	25	65	0.308
CCO^{GPPH}	2.26	2.26	62	0.097	25	995.51	0.583
GPPH	2.26	2.23	62(% L_{PP})	0.097	25(% L_{PP})	995.51	0.615

The still water simulations for all four hull shapes are conducted for various constant speed cases. For the VPS and CCO^{VPS} hulls, these speeds range from 2 to 15.5 m/s. For GPPH and CCO^{GPPH} hulls these speeds range from 2.5 to 20.5 m/s. The speeds correspond to beam Froude numbers, F_B ranging from 1 to 9 for each of the four hull geometries. The beam Froude number is calculated as $F_B = U/\sqrt{g \times B}$, where U is the speed, g is the acceleration due to gravity and B is the beam width (distance between the two keels at transom). Other simulation parameters adhered to the default values specified by Powersea, and the hydrodynamic characteristics are shown in Fig. 5.4. The hydrodynamic characteristics include: (*i*) friction drag experienced by the hull due to the shearing of the fluid boundary layer normalized by the hull weight, (*ii*) total resistance — a combination of friction drag, viscous pressure drag and wave making resistance, normalized by the hull weight, (*iii*) Rise of Center of Gravity (CG) — heave displacement relative to at-rest position normalized by the beam width, and (*iv*) trim — running angle of the hull about its lateral axis as it makes way in the water. While the friction drag, rise of CG, and pitch data are generated by Powersea, the weight normalized total resistance, R_T/W, is calculated as follows:

$$\frac{R_T}{W} = \tan(\eta_5) + \frac{D}{W} \times \frac{1}{\cos(\eta_5)} \ , \tag{5.5}$$

where D is the friction drag, W is the weight of the hull, and η_5 is the pitch (trim angle) [123]. It is important to note that the total resistance does not include whisker spray drag.

It is essential to note that the comparative analysis of CCO hulls with counterparts GPPH and VPS hulls reveals a consistent trend across various hydrodynamic characteristics. Both friction drag in Fig. 5.4 (a) and total resistance in Fig. 5.4 (b) exhibit a monotonically increasing trend as the planing speed increases. Additionally, the hull tends to rise on the water surface with an increase in planing speed, as evidenced by the rise in CG illustrated in Fig. 5.4 (c). Contrarily, the trim angle of the hull diminishes with an increase in the planing speed, as indicated in Fig. 5.4 (d). However, it is also imperative to acknowledge a slight disparity in the trim angle of the CCO hulls and their GPPH and VPS counterparts. The following reasons can explain this disparity. First, despite exhibiting a close match, the CCO hulls are not identical in shape to their GPPH and VPS counterparts. The minor differences in the shapes of planing surfaces and beam widths, as documented in Section 5.2.3 and Table 5.1, respectively, explain the deviations observed in sinkage and the trim angle at various speeds. Second, Powersea requires the forward perpendicular (FP) to be set at the intersection of the chine and keel lines along the hull's length. For both CCO^{GPPH} and

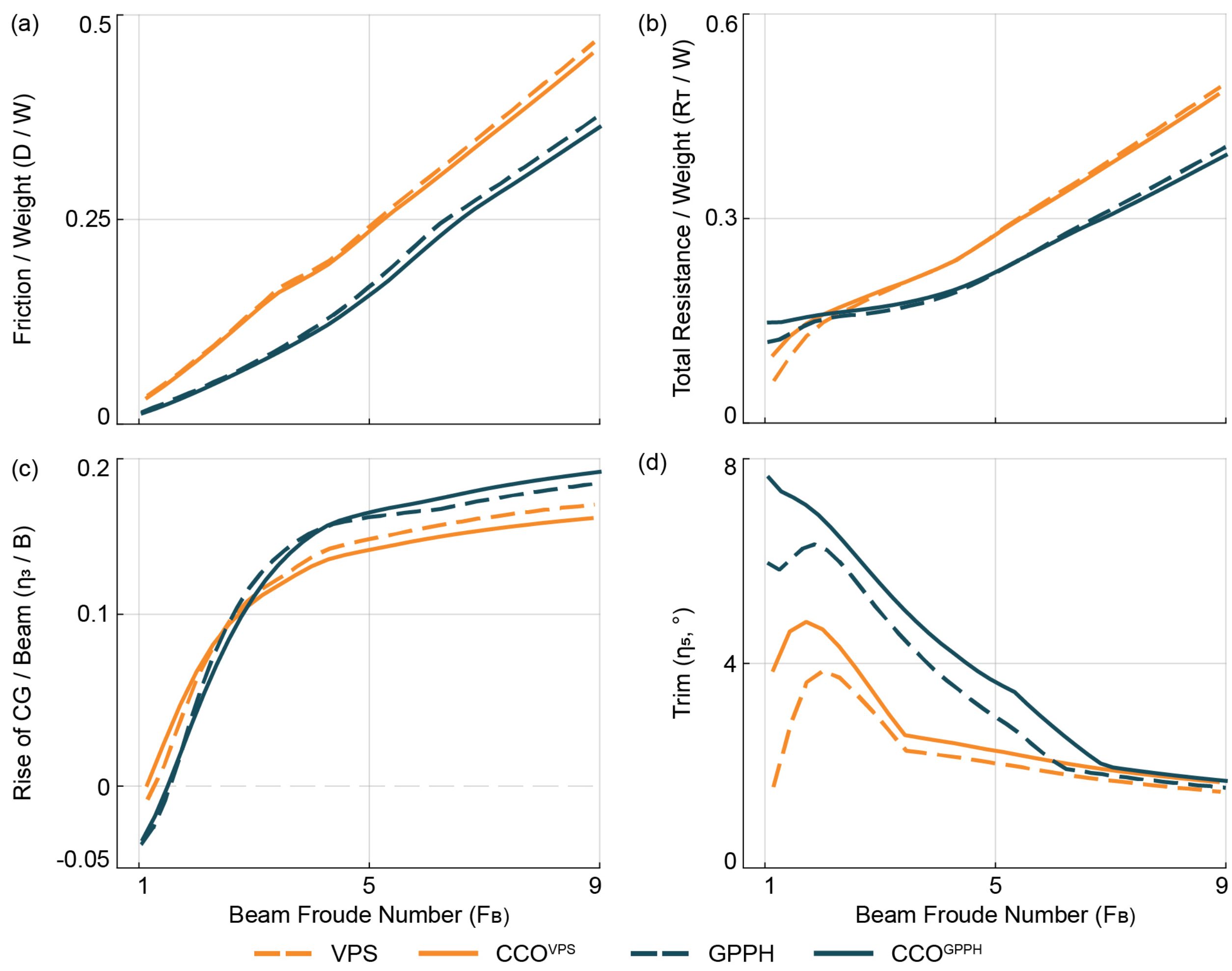


Figure 5.4: Comparison of VPS and GPPH hydrodynamics with their CCO counterparts in calm water: (a) Weight normalized friction drag; (b) Weight normalized total resistance; (c) Beam normalized rise of CG; and (d) Trim angle (positive bow up)

GPPH, this requirement results in differing hull length (L_{PP}) between the aft perpendicular (AP) and the forward perpendicular (FP), as shown in Fig. 5.3 (d). Furthermore, Powersea only utilizes the portion of the hull geometry between the AP and FP to simulate its behaviour in water. To ensure consistency across all simulations, we align the FP for all hull geometries with the origin at time t=0. Specifically for the GPPH, the LCG and K_{yy} are set based on the modeled length of the hull which corresponds to (L_{PP}) rather than (L_{OA}) (see Table 5.1), resulting in a slightly different LCG location compared to CCO^{GPPH}. These modeling differences lead to varying initial equilibrium positions of the hulls and, consequently, to the differences in the trim angle at various speeds. Lastly, the assumptions in Powersea's underlying theory make it less accurate for predicting hull motion in semi-planing regimes and when the beam Froude numbers are less than or equal to 1. Therefore, the significant deviation in the trim angle at lower speeds can be safely disregarded. At high speeds, when the hulls are advancing in the planing regime, and the bows of the hulls are no longer in contact with the water, the CCO hulls and their counterparts exhibit similar performance.

The investigation of hull motion in waves is carried out under constant speed, corresponding to a beam Froude number, $F_B = 4$ when navigating head seas. For VPS and CCO^{VPS} hulls, the speed equals 7 m/s, whereas for GPPH and CCO^{GPPH} hulls, the speed equals 10 m/s. Wave amplitudes of 0.02 m were applied to the VPS and CCO^{VPS} hulls, based on the tank test parameters for VPS hull from the [122] study. The GPPH and CCO^{GPPH} hulls were subjected to wave amplitudes of 0.04 m to maintain a wave amplitude to hull length ratio similar to that of the VPS and CCO^{VPS} hulls. Each of the four hull geometries was subjected to regular waves with wavelengths, λ equivalent to 4 and 10 times the overall length, L_{OA} of the planing hull. The graphical representation of these regular wave characteristics is provided in Fig. 5.5 (a-b). Figure 5.5 (c) and (e) display the final 6 seconds of the 100-second long time histories depicting the steady state, mean-subtracted heave (normalized by wave amplitude) and pitch (normalized by wave slope) motions of the hulls, corresponding to the non-dimensionalized wavelength represented as λ/L_{OA} ratios of 4 and 10, respectively. The mean values of normalized heave and pitch time histories for all hull geometries and λ/L_{OA} ratios

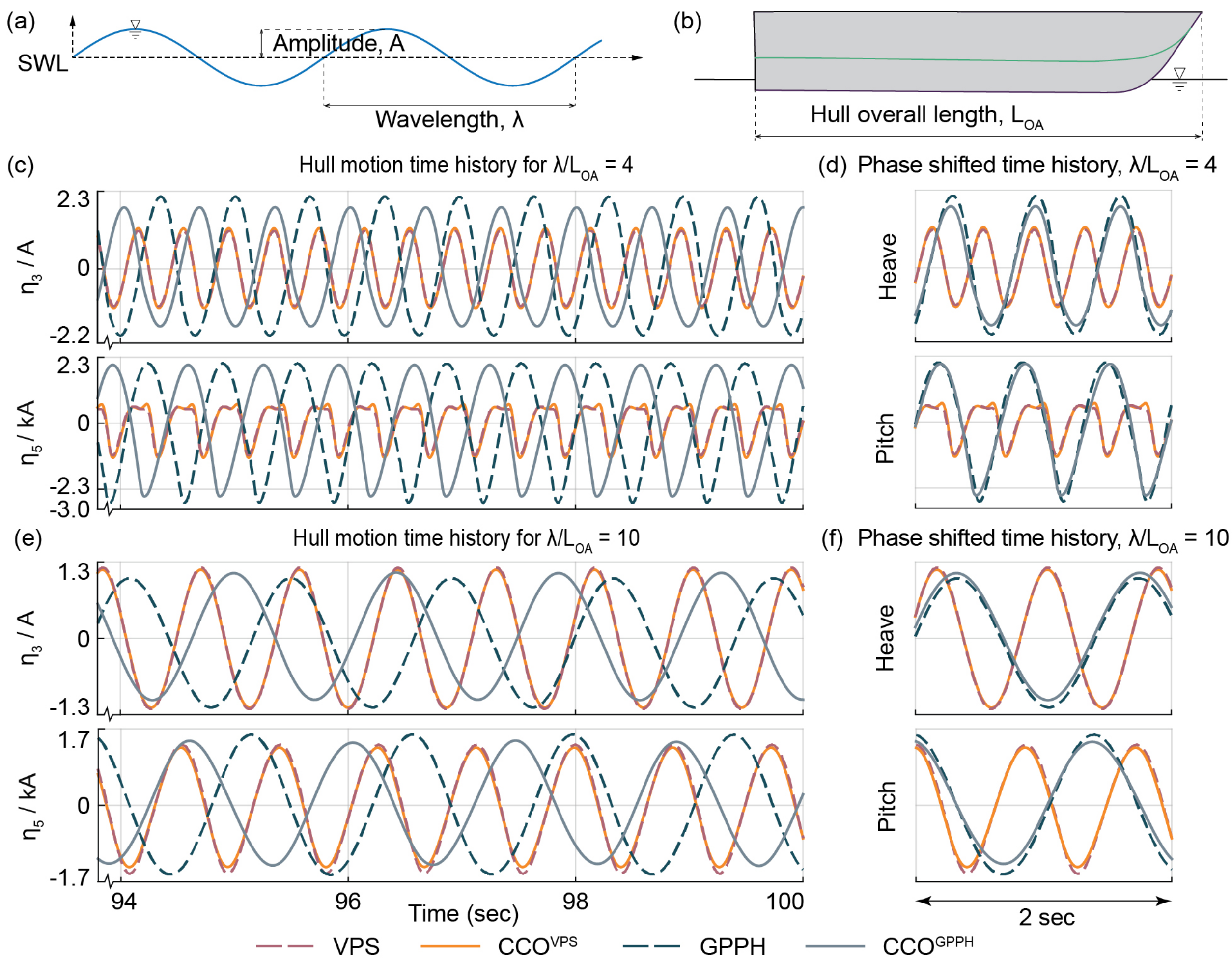


Figure 5.5: (a) Regular wave characteristics from a nominal Sea Water Level (SWL); (b) Overall length of the boat; Comparison of mean subtracted VPS and GPPH hull motion time histories with their CCO counterparts in head seas: (c) Six seconds of steady state time history for λ/L_{OA} = 4; (d) Phase shifted steady state time history over 2 second interval for λ/L_{OA} = 4; (e) Six seconds of steady state time history for λ/L_{OA} = 10; and (f) Phase shifted steady state time history over two second interval for λ/L_{OA} = 10. η_3/A is heave normalized by wave amplitude. η_5/kA is pitch normalized by wave slope.

are presented in Table 5.2. The small differences in the mean values stem from the variations in hull geometry and the modeling methods, as outlined earlier.

Table 5.2: Mean values of the normalized heave and pitch from Fig. 5.5

Hull geometry	**Mean heave, $\overline{\eta_3}/A$**		**Mean pitch, $\overline{\eta_5}/kA$**	
	$\lambda/L_{OA} = 4$	$\lambda/L_{OA} = 10$	$\lambda/L_{OA} = 4$	$\lambda/L_{OA} = 10$
CCO^{VPS}	2.846	2.127	2.252	3.354
VPS	2.807	2.199	1.933	2.988
CCO^{GPPH}	2.085	2.273	2.741	6.778
GPPH	2.788	2.494	2.854	5.939

The primary objective of these simulations was to acquire a deeper understanding of the similarities and differences in the hull motion of CCO hulls and their counterparts when subjected to waves. It is observed that the oscillatory hull motion for the GPPH and CCO^{GPPH} exhibits a significant phase shift. Nevertheless, the amplitudes and frequencies of the motion are nearly identical when evaluating the mean and phase-shifted data in Fig. 5.5 (d) and (f). This phase shift in hull motion can be explained by the modeling differences arising from the variations in L_{PP} and keel configurations between the GPPH and CCO^{GPPH} hull designs. In Powersea, the initial wave trough is positioned at the origin and coincides with the hull's FP at time t=0. Because the L_{PP} differs between the GPPH and CCO^{GPPH}, the LCG is located differently along each hull's length. This difference results in varying initial equilibrium positions and causes the hulls to encounter waves at different times at the same longitudinal position along the hull. The CCO^{GPPH} has an initial draft of 0.153 m and an initial trim of 4.389°. On the other hand, the GPPH has an initial draft of 0.150 m and an initial trim of 3.862°. Thus, the combined effects of the bow shape influencing the initiation of motion in wavy conditions, the differing initial equilibrium position of the hulls, and the time difference in wave encountered at the same longitudinal position along the hulls' length explain the phase shift between the motion histories of GPPH and CCO^{GPPH} hulls.

In contrast, for the VPS and CCO^{VPS} hulls, we observe no phase shift and nearly identical amplitude and frequency of the motion response for both λ/L_{OA} ratios, as shown in Fig. 5.5 (c-f). It is important to note that for the λ/L_{OA} ratio of 4, high amplitudes of heave and pitch motion are observed in all four hull geometries, indicating that a wave with λ/L_{OA} ratio of 4 excites the hulls more than a wave with λ/L_{OA} ratio of 10.

The findings from this section offer substantial evidence supporting the ability of CCO hulls to emulate the performance of their target hull shapes in both still and wavy water conditions.

5.4 Tunable Hydrodynamics With Shape-morphing Curved-origami Hulls

The curved-crease origami planing hull undergoes a gradual transformation from a flat sheet into the characteristic planing hull shape throughout its deployment. By controlling the resting angle of the fold lines located along the chine and keel of the curved-crease origami hull, we gain the ability to morph the hull shape on demand. This ability allows us to change the hull deadrise and grants control over various other geometric attributes, such as the overall length, beam width, bow curvature, center of gravity location, and the radius of gyration. These parameters collectively exert a significant influence on the hydrodynamic characteristics, allowing for on-demand shape morphing and tunability of hull hydrodynamics.

Here, we first discuss the influence of the rest angle of the fold lines on the hull deadrise and the overall hull geometry, which subsequently affects the hydrodynamic behavior. We will study three distinct hull configurations, all obtained from a single curved-crease pattern (starting configuration of flat sheets). As a starting point, we use the same curved-crease pattern as that developed for the VPS hull. The specific folded configurations correspond to deadrise angles of 10°, 20°, and 30° respectively, as graphically depicted in Fig. 5.6. A noticeable trend emerges as the hull's deadrise undergoes a transition from 10° to 30°. Specifically, this transition is marked by the following changes: an increase in the hull's overall length, a decrease in the beam width, a reduction in the bow's curvature, a forward displacement and vertical ascent of the center of gravity, along with an accompanying increase in the radius of gyration. In order to assess the influence of this morphable hull shape on the hull's hydrodynamic behavior, a series of simulations were conducted using the Powersea software in both calm waters and head seas. The geometric parameters that served as an input for these simulations corresponding to each configuration are presented in Table 5.3.

Table 5.3: Geometric parameters of the variable deadrise CCO hulls for hydrodynamic simulations

Hull geometry (deadrise)	**L$_{OA}$** (m)	**L$_{PP}$** (m)	**LCG** (% L$_{OA}$)	**VCG** (m)	**K$_{yy}$** (m)	**Weight** (N)	**B** (m)
10°	0.889	0.879	58.76	0.08	0.574	89.3	0.319
20°	0.945	0.935	60.53	0.096	0.623	89.3	0.304
30°	0.962	0.952	61.12	0.097	0.639	89.3	0.278

5.4.1 Calm Water Hydrodynamics

The simulations in still water for all three curved-crease origami hull configurations are conducted for various constant speed cases. The speeds (2 to 15.5 m/s) correspond to beam Froude numbers, F_B ranging from 1 to 9 for each of the three hull geometries. All simulation parameters adhere to the default values specified by Powersea, and the hydrodynamic characteristics are presented in Fig. 5.6. Similar to the preceding hydrodynamic investigation conducted in calm water, the attributes studied within this context include: (*i*) friction drag normalized by the hull weight, (*ii*) total resistance normalized by the hull weight, (*iii*) rise of CG normalized by the beam width, and (*iv*) trim angle of the hull.

The influence of shape change on the weight-normalized friction drag experienced by the hull across all deadrise variations is insignificant, except at very high speeds. As illustrated in Fig. 5.6 (a), the 20° deadrise hull exhibits slightly higher frictional resistance than 10° and 30° deadrise configurations at F_B close to 9. For a lower deadrise configuration, there is an increase in

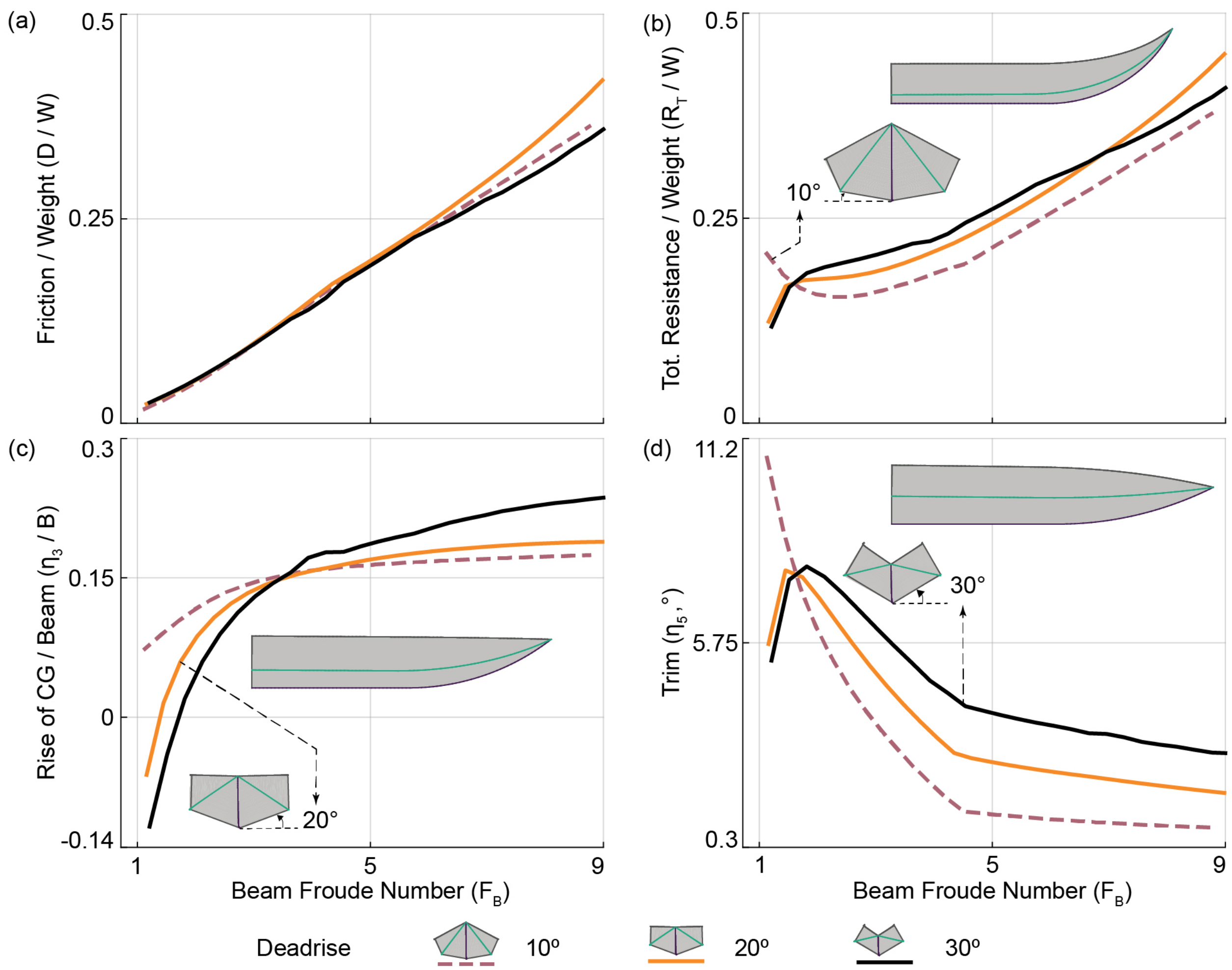


Figure 5.6: Effect of varying the CCO hull's deadrise on hydrodynamic performance in calm water: (a) Weight normalized friction drag; (b) Weight normalized total resistance; (c) Beam normalized rise of CG; (d) Trim angle (positive bow up).

the total resistance at lower speeds, but an interesting reversal of trends emerges at higher speeds. As seen in Fig. 5.6 (b), the lower deadrise configuration now exhibits lower total resistance. A similar trend is also observed for the trim angle of the hull, as shown in Fig. 5.6 (d). This change in trends for total resistance is expected because it is a derived quantity that depends upon the friction drag and trim angle. Since friction drag shows little variation across deadrise configurations, the trend of trim angle contributes to that of the total resistance. In the case of rise of CG, the low deadrise configuration tends to rise higher than the high deadrise configurations for F_B up to 4, as shown in Fig. 5.6 (c). A noticeable reversal in the hull rise trend emerges at high speeds. The variation in hull rise remains negligible across all deadrise configurations for practical considerations with the exception of the 30° deadrise configuration. These results can be interpreted in the following ways. In the case of a curved-crease origami hull configured with a lower deadrise, the planing surface typically takes on a flatter and broader profile. Consequently, with increasing surge speed, the hull tends to rise and glide over the water's surface. This phenomenon results in a reduced trim angle as the hull does not have to plow through the water, thereby leading to lower total resistance. In contrast, for a curved-crease origami hull configured with a higher deadrise, the planing surface is generally a shaper V-shape with a narrower profile. This hull tends to sink deeper into the water. Consequently, this hull has to overcome greater total resistance and exhibits increased trim angle at higher surge speeds.

5.4.2 Wavy Water Hydrodynamics

The investigation of hull motion in waves is carried out under the condition of a constant speed of 4.04 m/s when navigating head seas. The speed corresponds to F_B of 2.28, 2.34, and 2.44 for the 10°, 20°, and 30° deadrise configuration, respectively. Wave amplitudes of 0.02 m are applied to all three hull geometries. Each of the three hull geometries is subjected to regular waves with wavelengths equivalent to 4, 8, and 12 times the overall length of the planing hull. The mean subtracted steady-state hull motion is depicted in Fig. 5.7 (a-c). These figures display the final 10 seconds of the 100-second time histories and illustrate the heave normalized by wave amplitude and pitch normalized by wave slope of the hulls for the three λ/L_{OA} ratios.

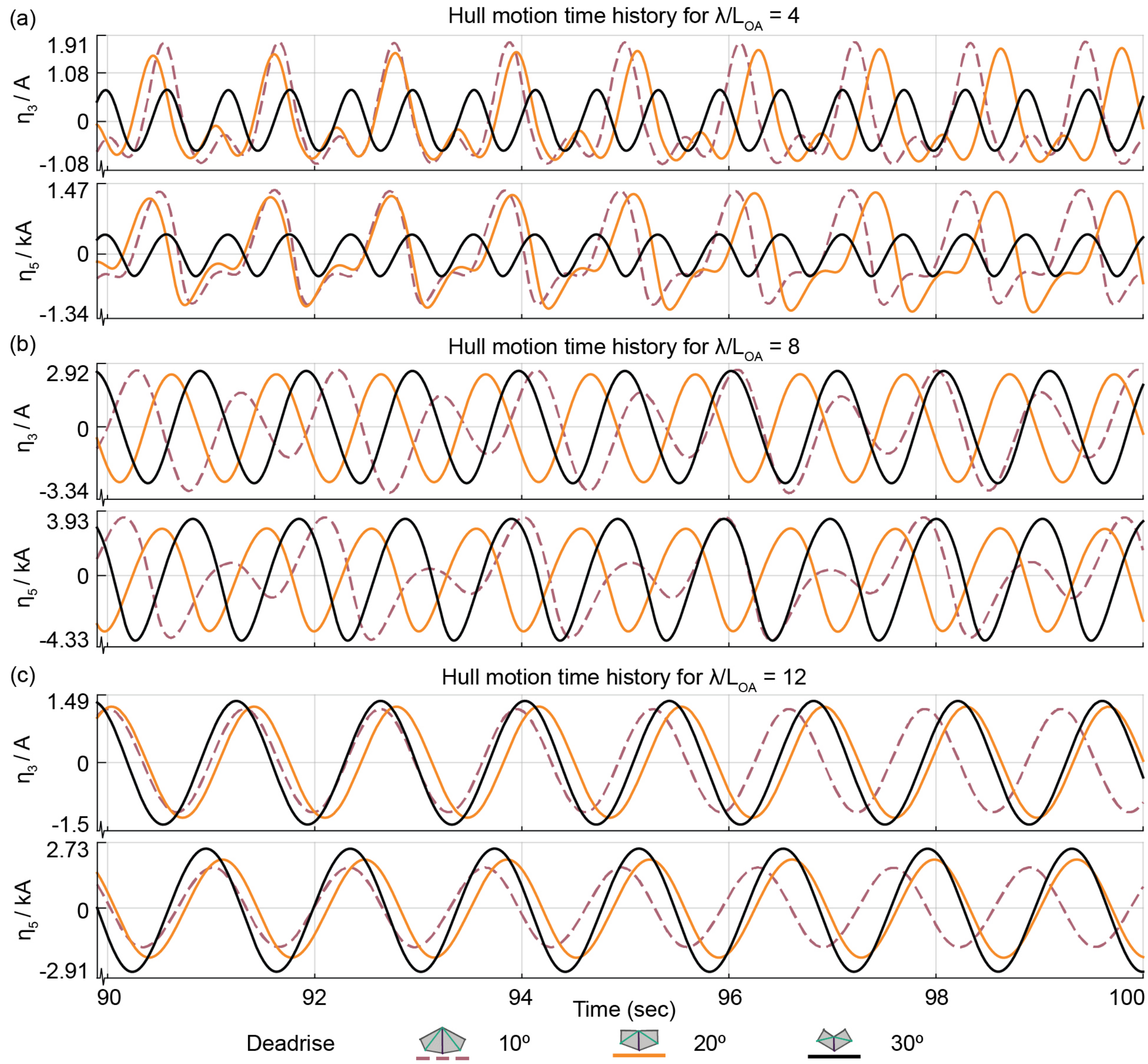


Figure 5.7: Effect of varying the CCO hull's deadrise on steady state hull motion in head seas for: (a) $\lambda/L_{OA} = 4$; (b) $\lambda/L_{OA} = 8$; (c) $\lambda/L_{OA} = 12$. Surge speed is 4.04 m/s, wave amplitude is 0.02 m and the last 10 seconds of the mean subtracted hull motion time history is shown. η_3/A is heave normalized by wave amplitude. η_5/kA is pitch normalized by wave slope.

The objective here is to understand the effects of varying deadrise configurations of the curved-crease origami hull on its performance under different wavy conditions. We observe that the hull configured with 30° deadrise exhibits strikingly smaller heave and pitch amplitudes for the λ/L_{OA} ratio of 4 compared to other configurations. As the λ/L_{OA} ratio increases to 8, the heave and pitch amplitudes increase for all hull shapes. Next, at λ/L_{OA} ratio of 12, there is a decrease in the motion amplitudes. The hull configured with 10° deadrise exhibits non-monochromatic hull motion, with higher amplitudes at shorter wavelengths, but reduces in amplitude and becomes monochromatic as the exciting wavelength increases. This hull shape has substantially higher amplitudes at shorter wave frequencies (i.e. λ/L_{OA} = 4), which can be expected for planing hulls with lower deadrise angles (flatter hull surface). Our observation here underscores the profound implications of altering the curved-crease origami hull's deadrise configuration on hull motion and overall hydrodynamics. Consequently, the shape-morphing ability of the hull can be harnessed to tune the hull response on-demand, making a single hull geometry adaptable to a wide range of sea conditions.

5.5 Physical Implementation

Translating these designs into practical, life-sized vessels offers several challenges. Critical consideration must be given to selecting appropriate materials for the panels and the flexible joints to ensure the hull's durability while meeting the demands of the naval industry. A preliminary examination of sheet curvature was conducted to identify suitable material options for curved origami panels. Deforming an initially flat sheet to conform to the curved-crease geometry generates significant bending stresses in the sheet itself. These bending stresses can be used to compute a maximum allowable thickness for the sheet, below which the material would remain elastic. The following formula was used to determine the maximum allowable sheet thickness for a given material to fabricate a curved crease origami hull: $t = 2\sigma_y/E\kappa$. Here, t represents the maximum sheet thickness, σ_y corresponds to the material's yield strength, E denotes the material's Young's modulus, and κ signifies the maximum sheet curvature, obtained from the Bar and Hinge model for a given length of the curved-crease origami hull. Fig. 5.8 (a) graphically depicts the maximum permissible sheet thickness for various materials, such as Aluminium, Glass fiber-reinforced polymer (GFRP), Polycarbonate, Acrylic, and Polyetherimide (Ultem®). This analysis, thus, serves as a crucial reference point for material selection and design considerations in fabricating the curved-crease origami hull.

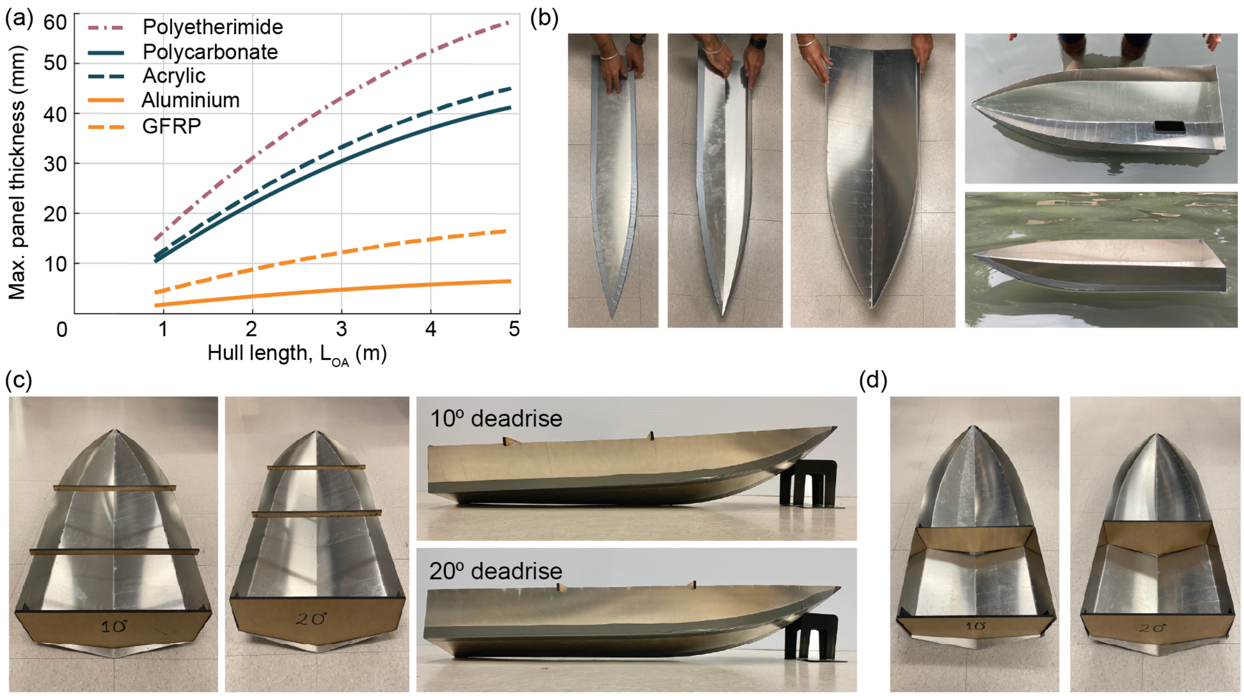


Figure 5.8: (a) Scalability analysis demonstrating the elastic limit (maximum thickness) for curved-crease origami hulls for different materials; (b) A 91 cm (36 in.) long curved-crease origami hull prototype fabricated using 1.5 mm thick aluminium sheets, joined together with duct tape for flexible joints — displaying the stowed (left) to deployed (right) configuration; (c) Aluminium CCO hull configured with deadrise configurations 10° and 20° strengthened with lateral stiffeners that snap onto the side of the hull; (d) Aluminium CCO hulls fitted with solid bulkhead.

A scaled model of the curved-crease origami planing hull is constructed by leveraging the results of the curvature analysis. Figure 5.8 (b) showcases a 91 cm (36 inch) long prototype afloat in a water pond, with a smartphone serving as a reference scale. This prototype is constructed using 1.5-millimeter-thick panels of 6061-grade aluminium. The aluminium panels are connected using heavy-duty duct tape along the curved creases, thus serving as the curved-origami fold joints. This prototype serves as an initial proof-of-concept demonstrating the material selection for prototypes to remain elastic. While simplistic, the joints of our prototype are watertight and allow for multiple fold-unfold cycles without degradation.

Post-deployment, the curved crease origami hulls require reinforcement to prevent collapse and torsional deformations. In Fig. 5.8 (c), we present the same aluminium CCO hull from Fig. 5.8 (b), but configured with two distinct deadrise angles — 10° and 20°. Each deadrise configuration is fortified with snap-fit lateral stiffeners, or girders, constructed from a mid-density fibreboard. These stiffeners effectively prevent collapse and provide sufficient reinforcement against torsional deformations, working in conjunction with the stiffness provided by the transom and curved aluminium sheets. The CCO hulls can also be fitted with solid bulkheads, as shown in Fig. 5.8 (d). In the future, the curved-crease origami hulls can be fitted with deployable bulkheads and stiffeners to strengthen the hull post-deployment.

A parallel fabrication effort was led by Jason P. Kelly and Daniel C. Hart at the Naval Surface Warfare Center, Carderock Division (NSWCCD) for a desktop-scale curved-crease origami-based planing hull [124]. This fabrication effort focused on the use of commonly available marine-composite material for the panels and flexible fiber-reinforced neoprene rubber for the curved folds to create an 18 in long prototype of an origami planing hull.

5.6 Concluding Remarks

This chapter introduces a hull fabrication approach based on curved-crease origami (CCO) principles. Employing the Bar and Hinge model for curved-origami folding and the Hausdorff distance metric, we explored the ability of two CCO hull variants to conform to conventional VPS and GPPH hull shapes. Obtaining conforming designs was achieved by optimizing the geometric parameters of the curved-crease pattern. Next, we demonstrated the ability of CCO hulls to emulate the hydrodynamic performance of conventional hull forms in still and wavy water conditions using the Powersea tool. Finally, we explored the shape-morphing ability of CCO hulls, which enables on-demand tunability of hydrodynamic performance to suit diverse sea conditions. Proof-of-concept prototypes are fabricated to demonstrate the feasibility of CCO for creating hulls.

CHAPTER 6

Conclusions and Future Work

This dissertation addressed the challenge of introducing reconfigurability into static structural systems without abandoning the load-bearing function that makes these systems useful. The central premise of the work is that reconfigurability in structural applications cannot be treated solely as a mechanism-design problem; reconfigurability must also be coupled with strength, stability, fabrication constraints, and predictable load transfer. This dissertation considered the inverse problem of starting with established structural forms and systematically introducing mobility, deployability, and programmable mechanical response. This philosophy was pursued through four objectives: developing a framework for transforming static trusses into reconfigurable systems using planar quadrilateral linkage principles; developing computational tools for evaluating the kinematic and mechanical response of the transformed systems; incorporating programmed hinge behavior to enable functional responses such as multistability and reusable energy-management structures; and broadening the framework beyond civil truss systems through curved-crease origami hulls that are offer rapid deployability and tunable hydrodynamic performance. Together, these objectives establish a foundation for the design and analysis of reconfigurable structural systems that can change shape while retaining meaningful structural function across multiple configurations.

6.1 Key Contributions of the Dissertation

The key contributions of this dissertation are summarized below.

A framework for transforming static trusses into reconfigurable systems and evaluating their kinematic behavior. This dissertation developed a systematic framework for converting static trusses into reconfigurable systems using principles of flat-foldable quadrilateral linkages. The method transforms triangular units of static trusses into quadrilateral linkages by introducing an additional node into selected members. The node location is determined using closed-form expressions, derived for all triangle geometries, that ensure the resulting quadrilateral linkage satisfies the Grashof flat-foldability criterion (Fig. 2.1). These nodes are placed on tensile members so that the transformed truss retains a stable load path and can carry the prescribed design load despite having mobility greater than one. The unit-level node-placement rules are extended to a system-level workflow that selects candidate members based on both the geometry of the triangular units and the corresponding member forces. The workflow applies to most statically determinate trusses with a continuous sequence of triangular units between the top and bottom chords. Its applicability was demonstrated using conventional truss layouts, including Scissor, Fink, and Warren trusses, as well as topology-optimized layouts, including cantilever, L-shaped, and serpentine trusses. A sequential kinematic simulator was also developed to visualize deployment paths and quantify packing performance as the transformed trusses morph through sequential actuation of their kinematic degrees of freedom. The resulting reconfigurable trusses can reduce their convex hull area by up to 93% and achieve compact storage while preserving the structural role of the original truss topology.

Proof-of-concept prototypes to demonstrate structural capacities of reconfigurable trusses. We validated the feasibility of constructing real-life reconfigurable trusses using our method by fabricating two proof-of-concept prototypes — a 30 cm reconfigurable cantilever truss and a 3 m Warren truss. Both prototypes employ a joint design strategy that distributes truss members in and out of the plane to ensure desired shape-morphing behavior without member contact or overlap. Load testing of the cantilever trusses confirmed that the reconfigurable truss exhibits force-stable behavior and maintains a peak load capacity and structural stiffness comparable to its static counterparts despite having mobility greater than one. The three-meter-long reconfigurable Warren truss weighing 38 kg supported a peak load of 19 kN, demonstrating the structural capacity and scalability of the proposed designs.

Computational tools for mechanical analysis of reconfigurable trusses. This dissertation developed computational tools for efficient nonlinear analysis of reconfigurable bar-linked structures whose global response is governed by joint rotational stiffness. The Three-Node Torsional Spring (3NTS) element provides a compact way to incorporate torsional springs into truss-like models without introducing rotational degrees of freedom or replacing joints with artificial beam elements. As a result, reconfigurable trusses with torsional springs can be analyzed using the same translational degrees of freedom as conventional truss models, while capturing large-displacement behavior, elastic buckling, and postbuckling response. This formulation provides the mechanical analysis capability needed to move beyond purely kinematic studies and evaluate how hinge stiffness influences force response, stability, and functional reconfiguration.

Programmable hinge mechanics for functional reconfigurable trusses. This dissertation extended the modeling capabilities of mechanical analysis tools to enable programmable mechanical response in reconfigurable trusses. An enhanced torsional spring law was introduced to bound the admissible rotation of the 3NTS element while preserving its intended linear response range. This law enables the simulation of sequential folding, locking, and load redistribution among the kinematic degrees of freedom of the truss. The framework was further expanded through magnetic multistable and idealized bistable hinge models, which introduce repeated energy barriers, multiple stable rotational states, and snap-through transitions in the reconfigurable trusses. These models show that local hinge behavior can be programmed to achieve a desired global structural response. In particular, reconfigurable trusses can manage impact energy through controlled geometric transformations rather than permanent material damage. This allows the primary members to remain elastic while the structure absorbs external work through recoverable reconfiguration. This capability provides a basis for reusable impact-energy management systems, controlled deployment paths, and intermediate stable configurations relevant to robotic structures, deployable space systems, and support frames for membrane or origami-based systems.

Rapidly deployable, shape-morphing curved-origami hulls with tunable hydrodynamic performance. Curved-crease origami hulls introduced in this dissertation can reproduce the hydrodynamic behavior of conventional planing hulls while also enabling geometry-based performance tuning. Because these hulls can closely conform to conventional hull geometries, their hydrodynamic response closely matches that of their conventional counterparts, including the VPS and GPPH hulls. Hydrodynamic simulations performed in Powersea show nearly equivalent trends in friction drag, total resistance, hull sinkage, and trim over a wide range of forward speeds in still water. In head-sea conditions, the curved-crease origami hulls also show close agreement with conventional hulls in the amplitude and frequency of hull motion time histories. These results indicate that hulls designed using curved-origami principles can preserve important performance characteristics associated with power efficiency, passenger comfort, and operational safety while providing the fabrication, storage, and deployment advantages of origami-based systems. In addition to replicating conventional hull behavior, the curved-crease origami hulls enable on-demand tuning of hydrodynamic performance by controlling the actuation of fold lines along the keel and chine. Changing the folding angle modifies key geometric features of the hull, especially the deadrise angle. Lower-deadrise configurations produce flatter planing surfaces that are better suited for calm water because they generally reduce sinkage, trim, and resistance. However, these configurations can produce larger heave and pitch in waves and are therefore less effective for seakeeping in wavy conditions. Higher-deadrise configurations produce sharper V-shaped planing surfaces that generally increase resistance, sinkage, and trim in still water but reduce heave and pitch amplitudes in waves. A single curved-crease origami hull can therefore transition between different deadrise configurations suited for different sea states, allowing the same structure to balance operational efficiency, passenger comfort, and seakeeping performance through controlled shape morphing.

6.2 Considerations for Future Work

While the reconfigurable trusses and origami-hulls introduced in this work offer exciting avenues for designing shape-morphing structures, several challenges must be overcome before translating these designs into practical structures. Future work can focus on tackling the following challenges:

General methods for designing reconfigurable trusses for three-dimensional structures. The node-introduction workflow described in Section 2.2.3 provides an effective method for transforming many two-dimensional statically determinate trusses into reconfigurable systems. However, the current implementation relies on prescribed rules for hinge placement and is best suited to trusses formed by a single sequence of triangular units. These assumptions limit its use for trusses with multiple layers of triangles, repeated triangular patterns, arbitrary polygonal cells, or three-dimensional connectivity. In such cases, introducing a new node can alter the mobility of several adjacent units. An unsuitable node or hinge location can therefore produce incompatible kinematics, reduce packing efficiency, or compromise structural stability. Future work should develop a more general design framework that identifies node locations, bar connectivity, and hinge-placement strategies without relying on case-specific rules. Graph-theoretic methods offer a promising direction for this generalization because trusses can be represented through nodes, edges, and connectivity constraints. Such a framework could identify the structural subgraphs requiring mobility, determine which bars should remain continuous for load transfer, and select hinge locations that preserve compatible deployment. This approach could also support the design of reconfigurable trusses generated by topology optimization, whose final layouts often include irregular connectivity and polygons with more than three sides. More broadly, a graph-based formulation would help extend the present framework from planar trusses to complex two-dimensional and three-dimensional truss systems while maintaining a direct connection between kinematics, load paths, and structural performance.

Mitigating structural overlap to optimize packing density. As the complexity and degrees of freedom for reconfigurable trusses increase, the potential for structural self-overlapping during actuation becomes increasingly evident. This issue is particularly highlighted in the Scissor truss example illustrated in Fig. 2.3 (a), where the actuation of the third degree of freedom results in portions of the structure overlapping. Real-life truss structures cannot achieve a compact state without providing special provisions for folding mechanisms. For such structures, it is crucial to understand their spatial movement patterns and identify the degrees of freedom whose actuation triggers a self-overlap. Addressing this challenge necessitates future investigation into varied sequences of degree-of-freedom actuation to mitigate structural overlap. Furthermore, automating the distribution of truss members out of the plane and incorporating additional methods for node placement (compressive members) will enhance the feasibility of reconfigurable structures. These practices can be explored to prevent self-overlapping and optimize the ability of the structure to minimize its footprint after shape transformations.

Coupled hydrodynamic and structural evaluation of shape-morphing curved-crease origami hulls. Future work should evaluate the coupled relationship between shape morphing, hydrodynamic loading, and structural response in curved-crease origami hulls. The shape change of these curved-origami hulls alters the deadrise angle and other geometric and inertial properties, including hull length, bow curvature, beam width, center of gravity, and radius of gyration. These changes can alter the hull surface pressure distribution, wave response, and local impact loads experienced by the hull. This effect is especially important near the bow, where planing hulls can experience high local pressures under wave-impact loading. The structural response must also be assessed because the shell thickness may be limited by the curvature and foldability requirements of the origami geometry, as shown in Fig. 5.8 (a). Future studies should therefore examine how payload, deadrise configuration, and hull geometry influence hydrodynamic pressures, global forces, stresses, and strains. Higher-fidelity computational models can be used for this purpose, including computational fluid dynamics and one-way coupled fluid-structure interaction analyses. These models can evaluate quantities that are not fully captured by simplified planning-hull tools, such as added wave resistance, spray resistance, local accelerations, and pressure variations over the hull surface. The relationship between shape morphing and seakeeping performance can also be studied using response amplitude operators, such as heave and pitch transfer functions, across different wavelengths and hull configurations. These analyses would clarify how changes in the curved-crease origami shape affect resonant behavior, impact loading, and structural demand. Two-tank experiments can further support the computational results by providing pressure, force, motion, and acceleration measurements for selected configurations.

APPENDIX A

Sequential Kinematic Simulation of Multi-DOF Reconfigurable Trusses

This section outlines the method for developing a sequential kinematic simulation program, which enables the movement of one degree of freedom at a time in any reconfigurable truss generated by the workflow in Section 2.2.3. The reconfigurable trusses in this study consist of two types of units, each with a single degree of freedom (DOF): (i) a Grashof quadrilateral linkage; and (ii) a unique configuration of a six-bar linkage. The kinematic simulator performs the position analysis of one DOF at a time, while applying rigid rotation and translation to the structure excluding DOFs that were actuated in previous steps. The position analysis for the two types of units is performed as follows.

A.1 Position Analysis of a General Quadrilateral Linkage

The Method of Projections [125, Ch. 4] is used to obtain the position of the nodes of a general quadrilateral linkage, as shown in Fig. A.1 (a-b). For a general quadrilateral linkage, θ_2 is the known angle made by the input link AB of length a. Link BC is the coupler link of length b and makes the angle θ_3 with the horizontal. Link CD is the output link of length c and makes the angle θ_4 with the horizontal. Link AD is the ground link with length d and is collinear with the X-axis

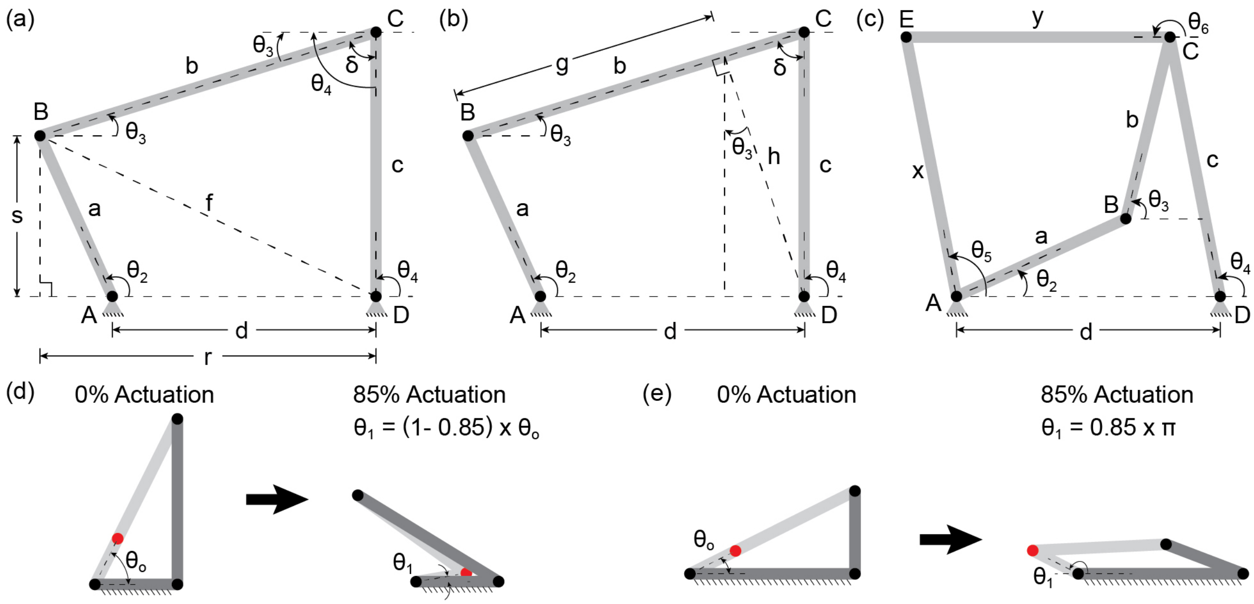


Figure A.1: Position analysis using the Method of Projections for: (a-b) a general quadrilateral linkage; (c) a unique configuration of a six-bar linkage; (d-e) Extent of actuation in terms of % actuation of the DOF explained for two different geometries of quadrilateral linkage. **Note:** A Ground link is not explicitly shown in (a-c) but is assumed to be present between nodes A and D.

(horizontal). The objective of the Method of Projection is to find the angles θ_3, θ_4, and coordinates of nodes B and C, given θ_2, a, b, c, and d.

In Fig. A.1 (a), f denotes the prime diagonal of the quadrilateral linkage and δ is the angle opposite to θ_2. The lengths r and s are determined using the projections of a as follows

$$r = d - a\cos\theta_2 \,, \tag{A.1}$$

$$s = a\sin\theta_2 \,. \tag{A.2}$$

Using Pythagoras' Theorem, we note that

$$f^2 = r^2 + s^2 \,. \tag{A.3}$$

From $\triangle$BCD, we also have

$$f^2 = b^2 + c^2 - 2bc\cos\delta \; , \tag{A.4}$$

$$\theta_4 = \theta_3 + \delta \; . \tag{A.5}$$

Simplifying Eq. A.3 and Eq. A.5 yields

$$\delta = \cos^{-1}\left(\frac{b^2 + c^2 - r^2 - s^2}{2bc}\right) \; . \tag{A.6}$$

Now consider the same quadrilateral linkage, as shown in Fig. A.1 (b). Here, we use δ and projections of c to find the lengths g and h as follows

$$h = c\sin\delta \; , \tag{A.7}$$

$$g = b - c\cos\delta \; . \tag{A.8}$$

Expressing r and s in terms of h and g, we get

$$r = g\cos\theta_3 + h\sin\theta_3 \; ,$$

$$s = h\cos\theta_3 - g\sin\theta_3 \; . \tag{A.9}$$

Simplifying further yields

$$\theta_3 = \tan^{-1}\left(\frac{hr - gs}{gr + hs}\right) \; . \tag{A.10}$$

With θ_3 known, we can now calculate θ_4, giving us sufficient information to determine the coordinates of each node relative to A (or the local origin) as well as use θ_4 to apply rigid rotation to the remainder of the structure. Points A and D remain fixed (as they are the ends of the ground link).

The coordinates of B are determined as follows

$$B_x = A_x + a\cos\theta_2 \ ,$$
$$B_y = A_y + a\sin\theta_2 \ . \tag{A.11}$$

Similarly, the coordinates of C are given by

$$C_x = D_x + c\cos\theta_4 \ ,$$
$$C_y = D_y + c\sin\theta_4 \ . \tag{A.12}$$

A.2 Position Analysis of Six-bar Linkage with a Single Degree of Freedom

The second type of unit observed in the reconfigurable trusses of this study is the 6-bar linkage with exactly one degree of freedom, as illustrated in Fig. A.1 (c). This type of linkage is formed when the two adjacent triangles are congruent and an additional node is introduced on the side shared by the two triangles. In this case, the objective is to find coordinates of B, C, and E, and angles θ_3, θ_4, θ_5, and θ_6, given θ_2, a, b, c, d, x, and y.

The analysis to obtain the coordinates of B and C, and angles θ_3 and θ_4 is identical to that performed in Appendix A.1. The coordinates of E are obtained by solving the following equations simultaneously, numerically.

$$(E_x - A_x)^2 + (E_y - A_y)^2 - x^2 = 0 \tag{A.13}$$
$$(E_x - C_x)^2 + (E_y - C_y)^2 - y^2 = 0 \tag{A.14}$$

There are two possible solutions for the above equations. We ensure the correctness of the solution by providing an initial guess for the coordinates E_x and E_y — coordinate location of E in the previous step of the simulation. With the coordinates of E known, we can determine angles θ_5 and θ_6. These angles, in addition to the displacement of C provide us enough information to determine the rigid rotation and translation that must be applied to the remaining structure, depending on the side the structure is connected to.

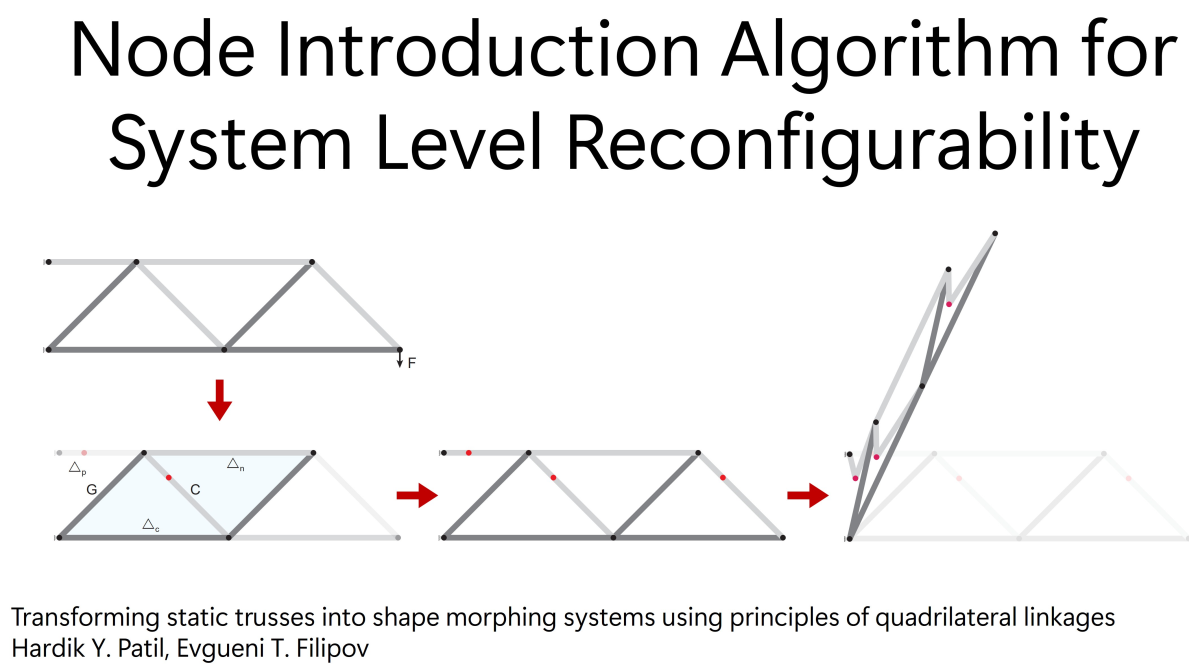
A.3 A Note on the Extent of Actuation of a Quadrilateral Linkage

In the sequential kinematic simulator, users can specify the extent of actuation for each DOF. The 0% actuation state represents the initial configuration, while the 100% actuation state corresponds to the flat-folded configuration of the quadrilateral linkage. At 0% actuation of the quadrilateral linkage, the decomposed links of the Candidate link are collinear, as shown in Fig. A.1 (d) and (e) on the left side. In this state, θ_o represents the angle between the shorter decomposed link and the Ground link. The transition to the flat-folded or 100% actuation state depends on the geometry of the linkage. The shorter link may rotate in either a clockwise or counterclockwise direction to achieve a flat-folded configuration. In the first case, the shorter link rotates clockwise until it makes an angle of $0°$ relative to the Ground link. For an intermediate state of 85% actuation, the shorter link is rotated to an angle of $\theta_1 = (1-0.85) \times \theta_o$ relative to the Ground link as shown in Fig. A.1 (d, right). In the second case, the shorter link rotates anti-clockwise until it makes an angle of $180°$ relative to the Ground link. For an intermediate state of 85% actuation, the shorter link is rotated to an angle of $\theta_1 = 0.85 \times \pi$ relative to the Ground link, as shown in Fig. A.1 (e, right).

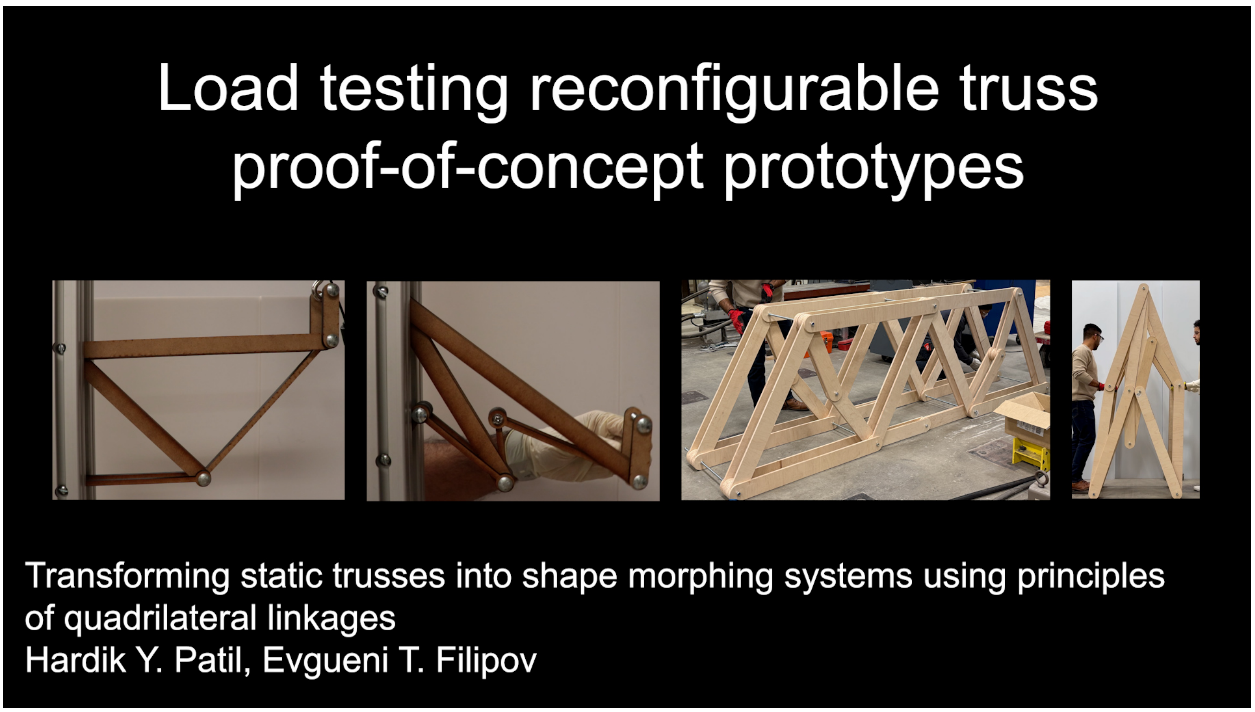
APPENDIX B

Supplementary Videos of Reconfigurable Trusses

The following supplementary videos accompanying Chapter 2 can be accessed at github.com/hardikyp/reconfigurable-trusses/tree/main/SupplementaryVideos.



Video B1: Node introduction workflow for system level reconfigurability in trusses.



Video B2: Load testing reconfigurable truss proof-of-concept prototypes.

APPENDIX C

Powersea: Low Aspect-ratio Strip Theory for Hull Hydrodynamics

The hydrodynamic performance of planing hulls discussed in this study is evaluated using Powersea, a simplified computational tool for rapidly estimating the motion and forces experienced by a hull. Powersea can estimate the heave, pitch, and surge motions of a hull shape, given a forward speed and water conditions, whether still or wavy. Therefore, it enables the investigation of the hull's porpoising behavior for seakeeping ability, effective power requirements, and dynamic natural frequencies in regular and irregular seas. This section explains the theoretical framework behind the low aspect-ratio strip theory algorithm and the underlying assumptions that make Powersea suitable for analyzing CCO planing hulls.

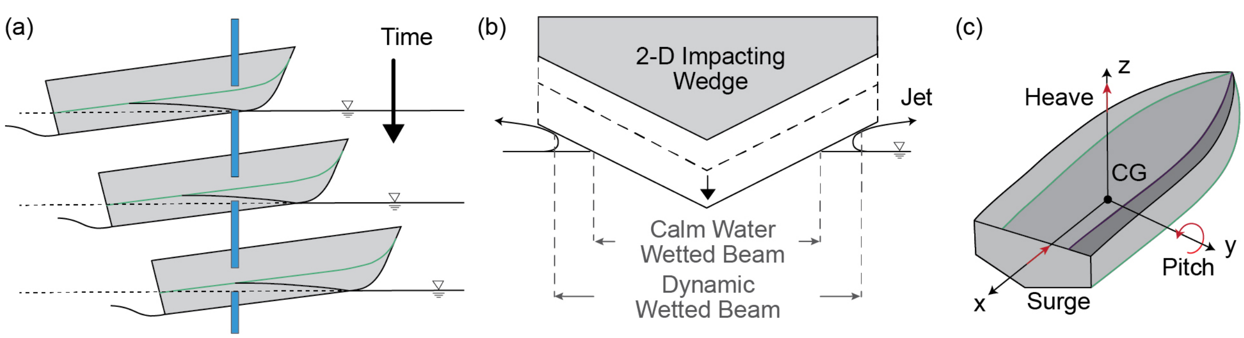


Figure C.1: Graphical representation of the assumptions of Powersea theory: (a) Observation plane indicating the section of the planing hull interacting with the water surface with the progression of time; (b) A single wedge impacting vertically on the water surface; (a) & (b) adapted from [126]; (c) Significant DOFs: Heave (η_3) – translation along the Z-axis, Pitch (η_5) – rotation about the Y-axis, and Surge (η_1) – translation along the X-axis.

The theoretical framework of Powersea is based on the Low-aspect-ratio strip theory for planing hulls developed by [127, 128]. This theory models the hull motion by discretizing the hull into a series of 2-D strips or wedges. In a water-fixed frame of observation, as illustrated in Fig. C.1 (a), these wedges impact the water surface with the progression of time. Figure C.1 (b) depicts a single wedge impacting the water surface and the assumptions used to calculate its hydrodynamic force per unit length. The total hydrodynamic forces acting on the hull body are then obtained by integrating the strip-wise hydrodynamic forces over the instantaneous wetted length of the hull. The 2-D strip theory makes three simplifying assumptions to reduce the number of active degrees of freedom of the hull and the total computational time as compared to a traditional full-scale computational fluid dynamics diffraction-radiation model. First, it is assumed that the perturbation velocities due to straight-line motion are considerably smaller in comparison to the velocity of the hull. Therefore, simplifying the equations of motion by setting the forward velocity of the hull to a constant. Second, only the vertical plane's force and pitch moment excitation are considered while formulating the equations of motion for each wedge, thereby reducing the total number of degrees of freedom. Third, the mathematical formulation of fluid flow is an empirical synthesis of theoretically derived flows. While this formulation is not explicit, there is sufficient confidence to expect promising results [129–131]. In addition to predicting hull motion and resistance in still water and waves, Powersea generates a longitudinal distribution of forces on the wedges and can be used to estimate pressures acting on the hull surface for the hull's structural integrity analysis. [132] outlines the details of the analysis procedure, strip-wise integration, solution of equations for Powersea, and a process to estimate sea loads on the planing surface based on Zarnick's formulation.

The assumptions of strip theory impose certain limitations on the hull geometry and hydrodynamics that powersea can simulate. Powersea assumes that the planing hull is hard-chined and has vertically extended sides. Furthermore, although Powersea predicts the heave motion, pitch motion, and calm water resistance accurately, the vertical acceleration of the hull is underpredicted by about 30% in regular and random seas [132] and the total resistance is underpredicted by about

15% due to not considering the whisker spray [133]. Nonetheless, the hydrodynamic properties are repeatable and can be tuned for engineering purposes. The above limitations are not of concern for our study, and in fact, the low aspect ratio strip theory is well suited for simulating origami-inspired planing hulls for the following three reasons. First, curved origami structures exhibit a distinctive change in topology or curved surface at the curved fold line, resembling the characteristics of a hard-chined vessel. Second, since the planing surface is the dominant geometric feature that influences the planing behavior of a hull, the shape or orientation of the side panels becomes less significant. Lastly, the computational simplicity facilitates rapid hydrodynamic simulation of various hull geometries within a short time frame. This capability is beneficial for investigating the relative change in hydrodynamic properties due to the effects of shape change resulting from origami folding. Powersea also assumes that the wavelengths encountered by the hull will be significantly larger than its length (wave slopes will be relatively small), and the hull speeds correspond to beam Froude numbers greater than 1 (that is, the vessel is advancing in semi-planing/planing regimes). These assumptions provide a reasonable approximation for many practical scenarios and ensure the applicability of Powersea for studying planing hulls in realistic water conditions.

BIBLIOGRAPHY

[1] Jörg Schneider. Introduction to safety and reliability of structures, volume 5. Iabse, 2006.

[2] Uta Hassler and Niklaus Kohler. Resilience in the built environment. Building Research & Information, 42(2):119–129, 2014.

[3] Martin Bechthold and James C Weaver. Materials science and architecture. Nature Reviews Materials, 2(12):17082, 2017.

[4] Seda Tosun and Feray Maden. Analysis of kinetic disaster relief shelters and a novel adaptive shelter proposal. Journal of Architectural Sciences and Applications, 8(1):438–455, 2023.

[5] Juan Pérez-Valcárcel, Manuel Muñoz-Vidal, Isaac López-César, Santiago Muñiz Gómez, Félix Suárez-Riestra, Manuel J Freire-Tellado, Jorge Aragón Fitera, and Emilio Mosquera-Rey. Deployable temporary housing for emergencies. The Journal of Architecture, 29(7-8):1157–1184, 2024.

[6] CG Foster and S Krishnakumar. A class of transportable demountable structures. International Journal of Space Structures, 2(3):129–137, 1987.

[7] Ashley P Thrall and CP Quaglia. Accordion shelters: A historical review of origami-like deployable shelters developed by the us military. Engineering structures, 59:686–692, 2014.

[8] Hua Yang, Xuan Liu, Weilin Lv, Yi Xia, Jiaming Liu, Jun Wei, Jianjun Zhang, Xu Han, and Wolfgang H Müller. Design and analysis of a reconfigurable origami tube with tunable load-bearing capacity. Thin-Walled Structures, 205:112452, 2024.

[9] Christopher B Churchill, David W Shahan, Sloan P Smith, Andrew C Keefe, and Geoffrey P McKnight. Dynamically variable negative stiffness structures. Science advances, 2(2):e1500778, 2016.

[10] Shannon A Zirbel, Robert J Lang, Mark W Thomson, Deborah A Sigel, Phillip E Walkemeyer, Brian P Trease, Spencer P Magleby, and Larry L Howell. Accommodating thickness in origami-based deployable arrays. Journal of mechanical design, 135(11):111005, 2013.

[11] Koryo Miura. Triangles and quadrangles in space. In Symposium of the International Association for Shell and Spatial Structures (50th. 2009. Valencia). Evolution and Trends in Design, Analysis and Construction of Shell and Spatial Structures: Proceedings. Editorial Universitat Politècnica de València, 2009.

[12] Suyeon Seo and Min-Woo Han. A review on deployable structures in space industry. International Journal of Precision Engineering and Manufacturing-Smart Technology, 3(1):83–96, 2025.

[13] Zhong You. Motion structures extend their reach. Materials Today, 10(12):52–57, 2007.

[14] Rainer Knäpper. Hebebuehne scissorlift.jpg. Wikimedia Commons, 2006. Licensed under CC BY-SA 2.0 Germany. Accessed 11 June 2026.

[15] Bromskloss and Andy Dingley. Ackermann simple design.svg. Wikimedia Commons, 2010. Derivative work by Andy Dingley from Ackermann.svg by User:Bromskloss. Licensed under CC BY-SA 3.0 Unported. Accessed 11 June 2026.

[16] Kenneth Snelson. The art of tensegrity. International journal of space structures, 27(2-3):71–80, 2012.

[17] Valter Böhm, Tobias Kaufhold, Florian Schale, and Klaus Zimmermann. Spherical mobile robot based on a tensegrity structure with curved compressed members. In 2016 IEEE international conference on advanced intelligent mechatronics (AIM), pages 1509–1514. IEEE, 2016.

[18] Evgueni T Filipov, Tomohiro Tachi, and Glaucio H Paulino. Origami tubes assembled into stiff, yet reconfigurable structures and metamaterials. Proceedings of the National Academy of Sciences, 112(40):12321–12326, 2015.

[19] Steven R Woodruff and Evgueni T Filipov. Curved creases redistribute global bending stiffness in corrugations: theory and experimentation. Meccanica, 56(6):1613–1634, 2021.

[20] Gary PT Choi, Levi H Dudte, and Lakshminarayanan Mahadevan. Programming shape using kirigami tessellations. Nature materials, 18(9):999–1004, 2019.

[21] Maria Redoutey, Ariel Roy, and Evgueni T Filipov. Pop-up kirigami for stiff, dome-like structures. International Journal of Solids and Structures, 229:111140, 2021.

[22] Neeraj Mandlekar, Mangala Joshi, and Bhupendra Singh Butola. A review on specialty elastomers based potential inflatable structures and applications. Advanced Industrial and Engineering Polymer Research, 5(1):33–45, 2022.

[23] Larry L Howell. Compliant mechanisms. In 21st century kinematics: The 2012 NSF Workshop, pages 189–216. Springer, 2013.

[24] Yunlan Zhang, David Restrepo, Mirian Velay-Lizancos, Nilesh D Mankame, and Pablo D Zavattieri. Energy dissipation in functionally two-dimensional phase transforming cellular materials. Scientific reports, 9(1):12581, 2019.

[25] Nahoko Yamamura, Daisuke Uriu, Mitsuru Muramatsu, Yusuke Kamiyama, Zendai Kashino, Shin Sakamoto, Naoki Tanaka, Toma Tanigawa, Akiyoshi Onishi, Shigeo Yoshida, et al. Social digital cyborgs: the collaborative design process of jizai arms. In Proceedings of the 2023 CHI conference on human factors in computing systems, pages 1–19, 2023.

[26] Dtom and Inductiveload. Landing gear schematic.svg. Wikimedia Commons, 2009. Original by Dtom; vectorization by Inductiveload. Public domain. Accessed 11 June 2026.

[27] Aarash YN Sofla, SA Meguid, KT Tan, and WK Yeo. Shape morphing of aircraft wing: Status and challenges. Materials & Design, 31(3):1284–1292, 2010.

[28] J Santiago-Prowald and Horst Baier. Advances in deployable structures and surfaces for large apertures in space. CEAS Space Journal, 5(3):89–115, 2013.

[29] Hardik Y. Patil and Evgueni T. Filipov. Transforming static trusses into shape morphing systems using principles of quadrilateral linkages. International Journal of Solids and Structures, 332:113913, 2026.

[30] M. Farshad and D. Isfahanian. Iranian plateau - the homeland of original truss structures. Journal of the American Oriental Society, 98(3):248–250, 1978.

[31] Robert Thorne. Structural Iron and Steel, 1850–1900. Routledge, 2017.

[32] Dario A Gasparini and Caterina Provost. Early nineteenth century developments in truss design in britain, france and the united states. Construction history, 5:21–33, 1989.

[33] Russell Charles Hibbeler and Kiang-Hwee Tan. Structural analysis. Pearson Prentice Hall Upper Saddle River, 2006.

[34] M Jon Turner, Ray W Clough, Harold C Martin, and LJ Topp. Stiffness and deflection analysis of complex structures. journal of the Aeronautical Sciences, 23(9):805–823, 1956.

[35] Ray W Clough. The finite element in plane stress analysis. Proc. 2^< nd> ASCE Confer. On Electric Computation, 1960, 1960.

[36] Mathias Stolpe. Truss optimization with discrete design variables: a critical review. Structural and Multidisciplinary Optimization, 53:349–374, 2016.

[37] Zhong You and Sergio Pellegrino. Foldable bar structures. International Journal of Solids and Structures, 34(15):1825–1847, 1997.

[38] Tae-Hyun Kim, Jong-Eun Suh, and Jae-Hung Han. Deployable truss structure with flat-form storability using scissor-like elements. Mechanism and Machine Theory, 159:104252, 2021.

[39] Jiten Patel and GK Ananthasuresh. A kinematic theory for radially foldable planar linkages. International journal of solids and structures, 44(18-19):6279–6298, 2007.

[40] Charles Hoberman. Radial expansion/retraction truss structures, June 18 1991. US Patent 5,024,031.

[41] Ichiro Ario, Masatoshi Nakazawa, Yoshikazu Tanaka, Izumi Tanikura, and Syuichi Ono. Development of a prototype deployable bridge based on origami skill. Automation in construction, 32:104–111, 2013.

[42] Greg Brouwer. Retractable arch will take center stage in salt lake. Civil Engineering, 72(1):15, 2002.

[43] Lara Alegria Mira, Ashley P Thrall, and Niels De Temmerman. Deployable scissor arch for transitional shelters. Automation in Construction, 43:123–131, 2014.

[44] PE Kassabian, Zhong You, and Sergio Pellegrino. Retractable roof structures. Proceedings of the Institution of Civil Engineers-Structures and Buildings, 134(1):45–56, 1999.

[45] Bo Han, Yundou Xu, Jiantao Yao, Dong Zheng, Yongjie Li, and Yongsheng Zhao. Design and analysis of a scissors double-ring truss deployable mechanism for space antennas. Aerospace Science and Technology, 93:105357, 2019.

[46] J Eddie Baker. An analysis of the bricard linkages. Mechanism and machine Theory, 15(4):267–286, 1980.

[47] SY Liu and Yan Chen. Myard linkage and its mobile assemblies. Mechanism and Machine Theory, 44(10):1950–1963, 2009.

[48] Yan Chen and Zhong You. Mobile assemblies based on the bennett linkage. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 461(2056):1229–1245, 2005.

[49] Xiaozhi Qi, Hailin Huang, Zhihuai Miao, Bing Li, and Zongquan Deng. Design and mobility analysis of large deployable mechanisms based on plane-symmetric bricard linkage. Journal of Mechanical Design, 139(2):022302, 2017.

[50] Kenneth D Snelson. Continuous tension, discontinuous compression structures, February 16 1965. US Patent 3,169,611.

[51] Gunnar Tibert. Deployable tensegrity structures for space applications. PhD thesis, KTH, 2002.

[52] Yixiang Liu, Qing Bi, Xiaoming Yue, Jiang Wu, Bin Yang, and Yibin Li. A review on tensegrity structures-based robots. Mechanism and Machine Theory, 168:104571, 2022.

[53] Gökhan Kiper, Eres Söylemez, and AU Özgür Kişisel. A family of deployable polygons and polyhedra. Mechanism and Machine Theory, 43(5):627–640, 2008.

[54] Yaxin Li and Sudarshan Krishnan. Geometric design and optimization of scissor-type deployable structures. Journal of Building Engineering, 65:105724, 2023.

[55] Suh In Kim and Yoon Young Kim. Topology optimization of planar linkage mechanisms. International Journal for Numerical Methods in Engineering, 98(4):265–286, 2014.

[56] Carlos José García-Mora and Jose Sánchez-Sánchez. Limitations in the design of deployable structures with straight scissors using identical elements. International Journal of Solids and Structures, 230:111171, 2021.

[57] Koryo Miura. Design and operation of a deployable truss structure. In NASA. Goddard Space Flight Center The 18th Aerospace Mech. Symp., 1984.

[58] AYN Sofla, DM Elzey, and HNG Wadley. Shape morphing hinged truss structures. Smart Materials and Structures, 18(6):065012, 2009.

[59] Yan Wang, Rongqiang Liu, Hui Yang, Qiang Cong, and Hongwei Guo. Design and deployment analysis of modular deployable structure for large antennas. Journal of Spacecraft and Rockets, 52(4):1101–1111, 2015.

[60] Louis R Giersch and Kevin Knarr. Deployment of a curved truss. Technical Report NPO-47269, National Aeronautics and Space Administration, 2010.

[61] Huaibo Yao, Yixin Huang, Wenlai Ma, Lei Liang, and Yang Zhao. Dynamic analysis of a large deployable space truss structure considering semi-rigid joints. Aerospace, 10(9):821, 2023.

[62] Heatherwick Studio. Rolling bridge. Retrieved May, 30:2020, 2002.

[63] R Packman. The rolling bridge, paddington basin. Structural Engineer, 83(2), 2005.

[64] Clark R Barker. A complete classification of planar four-bar linkages. Mechanism and Machine Theory, 20(6):535–554, 1985.

[65] Franz Grashof. Theoretische Maschinenlehre: Theorie der Getriebe und der mechanischen Meßinstrumente, volume 2. Voss, 1883.

[66] R Tyrrell Rockafellar. Convex analysis. Princeton mathematical series, 28, 1970.

[67] Tomás Zegard and Glaucio H Paulino. Grand—ground structure based topology optimization for arbitrary 2d domains using matlab. Structural and Multidisciplinary Optimization, 50:861–882, 2014.

[68] Zahur Ullah, Baseer Ullah, and Wajid Khan. Proportional topology optimisation with maximum entropy-based meshless method for minimum compliance and stress constrained problems. Engineering with Computers, 38(6):5541–5561, 2022.

[69] Ole Sigmund and Kurt Maute. Topology optimization approaches: A comparative review. Structural and multidisciplinary optimization, 48(6):1031–1055, 2013.

[70] Tomas Zegard, Christian Hartz, Arek Mazurek, and William F Baker. Advancing building engineering through structural and topology optimization. Structural and Multidisciplinary Optimization, 62(2):915–935, 2020.

[71] Robert J Ross et al. Wood handbook: wood as an engineering material. USDA Forest Service, Forest Products Laboratory, General Technical Report FPL-GTR-190, 2010: 509 p. 1 v., 190, 2010.

[72] KCP Lum, SL Ng, and I Black. Co2 laser cutting of mdf: 1. determination of process parameter settings. Optics & laser technology, 32(1):67–76, 2000.

[73] Ray Tahir Mushtaq, Yanen Wang, Mudassar Rehman, Aqib Mashood Khan, and Mozammel Mia. State-of-the-art and trends in co2 laser cutting of polymeric materials—a review. Materials, 13(17):3839, 2020.

[74] Hardik Y Patil and Evgueni T Filipov. Three-node torsional spring element formulation for the analysis of reconfigurable bar-linked structures. Journal of Applied Mechanics, 93(3):034502, 2026.

[75] Carlos A Felippa. A historical outline of matrix structural analysis: a play in three acts. Computers & Structures, 79(14):1313–1324, 2001.

[76] Bongsu Kim and Ashish D Deshpande. Design of nonlinear rotational stiffness using a noncircular pulley-spring mechanism. Journal of Mechanisms and Robotics, 6(4):041009, 2014.

[77] Jorge A Franco, Juan A Gallego, and Just L Herder. Static balancing of four-bar compliant mechanisms with torsion springs by exerting negative stiffness using linear spring at the instant center of rotation. Journal of Mechanisms and Robotics, 13(3):031010, 2021.

[78] Brian T Knox and James P Schmiedeler. A unidirectional series-elastic actuator design using a spiral torsion spring. Journal of Mechanical Design, 131(12), 2009.

[79] Claire Dumas, Stéphane Caro, Mehdi Cherif, Sébastien Garnier, and Benoît Furet. Joint stiffness identification of industrial serial robots. Robotica, 30(4):649–659, 2012.

[80] W Thomas Edwards. Effect of joint stiffness on standing stability. Gait & posture, 25(3):432–439, 2007.

[81] Cheng Fang, Arash Ajoudani, Antonio Bicchi, and Nikos G Tsagarakis. Online model based estimation of complete joint stiffness of human arm. IEEE Robotics and Automation Letters, 3(1):84–91, 2017.

[82] Andrew Gillman, Kazuko Fuchi, and PR Buskohl. Truss-based nonlinear mechanical analysis for origami structures exhibiting bifurcation and limit point instabilities. International Journal of Solids and Structures, 147:80–93, 2018.

[83] Yi Zhu, Mark Schenk, and Evgueni T Filipov. A review on origami simulations: from kinematics, to mechanics, toward multiphysics. Applied Mechanics Reviews, 74(3):030801, 2022.

[84] Lu Lu, Sophie Leanza, and Ruike Renee Zhao. Origami with rotational symmetry: a review on their mechanics and design. Applied Mechanics Reviews, 75(5):050801, 2023.

[85] Yunlan Zhang, Mirian Velay-Lizancos, David Restrepo, Nilesh D Mankame, and Pablo D Zavattieri. Architected material analogs for shape memory alloys. Matter, 4(6):1990–2012, 2021.

[86] Shaowei Zhu, Xiaojun Tan, Shuai Chen, Bing Wang, Li Ma, and Linzhi Wu. Quasi-all-directional negative stiffness metamaterials based on negative rotation stiffness elements. physica status solidi (b), 257(6):1900538, 2020.

[87] TAMER M WASFY. A torsional spring-like beam element for the dynamic analysis of flexible multibody systems. International Journal for Numerical Methods in Engineering, 39(7):1079–1096, 1996.

[88] ME Kartal, HB Basaga, A Bayraktar, and Murat Muvafık. Effects of semi-rigid connection on structural responses. Electronic journal of structural Engineering, 10:22–35, 2010.

[89] Juan Camilo Molina-Villegas and Jorge Eliecer Ballesteros Ortega. Closed-form solution of timoshenko frames with semi-rigid connections. In Structures, volume 48, pages 212–225. Elsevier, 2023.

[90] Douaa Raheem Mohammed and Murtada A Ismael. Effect of semi-rigid connection on post-buckling behaviour of frames using finite element method. Civil Engineering Journal, 5(7):1619–1630, 2019.

[91] Mark Schenk, Simon D Guest, et al. Origami folding: A structural engineering approach. Origami, 5:291–304, 2011.

[92] Ke Liu and Glaucio H Paulino. Nonlinear mechanics of non-rigid origami: an efficient computational approach. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 473(2206):20170348, 2017.

[93] ET Filipov, K Liu, Tomohiro Tachi, Mark Schenk, and Glaucio H Paulino. Bar and hinge models for scalable analysis of origami. International Journal of Solids and Structures, 124:26–45, 2017.

[94] Steven R Woodruff and Evgueni T Filipov. A bar and hinge model formulation for structural analysis of curved-crease origami. International Journal of Solids and Structures, 204:114–127, 2020.

[95] W. McGuire, R.H. Gallagher, and R.D. Ziemian. Matrix Structural Analysis. Wiley, 2015.

[96] Guoxing Lu and TX Yu. Energy absorption of structures and materials. Elsevier, 2003.

[97] Jongmin Shim, Sicong Shan, Andrej Košmrlj, Sung H Kang, Elizabeth R Chen, James C Weaver, and Katia Bertoldi. Harnessing instabilities for design of soft reconfigurable auxetic/chiral materials. Soft Matter, 9(34):8198–8202, 2013.

[98] Sahab Babaee, Jongmin Shim, James C Weaver, Elizabeth R Chen, Nikita Patel, and Katia Bertoldi. 3d soft metamaterials with negative poisson's ratio. Adv. Mater, 25(36):5044–5049, 2013.

[99] Ofek Peretz, Ezra Ben Abu, Anna Zigelman, Sefi Givli, and Amir D Gat. Multistable metafluid based energy harvesting and storage. Advanced Materials, 35(35):2301483, 2023.

[100] Ofek Peretz, Ezra Ben Abu, Anna Zigelman, Sefi Givli, and Amir D Gat. A metafluid with multistable density and internal energy states. Nature Communications, 13(1):1810, 2022.

[101] Chan Soo Ha, Roderic S Lakes, and Michael E Plesha. Design, fabrication, and analysis of lattice exhibiting energy absorption via snap-through behavior. Materials & Design, 141:426–437, 2018.

[102] Sicong Shan, Sung H Kang, Jordan R Raney, Pai Wang, Lichen Fang, Francisco Candido, Jennifer A Lewis, and Katia Bertoldi. Multistable architected materials for trapping elastic strain energy. Adv. Mater, 27(29):4296–4301, 2015.

[103] Gwanho Yoon, Inki Kim, and Junsuk Rho. Challenges in fabrication towards realization of practical metamaterials. Microelectronic Engineering, 163:7–20, 2016.

[104] Sarah CL Fischer, Leonie Hillen, and Chris Eberl. Mechanical metamaterials on the way from laboratory scale to industrial applications: Challenges for characterization and scalability. Materials, 13(16):3605, 2020.

[105] Hardik Y Patil, Kevin J Maki, and Evgueni T Filipov. Rapidly deployable hulls and on-demand tunable hydrodynamics with shape morphing curved crease origami. Journal of Fluids and Structures, 130:104176, 2024.

[106] Robert Carter. Boat remains and maritime trade in the persian gulf during the sixth and fifth millennia bc. Antiquity, 80(307):52–63, 2006.

[107] Alessandro Paoli and Armando V. Razionale. Large yacht hull measurement by integrating optical scanning with mechanical tracking-based methodologies. Robotics and Computer-Integrated Manufacturing, 28(5):592–601, 2012.

[108] George Bruce. Ship construction. Shipbuilding Management, pages 153–162, 2021.

[109] MP de Alwis, K Garme, Riccardo Lo Martire, JI Kåsin, and Björn Äng. Crew acceleration exposure, health and performance in high-speed operations at sea. In The 11th Symposium on High-Speed Marine Vehicles, Naples, Italy, 2017.

[110] Marco Meloni, Jianguo Cai, Qian Zhang, Daniel Sang-Hoon Lee, Meng Li, Ruijun Ma, Teo Emilov Parashkevov, and Jian Feng. Engineering origami: A comprehensive review of recent applications, design methods, and tools. Advanced Science, 8(13):2000636, 2021.

[111] Guanyu Chen, Dongsheng Chen, Jessica Weakly, and Cynthia Sung. Drag coefficient characterization of the origami magic ball. In ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (IDETC/CIE), 2023.

[112] Fuwen Hu, Zichuan Kou, Eyob Messele Sefene, and Tadeusz Mikolajczyk. An origami flexiball-inspired soft robotic jellyfish. Journal of Marine Science and Engineering, 11(4), 2023.

[113] Shannon A Zirbel, Brian P Trease, Mark W Thomson, Robert J Lang, Spencer P Magleby, and Larry H Howell. Hanaflex: a large solar array for space applications. In Micro- and Nanotechnology Sensors, Systems, and Applications VII, volume 9467, pages 179–187. SPIE, 2015.

[114] Kaori Kuribayashi, Koichi Tsuchiya, Zhong You, Dacian Tomus, Minoru Umemoto, Takahiro Ito, and Masahiro Sasaki. Self-deployable origami stent grafts as a biomedical application of ni-rich tini shape memory alloy foil. Materials Science and Engineering: A, 419(1–2):131–137, 2006.

[115] Landen Bowen, Brian Trease, Mary Frecker, and Timothy Simpson. Dynamic modeling and analysis of an origami-inspired optical shield for the starshade spacecraft. In Smart Materials, Adaptive Structures and Intelligent Systems, volume 50480, page V001T01A012. American Society of Mechanical Engineers, 2016.

[116] Levi H Dudte, Etienne Vouga, Tomohiro Tachi, and Lakshminarayanan Mahadevan. Programming curvature using origami tessellations. Nature Materials, 15(5):583–588, 2016.

[117] Deane L Harder, Olga Speck, Catriona L Hurd, and Thomas Speck. Reconfiguration as a prerequisite for survival in highly unstable flow-dominated habitats. Journal of Plant Growth Regulation, 23:98–107, 2004.

[118] K Miura. Science of miura-ori: A review. In Proceedings of 4th International Conference on Origami in Science, Mathematics, and Education (4OSME), 2008.

[119] Erik Demaine, Martin Demaine, Duks Koschitz, and Tomohiro Tachi. Curved crease folding a review on art, design and mathematics. In Proceedings of 35th Annual Symposium of IABSE/52nd Annual Symposium of IASS/6th International Conference on Space Structures: Taller, Longer, Lighter — Meeting growing demand with limited resources, page 1, London, United Kingdom, September 2011.

[120] Daniel Savitsky. Planing craft. Naval Engineers Journal, 97(2):113–141, 1985.

[121] Sasan Tavakoli, Mingyang Zhang, Aleksander A Kondratenko, and Spyros Hirdaris. A review on the hydrodynamics of planing hulls. Ocean Engineering, 303:117046, 2024.

[122] Dong Jin Kim, Sun Young Kim, Young Jun You, Key Pyo Rhee, Seong Hwan Kim, and Yeon Gyu Kim. Design of high-speed planing hulls for the improvement of resistance and seakeeping performance. International Journal of Naval Architecture and Ocean Engineering, 5(1):161–177, 2013.

[123] Lawrence J Doctors. Hydrodynamics of high-speed small craft. Technical report, 1985.

[124] Jason P. Kelly and Daniel C. Hart. Desktop-scale composite origami hull production demonstration and folding response prediction. Technical Report NSWCCD-65-TR-2024/02, Naval Surface Warfare Center, Carderock Division, West Bethesda, MD, April 2025. Distribution Statement A: Approved for public release; distribution is unlimited.

[125] Eric Constans and Karl B Dyer. Introduction to mechanism design: With computer applications. CRC Press, 2018.

[126] Christopher D Barry, Debabrata Ghosh, Richard Akers, and Andrew Ulak. Implementation, application and validation of the zarnick strip theory analysis technique for planing boats. In Proceedings of High Performance Yacht Design Conference, number P2002–7, Auckland, New Zealand, 2002.

[127] Ernest E. Zarnick. A nonlinear mathematical model of motions of a planing boat in regular waves. Technical Report DTNSRDC–78/032, David W. Taylor Naval Ship Research and Development Center, Bethesda, Maryland, 1978.

[128] Ernest E. Zarnick. A nonlinear mathematical model of motions of a planing boat in irregular waves. Technical Report DTNSRDC/SPD–0867–01, David W. Taylor Naval Ship Research and Development Center, Bethesda, Maryland, 1979.

[129] O Faltinsen and Rui Zhao. Numerical predictions of ship motions at high forward speed. Philosophical Transactions of the Royal Society of London. Series A: Physical and Engineering Sciences, 334(1634):241–252, 1991.

[130] Hui Sun and Odd M Faltinsen. The influence of gravity on the performance of planing vessels in calm water. Journal of Engineering Mathematics, 58:91–107, 2007.

[131] Alexander A Korobkin, Tatyana I Khabakhpasheva, and Kevin J Maki. Hydrodynamic forces in water exit problems. Journal of Fluids and Structures, 69:16–33, 2017.

[132] Richard H. Akers. Dynamic analysis of planing hulls in the vertical plane. In Proceedings of The Society of Naval Architects and Marine Engineers, New England, 1999.

[133] Daniel Savitsky, Michael F DeLorme, and Raju Datla. Inclusion of whisker spray drag in performance prediction method for high-speed planing hulls. Marine Technology and SNAME News, 44(01):35–56, 2007.

[134] Antonio Alessandro Deleo, James O'Neil, Hiromi Yasuda, Marco Salviato, and Jinkyu Yang. Origami-based deployable structures made of carbon fiber reinforced polymer composites. Composites Science and Technology, 191:108060, 2020.

[135] Peter Halswell, Philip A Wilson, Dominic J Taunton, and Stephen Austen. Design and performance of inflatable boats: flexibility and environmental considerations. 2011.

[136] A Deleo, J O'Neil, Hiromi Yasuda, Jinkyu Yang, and Marco Salviato. Composite origami: Foldable structures based on tachi-miura-polyhedron origami technique. Proceedings of SAMPE, 2018.

[137] Brittani R Russell, Ashley P Thrall, Joseph A Padula, and Jimmy E Fowler. Reconceptualization and optimization of a rapidly deployable floating causeway. Journal of Bridge Engineering, 19(4):04013013, 2014.

[138] Xiaofei Ma, Ning An, Qiang Cong, Jiang-Bo Bai, Minger Wu, Yan Xu, Jinxiong Zhou, Dayu Zhang, Taotao Zhang, Ruiwen Guo, et al. Design, modeling, and manufacturing of high strain composites for space deployable structures. Communications Engineering, 3(1):78, 2024.